

TW128: A Tree-Structured Duplex Authenticated Encryption Mode

Coda Hale

Hale Institute for Bicycle Appreciation

Abstract. AEAD tags are often treated as compact commitments to the full encryption context, but standard deployed modes such as AES-GCM and ChaCha20-Poly1305 do not provide this property. We present TW128, a nonce-based authenticated encryption mode with associated data designed around the stronger tag-centric requirement: the authentication tag should itself behave like a digest of the full (K, U, A, M) context. TW128 adapts the KangarooTwelve tree-hashing pattern to keyed duplex encryption over KECCAK- $p[1600, 12]$. A serial root duplex binds the key, nonce, associated data, and first message chunk, then aggregates tags from independently keyed leaf duplexes for the remaining chunks. The leaf decomposition is canonical, while the number of leaves evaluated at once is an implementation choice; scalar, SIMD, and multicore implementations therefore produce identical ciphertexts and tags without exposing the degree of parallelism as a mode parameter.

The final root tag is both the AEAD authentication tag and the object of the main security analysis. In the public permutation model, via an intermediate random oracle and the flat sponge lift, we prove that the wrapper returning this tag is a secure hash of the full input: collision-, second-preimage-, and preimage-resistant in the standard hash-function sense. The same root tag analysis yields AEAD-facing committing security, including full-input CMTDs-4. The root-centered structure also gives integrity under release of unverified plaintext in the sense of [ABL⁺14]. As baseline AEAD properties, we prove concrete multi-user indistinguishability and derive all-in-one authenticated encryption as the single-user corollary. At the implemented parameters, all stated notions meet a 128-bit security target. Vectorized `amd64` and `arm64` implementations recover the scalable performance profile of the tree design: in bulk they reach 48 Gbit/s, with the AVX-512 and `arm64` paths delivering 51–76% of hardware-accelerated AES-128-GCM throughput.

Keywords: authenticated encryption · committing security · sponge constructions · duplex constructions · Keccak

1 Introduction

Authenticated encryption with associated data (AEAD) is often used as though its tag were a compact digest of the whole encryption context: a short value binding the key, nonce, associated data, and message. Ordinary AEAD security does not give that property. AES-GCM and ChaCha20-Poly1305, for example, are not key-committing: an adversary can craft a single ciphertext that decrypts to valid plaintexts under two or more keys [DGRW18, ADG⁺22]. The problem is structural. These modes authenticate with algebraic MACs whose outputs can be made to validate under another context, while a digest is expected to resist both collisions and preimages.

This mismatch has practical consequences. The “invisible salamander” attack breaks Facebook’s attachment franking scheme [DGRW18]; related attacks construct ciphertexts that open to different well-formed files under different keys [ADG⁺22]; partitioning oracles

recover passwords from protocols built on non-committing AEAD [LGR21]; and context discovery attacks find valid decryption contexts for ciphertexts under standard modes such as CCM, EAX, and SIV, defeating the preimage side of the digest intuition [MLGR23]. These attacks differ in surface form, but they expose the same design gap: applications compare, store, or reason about the tag as a compact commitment, while the mode proves only ordinary authenticity.

Recent work formalizes parts of this expectation. Compactly committing AEAD was introduced for message franking [GLR17], encryption builds commitment into a dedicated primitive [DGRW18], and the committing hierarchy of [BH22] asks whether a ciphertext and tag can have another valid opening. Those are AEAD-facing notions over ciphertexts. The object applications often treat as the commitment, however, is the tag itself. This paper therefore takes the tag as the primary security object: we analyze the root tag with standard hash-function notions, then recover the AEAD committing games as consequences of the same root tag analysis.

TW128 (*Tree Wrap*, with 128 denoting the 128-bit security target) is designed around that tag-centric requirement. It adapts the KangarooTwelve [BDP⁺16] tree-hashing pattern to keyed duplex encryption over KECCAK- p [1600, 12]. A serial root duplex binds the key, nonce, associated data, and first message chunk. Independently keyed leaf duplexes encrypt the remaining chunks and return leaf tags, which the root aggregates before emitting the final root tag. The result is an AEAD mode whose authentication tag is also the natural digest point for the full (K, U, A, M) context.

The tree shape also separates transcript compatibility from implementation parallelism. The root/leaf decomposition and leaf order are fixed by the mode, but the number of leaves evaluated at once is not. Scalar, SIMD, and multicore implementations therefore produce identical ciphertexts and tags without negotiating a parallelism degree or exposing it on the wire. Single-chunk messages use only the root duplex, matching KangarooTwelve’s short-input behavior and avoiding tree overhead (Section 8.4).

The main security result is that the wrapper H_{TW128} returning the final root tag is a secure hash of the full input in the sense of [RS04]: collision-, second-preimage-, and preimage-resistant. The same proof spine yields AEAD-facing committing security, including full-input CMTDs-4. The root-centered structure also gives INT-RUP integrity [ABL⁺14]: even when an adversary can obtain unverified stream plaintext, fresh accepting ciphertexts remain bounded by the same tag-guessing and collision terms. We also prove the baseline AEAD security bounds needed of a nonce-based mode: concrete multi-user indistinguishability [BT16] in the public permutation model, with all-in-one authenticated encryption [RS06] as the single-user corollary.

At the implemented parameters ($k = c = \nu = \tau_{\text{root}} = \tau_{\text{leaf}} = 256$), every stated advantage is at most 2^{-128} for adversaries running up to $N_{\text{tot}} \leq 2^{63}$ KECCAK- p [1600, 12] calls, on the order of 2^{71} bytes of processed data. The cap is a data-scale choice rather than a security ceiling: the dominant terms are birthday-type on the 256-bit capacity and become vacuous only near 2^{128} calls (Section 7.1). Optimized `amd64` and `arm64` implementations preserve the canonical transcript while evaluating leaf duplexes in batches; in bulk, TW128 reaches 48 Gbit/s with AVX-512 and 39 Gbit/s on an Apple M4 Pro, 51–76% of hardware-accelerated AES-128-GCM throughput (Section 8).

1.1 Our Contributions

- **A tag-first tree AEAD mode.** We specify TW128, a nonce-based AEAD with associated data built as a two-level tree of strict [BDPVA11] duplexes over KECCAK- p [1600, 12] (Section 3). A root duplex emits the authentication tag and acts as the digest point for the full encryption context, while indexed leaf duplexes provide scalable message processing. The mode fixes the root/leaf transcript but

- 94 leaves the number of leaves evaluated simultaneously to the implementation, so scalar
95 and vectorized implementations interoperate without changing ciphertexts or tags.
- 96 • **Root tag hash security.** We prove that the wrapper H_{TW128} returning the root
97 tag is a secure hash of the full (K, U, A, M) input in the sense of [RS04]: collision-,
98 second-preimage-, and preimage-resistant (Theorems 4 to 6 and Corollary 2). Thus
99 the authentication tag has the digest-like binding and preimage properties often
100 expected of a compact AEAD tag.
 - 101 • **Committing and RUP consequences.** We instantiate the same root tag analysis
102 in the existing AEAD vocabulary. The all-bodies-equal structure of CMTDs-4
103 gives the full-input s -way committing bound without the leaf-collision slack of the
104 general hash collision theorem [BH22] (Corollary 4), and the remaining CMT/CMTD
105 notions for $\ell \in \{1, 3, 4\}$ follow from the committing hierarchy and direct source-
106 game arguments (Corollary 5). We also prove INT-RUP integrity [ABL⁺14], with
107 plaintext-computing oracle work charged through the same transcript resources
108 (Theorem 11).
 - 109 • **Baseline AEAD security.** We give a concrete multi-user indistinguishability
110 bound [BT16] for D users in the public permutation model (Theorem 9). The $D = 1$
111 case yields a concrete all-in-one AE bound [RS06] (Corollary 6). The proof proceeds
112 through a dedicated TW128 random oracle model and is transferred to the public
113 permutation setting by the [BDPVA08] flat sponge lift.
 - 114 • **Optimized implementations.** We describe and evaluate vectorized amd64 (AVX-
115 512 and AVX2) and arm64 (NEON with the SHA-3 extension) implementations whose
116 batched leaf cores realize the mode’s scalable parallelism in practice (Section 8).

117 1.2 Related Work

118 **Committing AEAD and tag commitment.** The study of committing authenticated
119 encryption was initiated by [GLR17], who introduced compactly committing AEAD for
120 message franking, and by [DGRW18], who built it from a dedicated primitive (encryptment).
121 [BH22] organized collision-style committing notions into the CMT-1/3/4 and CMTD
122 hierarchy and gave the UtC, RtC, and HtE transforms; [CR22] gave the CTX transform,
123 which commits a nonce-based AE scheme to its full context by adding a hash. These
124 transforms are efficient and broadly applicable, but commitment remains an external
125 layer or an AEAD game property. TW128 instead makes the authentication tag itself the
126 hash-like commitment: the root tag is the compact object, and the root tag analysis gives
127 the [BH22] full-input committing consequence. Closest in spirit is [DHM⁺24], which also
128 builds nonce-based AE natively on SHA-3 functions and obtains CMT-4 from collision
129 resistance; it targets session-based communication through a sequential duplex chain,
130 whereas TW128 is a standalone tree AEAD with implementation-selected leaf parallelism
131 (Section 9.3).

132 **Sponge, duplex, and tree AEADs.** Permutation-based authenticated encryption built on
133 the sponge and duplex constructions of [BDPVA08, BDPVA11] is well established. Keyak
134 [BDP⁺15] uses the Motorist mode over round-reduced KECCAK- p and admits parallel
135 duplex instances; Ascon [DEMS21] and Xoodyak [DHP⁺20] are serial duplex AEADs
136 over smaller permutations. TW128 shares the single-permutation, constant-state duplex
137 foundation but makes the root of a tree the authentication tag. Keyak fixes the degree of
138 parallelism as a mode parameter, while TW128 fixes only the indexed leaf decomposition
139 and lets the implementation choose how many leaves to evaluate at once (Section 9.2).
140 Farfalle [BDH⁺17] reaches implementation-chosen, wire-invisible parallelism through a

parallel PRF, instantiated over KECCAK- p as Kravatte, with AE modes layered on top. TW128 retains the strict [BDPVA11] duplex object and draws its parallelism from the tree topology; the committing behavior of the Farfalle modes has not been analyzed.

Release of unverified plaintext. Andreeva et al. [ABL⁺14] formalized authenticated encryption in the release-of-unverified-plaintext setting by separating plaintext computation from verification. They define INT-RUP for integrity and plaintext awareness notions for privacy, and show that ordinary ciphertext integrity does not imply INT-RUP. TW128 targets the integrity side of this model: the released stream plaintext is not claimed to be private, but it does not reveal the hidden tag values needed to make a fresh ciphertext verify.

Tree hashing. TW128’s two-level shape is inspired by KangarooTwelve [BDP⁺16], a tree hash over the same permutation. TW128 adopts its choice of permutation and final-node/leaf split and adapts them to authenticated encryption: each leaf chunk is encrypted and tagged independently, leaf tags replace unkeyed chaining values, and the root is keyed by (K, U) . The construction also inherits KangarooTwelve’s short-message behavior: inputs fitting in one chunk use only the root duplex. Because TW128 has a fixed two-level topology with independently keyed, indexed leaves and phase-separated duplex calls, it does not need KangarooTwelve’s Sakura tree encoding [BDPVA14] to make its transcripts unambiguous (Section 3).

1.3 Paper Organization

Section 2 recalls the notation, sponge and duplex constructions, nonce-based AEAD and RUP integrity notions, committing security [BH22], and the hash notions of [RS04]. Section 3 specifies the TW128 construction and parameters. Section 4 fixes the scope of the security claims, the public permutation games, intermediate random oracle model, resource accounting, and loss terms. Section 5 proves the root tag hash and committing security results that drive the tag-centric claims, and Section 6 proves multi-user indistinguishability, all-in-one AE security, and INT-RUP integrity. Section 7 summarizes the concrete bounds. Section 8 reports implementation performance, and Section 9 compares TW128 to prior AEAD and committing constructions and discusses limitations outside the formal scope. The appendices collect structural lemmas, the flat sponge lift, vectorized kernel details, and test vectors.

2 Preliminaries

2.1 Notation

Strings. Write $\{0, 1\}$ for the binary alphabet. For an integer $n \geq 0$, $\{0, 1\}^n$ denotes the set of n -bit strings, $\{0, 1\}^* = \bigcup_{n \geq 0} \{0, 1\}^n$ the set of all finite-length bit strings, and ε the empty string. For $x \in \{0, 1\}^*$, $|x|$ is the bit length and $x \parallel y$ is the concatenation; the bitwise XOR of two equal-length strings $x, y \in \{0, 1\}^n$ is $x \oplus y$. A *byte string* is an element of $(\{0, 1\}^8)^*$; its byte length is denoted $\|x\| = |x|/8$. For an integer $0 \leq n < 2^{64}$, $\langle n \rangle_8 \in \{0, 1\}^{64}$ is its eight-byte little-endian encoding. For a finite or infinite bit string x and $0 \leq \tau \leq |x|$ when x is finite, $[x]_\tau \in \{0, 1\}^\tau$ denotes the leading τ bits of x .

Sampling and probability. We write $x \leftarrow \$ S$ for sampling x uniformly at random from a finite set S , with implicit coins independent across samples. For an event E in a probability experiment, $\Pr[E]$ denotes its probability.

184 **Sets of maps.** For a set S , $\text{Perm}(S)$ denotes the set of permutations on S ; for a positive
 185 integer b , $\text{Perm}(\{0, 1\}^b)$ is the set of b -bit permutations.

186 **Games and advantages.** We write $\mathcal{A}^{O_1, \dots, O_k}$ for a (possibly randomized) adversary with
 187 oracle access to $O_1, \dots, O_k$, and $\Pr[\mathbf{G}^{\mathcal{A}} \Rightarrow 1]$ for the probability that a game $\mathbf{G}$ —an
 188 experiment that runs $\mathcal{A}$ against specified oracles and returns a Boolean outcome—returns
 189 1. For a distinguishing experiment between two games $\mathbf{G}_0, \mathbf{G}_1$, the advantage of $\mathcal{A}$ against
 190 a scheme Π in notion $\mathbf{X}$ is the absolute difference

$$191 \quad \text{Adv}_{\Pi}^{\mathbf{X}}(\mathcal{A}) = |\Pr[\mathbf{G}_0^{\mathcal{A}} \Rightarrow 1] - \Pr[\mathbf{G}_1^{\mathcal{A}} \Rightarrow 1]|.$$

192 For a win-event experiment without a hidden challenge bit it is instead $\Pr[\mathbf{G}^{\mathcal{A}} \text{ wins}]$, the
 193 probability the game returns 1; a bit-guessing experiment with a hidden challenge bit uses
 194 the rescaled form $|2 \Pr[\mathbf{G}^{\mathcal{A}} \text{ wins}] - 1|$, as in the multi-user notion of Section 2.7.

195 **Concrete security.** All bounds in this paper are concrete: each advantage is an explicit
 196 function of the resource caps introduced in Section 4.3. We make no use of asymptotic or
 197 negligibility conventions.

198 2.2 Sponges and Duplexes over Keccak- p

199 **The Keccak- p permutation.** The KECCAK- p family of permutations, defined in
 200 FIPS 202 [Nat15] as the generalization of the Keccak design’s Keccak- f permutations
 201 to a free round count, is parameterized by a state width b and a round count n_r ; we
 202 write $\text{KECCAK} - p[b, n_r] \in \text{Perm}(\{0, 1\}^b)$ for the corresponding permutation. TW128 uses
 203 KECCAK- $p[1600, 12]$, the same primitive as KangarooTwelve [BDP⁺16], fixed throughout.
 204 In the public permutation security model of Section 4.1 and the flat sponge lift of Section B,
 205 this permutation is modeled as a uniformly random element of $\text{Perm}(\{0, 1\}^{1600})$.

206 **The sponge construction.** Following [BDPVA08], fix a permutation $\pi \in \text{Perm}(\{0, 1\}^b)$,
 207 a *sponge-compliant* padding rule pad , and a rate $1 \leq r < b$ with capacity $c = b - r$. A
 208 padding rule appends $\text{pad}[r](|M|)$, a bit string determined by the input length and the
 209 rate, to an input M , extending it to a sequence of r -bit blocks; it is sponge-compliant
 210 [BDPVA11, Definition 1] if $M \parallel \text{pad}[r](|M|)$ is never empty and, for all $M \neq M'$ and all
 211 $n \geq 0$, $M \parallel \text{pad}[r](|M|) \neq M' \parallel \text{pad}[r](|M'|) \parallel 0^{nr}$. The sponge function $\text{Sponge}[\pi, \text{pad}, r] :$
 212 $\{0, 1\}^* \times \mathbb{N} \rightarrow \{0, 1\}^*$ maps an input M and an output length ℓ to an ℓ -bit string by
 213 absorbing $M \parallel \text{pad}[r](|M|)$ into the all-zero b -bit state in r -bit blocks (each separated by an
 214 application of π) and squeezing ℓ bits in r -bit blocks from the resulting state. [BDPVA08,
 215 Theorem 2] shows that $\text{Sponge}[\pi, \text{pad}10^*, r]$, where the simple rule $\text{pad}10^*$ appends a single
 216 1 bit followed by the minimum number of 0 bits reaching a multiple of r , is indistinguishable
 217 from a random oracle when π is uniformly random. We bound its distinguishing advantage
 218 by $\text{Lift}_c(N_{\text{tot}}) = N_{\text{tot}}(N_{\text{tot}} + 1)/2^{c+1}$, derived in Lemma 20 from the product upper bounds
 219 of [BDPVA08, Lemmas 4 and 5].

220 **The duplex construction.** The *duplex* construction of [BDPVA11] extends the sponge
 221 to a stateful interface. A duplex object $\text{Duplex}[\pi, \text{pad}, r]$ is initialized to the all-zero state.
 222 Each call $\text{duplexing}(\sigma, \ell)$, with $\ell \leq r$, absorbs an input string σ of at most $\rho_{\max}(\text{pad}, r)$
 223 bits via π and returns the first ℓ bits of the resulting state. Here $\rho_{\max}(\text{pad}, r) = \min\{x :$
 224 $x + |\text{pad}[r](x)| > r\} - 1$ is the largest input length for which the padded input fits in a single r -
 225 bit block; for the $\text{pad}10^*1$ rule, $\rho_{\max}(\text{pad}10^*1, r) = r - 2$ [BDPVA11]. [BDPVA11, Lemma 3]
 226 establishes the duplex-to-sponge equality: for any call sequence $((\sigma_0, \ell_0), \dots, (\sigma_i, \ell_i))$, the
 227 i -th output equals

$$228 \quad \text{Sponge}[\pi, \text{pad}, r](\sigma_0 \parallel \text{pad}_0 \parallel \dots \parallel \sigma_i, \ell_i),$$

where $\text{pad}_j = \text{pad}[r](|\sigma_j|)$ is the padding the duplex appends in its j -th call; the flattened input carries these interior paddings explicitly, and the sponge appends only the final pad_i . The flattening map is injective by [BDPVA11, Lemma 4]: the duplex input-string sequence is recoverable from the flattened sponge input.

TW128 instantiation. TW128 fixes the padding rule pad10^*1 and the rate $r = 1344$, so the capacity is $c = 256$ bits. The largest input length per duplex call is therefore $\rho_{\max}(\text{pad10}^*1, 1344) = 1342$ bits.

Bridge to the H -input domain. The indistinguishability theorem of [BDPVA08] is stated for the simpler pad10^* padding over its unpadded H -input domain, while TW128 uses pad10^*1 internally. [BDPVA11, Theorem 3] supplies an injective preprocessing bridge $I[r, r_{\max}]$ between the two domains. For TW128, $r = r_{\max} = 1344$, so the bridge reduces to appending $1 \parallel 0^q$ with q determined by the input length modulo $r_{\max}$; on the bridge image, q is recovered as the trailing-zero count. Section 4.2 instantiates this bridge on typed TW128 duplex transcript prefixes as the map $\text{flat}(\cdot)$ used by the random oracle proofs throughout the paper.

2.3 AEAD Syntax and Verification Interfaces

Syntax. A *nonce-based authenticated encryption scheme with associated data* (AEAD scheme) is a pair $\text{SE} = (\text{Enc}, \text{Dec})$ of deterministic algorithms with associated key space $\mathcal{K} \subseteq (\{0, 1\}^8)^*$, nonce space $\mathcal{U} \subseteq (\{0, 1\}^8)^*$, associated data space $\mathcal{A} \subseteq (\{0, 1\}^8)^*$, message space $\mathcal{M} \subseteq (\{0, 1\}^8)^*$, and ciphertext space $\mathcal{C} \subseteq (\{0, 1\}^8)^*$. The encryption algorithm $\text{Enc} : \mathcal{K} \times \mathcal{U} \times \mathcal{A} \times \mathcal{M} \rightarrow \mathcal{C}$ produces a ciphertext $C = \text{Enc}_K(U, A, M)$. The decryption algorithm $\text{Dec} : \mathcal{K} \times \mathcal{U} \times \mathcal{A} \times \mathcal{C} \rightarrow \mathcal{M} \cup \{\perp\}$ returns either a plaintext or the rejection symbol $\perp$. The *tag length in bytes* is the constant $\|\text{Enc}_K(U, A, M)\| - \|M\|$, fixed by the scheme. We write nonce values as U to keep N available for the construction-cost resource variable used in the concrete bounds. In this generic syntax, C denotes the full AEAD ciphertext, including any tag material. The TW128 algorithm section (Section 3) writes this same value as $C \parallel T$, with C the ciphertext body and T the root tag.

Correctness. For every $(K, U, A, M) \in \mathcal{K} \times \mathcal{U} \times \mathcal{A} \times \mathcal{M}$, $\text{Dec}_K(U, A, \text{Enc}_K(U, A, M)) = M$. An AEAD scheme is *tidy* if every accepting decryption is consistent with encryption: whenever $\text{Dec}_K(U, A, C) = M \neq \perp$, one has $\text{Enc}_K(U, A, M) = C$.

All-in-one AE. We use the all-in-one authenticated encryption notion of [RS06], instantiated for nonce-based AE following [BT16]. The experiment grants either the real pair $(\text{Enc}_K, \text{Ver}_K)$ or the ideal pair $(\text{Rand}, \text{Replay})$, where Rand returns a fresh uniform byte string of the ciphertext length on each encryption query and Replay accepts exactly the tuples Rand produced. The advantage $\text{Adv}_{\text{SE}}^{\text{ae}}(\mathcal{A})$ is the distinguishing advantage between these two worlds over nonce-respecting adversaries. This all-in-one notion is instantiated for TW128 in Section 6.1.

2.4 Root Tag Hash Security

Definition 1 (Root tag wrapper). For every TW128 encryption input $X = (K, U, A, M)$ in the scheme domain, define the root tag wrapper by

$$\text{H}_{\text{TW128}}(K, U, A, M) = T \quad \text{where} \quad \text{Enc}_K(U, A, M) = C \parallel T.$$

We also write $\text{H}_{\text{TW128}}(X)$ for $\text{H}_{\text{TW128}}(K, U, A, M)$. Equivalently, H_{TW128} runs TW128 encryption and discards the ciphertext body.

The root tag as a hash. Many uses of commitment store, transmit, or compare a short tag rather than the full ciphertext, and ask that the short string alone be binding—the compactly committing AEAD goal introduced by [GLR17] for message franking. TW128 ciphertexts are written $C \parallel T$ with T the τ_{root} -bit root tag, and we capture this goal directly by treating Definition 1 as a hash function. Keyed by the permutation, this is a hash family, and we ask that it satisfy the hash-function security notions of [RS04], over the TW128 input domain, together with the local s -way multicollision extension used for the committing theorems. For fixed byte lengths a and μ , write

$$\mathcal{X}_{a,\mu} = \{0,1\}^k \times \{0,1\}^\nu \times (\{0,1\}^8)^a \times (\{0,1\}^8)^\mu$$

for the fixed-length slice of valid inputs, with bit length $m_{a,\mu} = k + \nu + 8a + 8\mu$.

Collision resistance (Coll). No adversary finds distinct inputs $X_1, \dots, X_s$ (any $s \geq 2$) with $H_{\text{TW128}}(X_1) = \dots = H_{\text{TW128}}(X_s)$. The case $s = 2$ is the RS04 Coll notion; larger s is the corresponding multicollision extension.

Preimage resistance (Pre). For fixed a, μ , given $H_{\text{TW128}}(X^*)$ for a uniform $X^* \leftarrow \$ \mathcal{X}_{a,\mu}$, no adversary finds any valid input X with $H_{\text{TW128}}(X) = H_{\text{TW128}}(X^*)$.

Second-preimage resistance (Sec/eSec). The standard and everywhere forms of second-preimage resistance of [RS04], on each fixed input slice $\mathcal{X}_{a,\mu}$, are reduced explicitly to two-way collision resistance in Corollary 2.

Everywhere preimage resistance (ePre). For a target tag T^* fixed in advance, no adversary finds X with $H_{\text{TW128}}(X) = T^*$.

The advantages $\text{Adv}_{\text{TW128}}^{\text{Coll}}$, $\text{Adv}_{\text{TW128}}^{\text{Pre}[a,\mu]}$, and $\text{Adv}_{\text{TW128}}^{\text{ePre}}$ are the respective success probabilities, instantiated for TW128 in Section 5.1. The Pre bound is proved directly in Theorem 5, while the second-preimage bounds are inherited from Coll in Corollary 2. Coll is the binding property of the tag taken as a compact commitment to (K, U, A, M) , Pre is the random-opening preimage property on fixed input slices, and ePre is the fixed-target preimage property for query-independent tags. The [BH22] committing notions are the collision-facing AEAD projection of the same tag analysis: a winning opening gives distinct inputs sharing a tag, while the dedicated CMTDs-4 proof uses the all-bodies-equal game structure to avoid the leaf-collision slack of the general hash collision theorem (Section 5).

2.5 AEAD Committing Notions

[BH22] formalize a family of committing security notions for nonce-based AEAD schemes, parameterized by how much of the encryption input is committed and by whether commitment is established through decryption (the D-notions) or through encryption agreement (the E-notions). We recall their definitions and the s -way generalization because the root tag hash bounds of Section 5 are later instantiated in this AEAD vocabulary.

What-is-committed functions. For $\ell \in \{1, 3, 4\}$, the *what-is-committed* function WiC_ℓ projects an encryption input $(K, U, A, M) \in \mathcal{K} \times \mathcal{U} \times \mathcal{A} \times \mathcal{M}$ to the portion it should commit:

$$\begin{aligned} \text{WiC}_1(K, U, A, M) &= K, \\ \text{WiC}_3(K, U, A, M) &= (K, U, A), \\ \text{WiC}_4(K, U, A, M) &= (K, U, A, M). \end{aligned}$$

CMTDs- ℓ and CMTs- ℓ games. For $s \geq 2$, [BH22, Fig. 5] defines the s -way committing games on tuples $(K_i, U_i, A_i, M_i)_{i=1}^s$ with pairwise distinct WiC_ℓ -images, all entries in the scheme's syntactic domain and none equal to $\perp$. The s -way CMTDs- ℓ (decryption) game has the adversary output a ciphertext C alongside the tuples and returns 1 if $\text{Dec}_{K_i}(U_i, A_i, C) = M_i$ for every i ; the printed return line of [BH22, Fig. 5] writes C_i , but the game outputs a single ciphertext C . The s -way CMTs- ℓ (encryption) game has the adversary output the tuples alone and returns 1 if all encryptions $\text{Enc}_{K_i}(U_i, A_i, M_i)$ are equal. Advantages $\text{Adv}_{\text{SE}}^{\text{CMTDs-}\ell}(\mathcal{A})$ and $\text{Adv}_{\text{SE}}^{\text{CMTs-}\ell}(\mathcal{A})$ are the probabilities that the adversary wins the respective game; the parameter s is suppressed from the notation when clear from context. The two-tuple case $s = 2$ is the original CMTD- ℓ /CMT- ℓ pair of [BH22, Fig. 4]. These definitions are instantiated for TW128 in Section 5.1, with the public permutation lift supplied by Section B.

Relations. [BH22, Figs. 4 and 5] record the implications among the notions. For the D-notions we use:

$$\text{CMTDs-4} \Rightarrow \text{CMTDs-}\ell \text{ for } \ell \in \{1, 3\}$$

(both immediate: pairwise distinct keys ($\ell = 1$) or pairwise distinct (K, U, A) triples ($\ell = 3$) are *a fortiori* pairwise distinct full tuples, so every CMTDs- ℓ winner is a CMTDs-4 winner). The E-notions are bounded directly in Corollary 5 by the same final-root candidate-collision argument, rather than through an encryption reduction.

2.6 Integrity under Release of Unverified Plaintext

For the RUP setting of [ABL⁺14], a conventional decryption algorithm is separated into a plaintext-computing algorithm and a verification algorithm. The integrity notion INT-RUP gives the adversary access to encryption, plaintext computation, and verification, and asks whether verification accepts a fresh ciphertext and tag not returned by encryption. As in the nonce-IV setting of [ABL⁺14], encryption queries are nonce-respecting, while plaintext computation and verification may be queried on arbitrary nonces. This notion protects only ciphertext integrity; privacy under release of unverified plaintext requires plaintext awareness in addition to ordinary encryption privacy.

2.7 Multi-User Indistinguishability

The multi-user indistinguishability notion of [BT16, Fig. 3] extends the same real-vs-ideal oracle pair to an adversary-selected set of users. The number of users u is adversary-determined: it is the number of **New** queries the adversary makes, i.e., the final value of the user counter v in the game below. A challenge bit $b \leftarrow \{0, 1\}$ selects between the real construction interface (real ciphertexts, real verification) and an ideal interface (uniform strings of matching length, verification that accepts only previously-issued encryption tuples). The adversary also accesses the public primitive interface (π, π^{-1}) , exposed identically in both worlds. This follows the same-oracle convention of [BT16], who work in the ideal-cipher model and give both worlds the same ideal-cipher oracles; here the shared primitive is the random permutation. The adversary outputs a guess b' and wins if $b' = b$.

Game and advantage. The game $\text{G}_{\text{SE}}^{\text{mu-ind}}(\mathcal{A})$ of [BT16, Fig. 3] samples $b \leftarrow \{0, 1\}$ and maintains a user counter v , a set V of (d, U) pairs already used by encryption, and a set W of recorded encryption tuples; its **New**, **Enc**, and **Vf** oracles are instantiated for TW128 in Section 6.2, together with the public primitive interface (π, π^{-1}) , which does not depend on the challenge bit. We write the user index as d and the associated data input as A ,

where [BT16] writes i and H , and take the absolute value of their signed advantage so that

$$\begin{aligned} \text{Adv}_{\text{SE}}^{\text{mu-ind}}(\mathcal{A}) &= |2 \Pr[\text{G}_{\text{SE}}^{\text{mu-ind}}(\mathcal{A}) \text{ returns } 1] - 1| \\ &= |\Pr[\mathcal{A}^{\text{Enc}, \text{Vf}, b=1} \Rightarrow 1] - \Pr[\mathcal{A}^{\text{Enc}, \text{Vf}, b=0} \Rightarrow 1]| \end{aligned}$$

the second equality holding because b is uniform. Thus $\text{Adv}_{\text{SE}}^{\text{mu-ind}}$ is the two-game distinguishing advantage between the real (Enc, Ver) pair and the ideal ($\text{Rand}, \text{Replay}$) pair instantiated per user. The single-user case $u = 1$ recovers the all-in-one AE notion of Section 2.3, augmented with the public primitive interface (the real permutation in both worlds).

Per-user nonce uniqueness. Nonce uniqueness is enforced per user via the set V : the same nonce may appear under distinct users but not twice under the same user. This is strictly weaker than the global nonce-uniqueness requirement that would force each nonce value U to appear under at most one user.

3 The TW128 Construction

3.1 Tree Structure

TW128 is a nonce-based AEAD built as a two-level tree of duplex objects over KECCAK- $p[1600, 12]$. The *root* duplex is initialized with (K, U) , absorbs the associated data and the first ciphertext chunk, and emits the final τ_{root} -bit root tag T . The remaining plaintext is partitioned into *leaf* chunks $M_1, \dots, M_n$; leaf j is encrypted by an independent duplex initialized with (K, U, j) and returns a τ_{leaf} -bit leaf tag T_j . For multi-chunk messages, the root aggregates $T_1 \parallel \dots \parallel T_n \parallel \langle n \rangle_8$ before emitting T . Thus the authentication tag is also the digest point for the full (K, U, A, M) context. The construction is depicted in Figure 1; the exact edge-case conventions are specified in Section 3.4.

The tree fixes the transcript but not the implementation’s parallelism. Leaves operate on disjoint states after (K, U) is fixed, so scalar, SIMD, and multicore implementations can evaluate different numbers of leaves at once while producing identical ciphertexts and tags. The root state and each leaf state are bounded 1600-bit objects, and leaf tags can be aggregated as a stream, giving constant working memory at any fixed degree of parallelism. This is the AEAD analog of KangarooTwelve’s final-node hashing of per-chunk chaining values, with leaf tags as the chaining values and the root tag as the final digest.

3.2 Parameters and Domain

Table 1 lists the user-visible and derived scheme parameters. The implementation uses the same byte length for the root tag and each leaf tag, but the proofs of Sections 5 and 6 keep τ_{root} and τ_{leaf} as separate symbols.

The bound $M_{\text{max}} = 8183 \cdot 2^{64}$ bytes is the largest plaintext whose leaf index fits in $\langle \cdot \rangle_8$: writing $\text{leaves}(M) = \max(0, \lceil \|M\|/\beta \rceil - 1)$, every supported plaintext satisfies $0 \leq \text{leaves}(M) \leq 2^{64} - 1$, so leaf indices lie in $\{1, \dots, 2^{64} - 1\}$. The associated data bound A_{max} is a fixed scheme parameter, taken equal to M_{max} . Inputs exceeding these bounds are outside the scheme domain.

The derived per-call absorption bound ρ follows the [BDPVA11] analysis: with $r = 1344$ and pad10^*1 padding, $\rho_{\text{max}}(\text{pad10}^*1, r) = r - 2 = 1342$ bits. The byte-aligned data field shares this allotment with the four-bit phase suffix of Section 3.3, so ρ is the largest integer satisfying $8\rho + 4 \leq 1342$, giving $\rho = 167$ data bytes (and $8\rho + 4 = 1340 \leq 1342$).

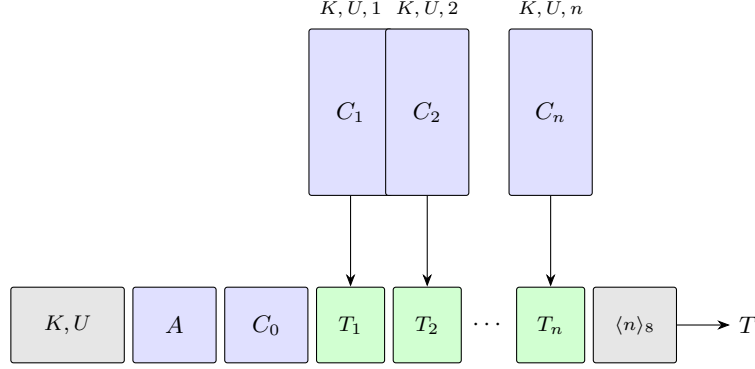


Figure 1: Schematic of TW128 for $n \geq 1$ leaves. The root duplex (bottom row) absorbs the associated data, root ciphertext chunk, leaf tags, and leaf count before emitting the root tag T . Each independent leaf duplex (top row) encrypts one leaf chunk and emits a leaf tag.

Table 1: TW128 parameters.

Parameter	Value	Description
k	256	Key length (bits)
ν	256	Nonce length (bits)
τ_{root}	256	Root tag length (bits)
τ_{leaf}	256	Leaf tag length (bits)
β	8183	Plaintext chunk size (bytes)
M_{max}	$8183 \cdot 2^{64}$ bytes	Maximum plaintext length
A_{max}	$8183 \cdot 2^{64}$ bytes	Maximum associated data length
b	1600	Permutation state width (bits)
n_r	12	Permutation rounds
r	1344	Sponge rate (bits)
c	256	Sponge capacity (bits)
ρ	167	Maximum byte-aligned absorption per duplex call (bytes)

Parameter rationale. The capacity, key length, and tag lengths are all set to 256 bits so the flat sponge loss, key-guessing terms, tag-guessing terms, and root/leaf collision terms meet a 128-bit target at the concrete work cap of Section 7.1. With $b = 1600$, this fixes the rate at $r = 1344$. The 12-round KECCAK- $p[1600, 12]$ permutation matches KangarooTwelve [BDP⁺16]; Section 9.4 discusses the cryptanalytic record for this round count. The chunk size is $\beta = 49\rho = 8183$ bytes, the largest multiple of the byte-aligned per-call absorption bound not exceeding 2^{13} bytes; this fully utilizes message duplex calls inside full leaves while keeping the leaf working set small. The 256-bit nonce is not required by the nonce-respecting bounds, but it is essentially free in the root initialization call and supports both random nonces and narrower counter nonces by zero-padding.

3.3 Domain Separation

Distinct duplex instances and phases are separated at the transcript level by initialization prefixes and phase suffixes. The initialization strings carry a 6-byte ASCII prefix and, for leaves, an eight-byte little-endian chunk index:

$$p_{\text{root}} = \text{"TW128R"} \in \{0, 1\}^{48},$$

$$p_{\text{leaf}} = \text{"TW128L"} \in \{0, 1\}^{48}.$$

412 The root’s first absorbed string is $p_{\text{root}} \parallel K \parallel U$, and leaf j ’s first absorbed string is
 413 $p_{\text{leaf}} \parallel K \parallel U \parallel \langle j \rangle_8$. Every byte-aligned absorption also carries one of the four-bit phase
 414 suffixes in Table 2, written least significant bit first: absorbing a byte block σ under suffix
 415 σ_{ph} feeds $\sigma \parallel \sigma_{\text{ph}}$ to the duplex interface of Section 2.2 before pad_{10*1} padding.

Table 2: Phase suffixes (4 bits each, appended LSB-first).

Name	Hex	Use
σ_{init}	0xC	End of initialization (the sole init call)
σ_{ad}^-	0x9	Non-final associated data block
σ_{ad}^+	0xD	Final associated data block
σ_{msg}^-	0xA	Non-final message block
σ_{msg}^+	0xE	Final message block
σ_{agg}^-	0xB	Non-final aggregation block
σ_{agg}^+	0xF	Final aggregation block (emits root tag)

416 Table 3 summarizes the root and leaf transcript schedule. The algorithms of Sections 3.4
 417 and 3.5 give the normative byte-level specification; the table records which phase is present
 418 and what output the final call of that phase supplies.

Table 3: TW128 transcript schedule. “First M_i keystream” means $8 \min(\|M_i\|, \rho)$ output bits for the first duplex block of that message chunk; subsequent message-phase outputs have the length of the next duplex block, as in Algorithm 1.

Instance	Phase	Suffixes	Absorbed string	Output supplied	When
Root	Init	σ_{init}	$p_{\text{root}} \parallel K \parallel U$	0 if $A \neq \varepsilon$; otherwise first M_0 keystream	Always
Root	AD	$\sigma_{\text{ad}}^-, \sigma_{\text{ad}}^+$	ρ -byte block partition of A	Final call emits first M_0 $A \neq \varepsilon$ keystream	$A \neq \varepsilon$
Root	Message	$\sigma_{\text{msg}}^-, \sigma_{\text{msg}}^+$	ρ -byte block partition of C_0	Non-final calls emit next keystream; final call emits T if $n = 0$, and 0 otherwise	Always
Leaf j	Init	σ_{init}	$p_{\text{leaf}} \parallel K \parallel U \parallel \langle j \rangle_8$	First M_j keystream	$1 \leq j \leq n$
Leaf j	Message	$\sigma_{\text{msg}}^-, \sigma_{\text{msg}}^+$	ρ -byte block partition of C_j	Non-final calls emit next keystream; final call emits T_j	$1 \leq j \leq n$
Root	Aggregation	$\sigma_{\text{agg}}^-, \sigma_{\text{agg}}^+$	$T_1 \parallel \dots \parallel T_n \parallel \langle n \rangle_8$	Final call emits T	$n \geq 1$

419 The combined effect of the initialization prefixes and the phase suffixes is that the
 420 flattened sponge input of any TW128 duplex call uniquely determines the duplex instance,
 421 the leaf index when applicable, the phase, and the duplex-call position within that phase.
 422 Definition 2 (Well-formed TW128 transcript prefixes), Lemma 6 (Transcript Injectivity),
 423 and its structural corollaries formalize this parsing fact for the proofs of Sections 5 and 6
 424 and Corollary 5.

3.4 Encryption

425 Algorithm 2 specifies $\text{Enc}_K(U, A, M)$ using the helper procedures of Algorithm 1. The
 426 helpers sit below the tree layer: the encryption and decryption algorithms split bodies
 427 into β -byte chunks, while the helpers split a chosen byte string into ρ -byte duplex blocks.
 428 STREAMENC XORs each plaintext block with the current keystream, absorbs the resulting
 429 ciphertext block under the message suffix, and uses the duplex output as the next keystream.
 430 This is the duplex-level presentation of the overwrite mechanism of [BDPVA11, §6.2 and
 431 Algorithm 5]. The decryption helper STREAMDEC uses the same schedule with ciphertext
 432 blocks as input: it recovers $m_i = c_i \oplus Z$, absorbs the same c_i , and returns the plaintext
 433 string and final output value.
 434

Algorithm 1 TW128 helper procedures.

```

1: procedure ABSORB( $\mathcal{D}, s, \sigma^-, \sigma^+, \ell$ )
2:   Partition  $s$  into duplex blocks  $s_1, \dots, s_t$  of exactly  $\rho$  bytes each, except that the
   last block may be shorter.
3:   Require  $t \geq 1$ ; the empty case yields the single empty duplex block  $s_1 = \varepsilon$ .
4:   for  $i \leftarrow 1$  to  $t - 1$  do
5:      $\mathcal{D}.\text{duplexing}(s_i \parallel \sigma^-, 0)$ 
6:   end for
7:   return  $\mathcal{D}.\text{duplexing}(s_t \parallel \sigma^+, \ell)$ 
8: end procedure

9: procedure STREAMENC( $\mathcal{D}, m, Z, \sigma^-, \sigma^+, \ell$ )
10:  Partition  $m$  into duplex blocks  $m_1, \dots, m_t$  of exactly  $\rho$  bytes each, except that the
   last block may be shorter.
11:  Require  $t \geq 1$ ; if  $m = \varepsilon$ , set  $t = 1$  and  $m_1 = \varepsilon$ .
12:  for  $i \leftarrow 1$  to  $t - 1$  do
13:     $c_i \leftarrow m_i \oplus Z$ 
14:     $Z \leftarrow \mathcal{D}.\text{duplexing}(c_i \parallel \sigma^-, 8\|m_{i+1}\|)$ 
15:  end for
16:   $c_t \leftarrow m_t \oplus Z$ 
17:   $Z' \leftarrow \mathcal{D}.\text{duplexing}(c_t \parallel \sigma^+, \ell)$ 
18:  return  $(c_1 \parallel \dots \parallel c_t, Z')$ 
19: end procedure

```

Edge-case convention. The helper partition always has at least one block: an empty input is the single empty final block carrying the final phase suffix. The associated-data phase is present exactly when $A \neq \varepsilon$; when $A = \varepsilon$, the root σ_{init} call emits the first root-chunk keystream. The first keystream request for a message string X is $8 \min(\|X\|, \rho)$ bits. If $n = 0$, the root message phase requests τ_{root} bits on its final σ_{msg}^+ call and no aggregation phase occurs. If $n \geq 1$, that root message call requests zero bits, each leaf final message call requests τ_{leaf} bits, and the final σ_{agg}^+ aggregation call requests τ_{root} bits. These conventions apply identically to decryption.

The leaf loop is annotated **in parallel** to make explicit that the n leaf computations are mutually independent; Section 8.1 discusses the SIMD realizations. The annotation is an implementation freedom, not a mode parameter: evaluating leaves serially, in SIMD batches, or across cores leaves the transcript and wire format unchanged.

Streaming aggregation. The string $G = T_1 \parallel \dots \parallel T_n$ in Algorithms 2 and 3 names the leaf tags only for specification. The closing ABSORB consumes $G \parallel \langle n \rangle_8$ as ρ -byte duplex blocks in leaf order and advances the root state after each, so an implementation need not hold G in full. It can absorb leaf tags in order as they become available, or buffer completed out-of-order leaves until the next in-order tag or SIMD batch can be absorbed, keeping only a bounded number of tags at a time. With the bounded 1600-bit state of every duplex, this keeps the working memory constant in $\|M\|$ for a fixed degree of parallelism, growing only with the number of leaves evaluated at once rather than with the message length; Section 8.1 describes the batched realization.

3.5 Decryption and Correctness

Algorithm 3 specifies $\text{Dec}_K(U, A, C \parallel T)$. Decryption mirrors encryption at the duplex block level: each ciphertext block is absorbed under the same phase suffix, and the plaintext

Algorithm 2 $\text{Enc}_K(U, A, M)$.**Require:** $K \in \{0, 1\}^k$, $U \in \{0, 1\}^\nu$, A with $\|A\| \leq A_{\max}$, M with $\|M\| \leq M_{\max}$.**Ensure:** $C \parallel T$ with $\|C\| = \|M\|$ and $T \in \{0, 1\}^{\tau_{\text{root}}}$.

```

1:  $M_0 \leftarrow$  first  $\min(\|M\|, \beta)$  bytes of  $M$  (possibly  $\varepsilon$ ).
2: Partition the remaining suffix into chunks  $M_1, \dots, M_n$  of exactly  $\beta$  bytes each, except
   that the last chunk may be shorter but nonempty, so  $n = \text{leaves}(M)$ .
3:  $\mathcal{D}_{\text{root}} \leftarrow$  new Duplex
4: if  $A = \varepsilon$  then
5:    $Z \leftarrow \mathcal{D}_{\text{root}}.\text{duplexing}(p_{\text{root}} \parallel K \parallel U \parallel \sigma_{\text{init}}, 8 \min(\|M_0\|, \rho))$ 
6: else
7:    $\mathcal{D}_{\text{root}}.\text{duplexing}(p_{\text{root}} \parallel K \parallel U \parallel \sigma_{\text{init}}, 0)$ 
8:    $Z \leftarrow \text{ABSORB}(\mathcal{D}_{\text{root}}, A, \sigma_{\text{ad}}^-, \sigma_{\text{ad}}^+, 8 \min(\|M_0\|, \rho))$ 
9: end if
10: if  $n = 0$  then
11:    $(C_0, T) \leftarrow \text{STREAMENC}(\mathcal{D}_{\text{root}}, M_0, Z, \sigma_{\text{msg}}^-, \sigma_{\text{msg}}^+, \tau_{\text{root}})$ 
12:   return  $C_0 \parallel T$ 
13: end if
14:  $(C_0, \_) \leftarrow \text{STREAMENC}(\mathcal{D}_{\text{root}}, M_0, Z, \sigma_{\text{msg}}^-, \sigma_{\text{msg}}^+, 0)$ 
15: for  $j \leftarrow 1$  to  $n$  in parallel do
16:    $\mathcal{D}_j \leftarrow$  new Duplex
17:    $Z_j \leftarrow \mathcal{D}_j.\text{duplexing}(p_{\text{leaf}} \parallel K \parallel U \parallel \langle j \rangle_8 \parallel \sigma_{\text{init}}, 8 \min(\|M_j\|, \rho))$ 
18:    $(C_j, T_j) \leftarrow \text{STREAMENC}(\mathcal{D}_j, M_j, Z_j, \sigma_{\text{msg}}^-, \sigma_{\text{msg}}^+, \tau_{\text{leaf}})$ 
19: end for
20:  $G \leftarrow T_1 \parallel \dots \parallel T_n$ .
21:  $T \leftarrow \text{ABSORB}(\mathcal{D}_{\text{root}}, G \parallel \langle n \rangle_8, \sigma_{\text{agg}}^-, \sigma_{\text{agg}}^+, \tau_{\text{root}})$ 
22: return  $C_0 \parallel C_1 \parallel \dots \parallel C_n \parallel T$ 

```

block is recovered as $m_i = c_i \oplus Z^{(i)}$ from the keystream value that encryption would have used at that position. The edge cases are those of Section 3.4. The procedure absorbs all in-domain ciphertext bytes before comparing the final tag, so the verification transcript is determined by (K, U, A, C) whether or not the submitted tag accepts.

Correctness. For every $K \in \{0, 1\}^k$, $U \in \{0, 1\}^\nu$, A with $\|A\| \leq A_{\max}$, and M with $\|M\| \leq M_{\max}$, $\text{Dec}_K(U, A, \text{Enc}_K(U, A, M)) = M$. Encryption and decryption absorb the same ciphertext blocks under the same duplex instances and phase suffixes, so their duplex states and final root tags agree. Each recovered block satisfies $c_i \oplus Z^{(i)} = (m_i \oplus Z^{(i)}) \oplus Z^{(i)} = m_i$.

Lemma 1 (TW128 tidiness). *For every valid TW128 decryption input, if $\text{Dec}_K(U, A, C \parallel T) = M \neq \perp$, then $\text{Enc}_K(U, A, M) = C \parallel T$.*

Proof. An accepting decryption has $\|M\| = \|C\|$ and therefore the same canonical chunking as re-encryption of M . The decryption transcript absorbs the same (K, U, A) values and ciphertext blocks that this re-encryption absorbs, under the same duplex instances and phase suffixes. Its stream outputs recover $m_i = c_i \oplus Z^{(i)}$, so re-encryption emits the submitted ciphertext blocks again. The recomputed leaf tags and final root candidate tag are therefore the same values; since decryption accepts only when the final candidate equals the submitted tag T , re-encryption returns $C \parallel T$. $\square$

Algorithm 3 $\text{Dec}_K(U, A, C \parallel T)$.

Require: K, U, A as in Algorithm 2, ciphertext C with $\|C\| \leq M_{\max}$, candidate tag $T \in \{0, 1\}^{\tau_{\text{root}}}$.

Ensure: M with $\|M\| = \|C\|$, or $\perp$.

```

1:  $C_0 \leftarrow$  first  $\min(\|C\|, \beta)$  bytes of  $C$  (possibly  $\varepsilon$ ).
2: Partition the remaining suffix into chunks  $C_1, \dots, C_n$  of exactly  $\beta$  bytes each, except
   that the last chunk may be shorter but nonempty, so  $n = \text{leaves}(C)$ .
3:  $\mathcal{D}_{\text{root}} \leftarrow$  new Duplex
4: if  $A = \varepsilon$  then
5:    $Z \leftarrow \mathcal{D}_{\text{root}}.\text{duplexing}(p_{\text{root}} \parallel K \parallel U \parallel \sigma_{\text{init}}, 8 \min(\|C_0\|, \rho))$ 
6: else
7:    $\mathcal{D}_{\text{root}}.\text{duplexing}(p_{\text{root}} \parallel K \parallel U \parallel \sigma_{\text{init}}, 0)$ 
8:    $Z \leftarrow \text{ABSORB}(\mathcal{D}_{\text{root}}, A, \sigma_{\text{ad}}^-, \sigma_{\text{ad}}^+, 8 \min(\|C_0\|, \rho))$ 
9: end if
10: if  $n = 0$  then
11:    $(M_0, T') \leftarrow \text{STREAMDEC}(\mathcal{D}_{\text{root}}, C_0, Z, \sigma_{\text{msg}}^-, \sigma_{\text{msg}}^+, \tau_{\text{root}})$ 
12:   return  $M_0$  if  $T' = T$  (compared in constant time), else  $\perp$ .
13: end if
14:  $(M_0, \_) \leftarrow \text{STREAMDEC}(\mathcal{D}_{\text{root}}, C_0, Z, \sigma_{\text{msg}}^-, \sigma_{\text{msg}}^+, 0)$ 
15: for  $j \leftarrow 1$  to  $n$  in parallel do
16:    $\mathcal{D}_j \leftarrow$  new Duplex
17:    $Z_j \leftarrow \mathcal{D}_j.\text{duplexing}(p_{\text{leaf}} \parallel K \parallel U \parallel \langle j \rangle_8 \parallel \sigma_{\text{init}}, 8 \min(\|C_j\|, \rho))$ 
18:    $(M_j, T_j) \leftarrow \text{STREAMDEC}(\mathcal{D}_j, C_j, Z_j, \sigma_{\text{msg}}^-, \sigma_{\text{msg}}^+, \tau_{\text{leaf}})$ 
19: end for
20:  $G \leftarrow T_1 \parallel \dots \parallel T_n$ .
21:  $T' \leftarrow \text{ABSORB}(\mathcal{D}_{\text{root}}, G \parallel \langle n \rangle_8, \sigma_{\text{agg}}^-, \sigma_{\text{agg}}^+, \tau_{\text{root}})$ 
22: if  $T' = T$  (compared in constant time) then
23:   return  $M_0 \parallel M_1 \parallel \dots \parallel M_n$ 
24: else
25:   return  $\perp$ 
26: end if

```

4 Security Model

This section fixes the shared proof framework for TW128: the public permutation convention, input-validity convention, intermediate random oracle model, resource accounting, and recurring loss terms. The formal game definitions live with the theorem sections that use them: tag hash and committing games in Section 5.1, and AEAD-facing games in Sections 6.1 and 6.2.

Table 4 summarizes the proof obligations before those formal games. The loss terms named there are defined in Section 4.4, and the implemented-parameter instantiations appear in Section 7.

Proof worlds. All final statements are public-permutation claims for TW128 with $\text{KECCAK-}p[1600, 12]$ modeled as a uniformly random permutation. The proofs first establish random oracle bounds for the flat duplex interface of Section 4.2; the flat sponge lift of Section B then transfers those bounds to the public-permutation games of Sections 5.1, 6.1 and 6.2.

Scope of the security claims. In the AE-style distinguishing games, encryption queries are nonce-respecting and the verification oracle returns only acceptance or rejection. The

Table 4: Proof obligations and the dominant loss shapes used for TW128. The table is only a roadmap; the formal games and resource accounting are defined in Sections 4.3, 5.1, 6.1 and 6.2.

Notion	Adversary goal	Main result	Loss shape
H_{TW128} collision	Find distinct full inputs (K, U, A, M) with the same root tag.	Theorem 4	Capacity lift, root collision, leaf collision
H_{TW128} preimage	Find an input hitting a fixed target tag.	Theorem 6	Capacity lift, root preimage
H_{TW128} second preimage	Find a second preimage in the standard or everywhere [RS04] sense.	Corollary 2	Inherited from collision resistance
CMT / CMTD	Produce equal encryptions or one valid ciphertext opening under distinct committed inputs.	Corollaries 4 and 5	Capacity lift, root collision
INT-RUP	Produce a fresh accepting ciphertext after encryption and plaintext-computing queries.	Theorem 11	Capacity lift, key guess, leaf collision, tag guess
AE / mu-ind	Distinguish real encryption and verification from random ciphertexts and replay-only verification, with nonce-respecting encryption queries.	Theorem 9 and Corollary 6	Capacity lift, key guess, leaf collision, tag guess

INT-RUP game gives integrity under a plaintext-computing oracle, but it does not give privacy under release of unverified plaintext or plaintext-awareness security. None of the AE-style games gives nonce-misuse resistance. The hash and committing games are adversarial-initialization games over submitted inputs (K, U, A, M) or ciphertext openings and impose no nonce uniqueness condition. Across all games, the stated bounds are birthday-type, single-stage, and leakage-free: beyond-birthday and side-channel security are outside the model. Section 9.5 discusses the consequences of these exclusions and the corresponding open problems, and Section 9.4 reviews the third-party cryptanalysis supporting the random-permutation modeling of KECCAK- p [1600, 12].

The proof dependency spine is as follows. Section A establishes transcript decoding, bridge injectivity, output-point separation, and the local probabilistic accounting lemmas for key tests, leaf tag collisions, and root tag guesses. Section 5 uses those structural facts to build a chronological sequence of final root candidates and apply a single root tag hash bound; the hash and committing theorems differ only in which collision or preimage event is reduced to that candidate sequence. Section 6 uses the same facts for the random oracle mu-ind and INT-RUP bounds. Section B transfers both branches to the public permutation games defined in the theorem sections.

4.1 Public Permutation Framework

Primitive convention. In every public permutation game considered in this paper, the challenger samples $\pi \leftarrow \$ \text{Perm}(\{0, 1\}^{1600})$ and exposes the direct primitive interface (π, π^{-1}) to the adversary. The real-vs-ideal AE-style games compare the real construction interface with the ideal interface specified by the game over this shared primitive, following the same-oracle convention of [BT16]; the hash and committing games of Section 5.1 are one-world success probabilities over the same sampled permutation. The simulator $(\mathcal{S}^{\text{RO}_{\text{flat}}}, \mathcal{S}^{-1, \text{RO}_{\text{flat}}})$ of Section B is a proof device, entering through the lift from the intermediate random oracle model.

Validity convention. All construction interface inputs are checked against TW128’s public syntax and domain before any construction computation or construction-internal primitive lookup is performed. An encryption input is valid when $U \in \{0, 1\}^\nu$, A is a byte string with

522 $\|A\| \leq A_{\max}$, and M is a byte string with $\|M\| \leq M_{\max}$. A verification or decryption input
 523 is valid when $U \in \{0, 1\}^\nu$, A is a byte string with $\|A\| \leq A_{\max}$, and the full ciphertext parses
 524 as $C \parallel T$ with $\|C\| \leq M_{\max}$ and $T \in \{0, 1\}^{\tau_{\text{root}}}$. In the hash and committing games, each
 525 submitted input (K, U, A, M) must be a valid encryption input including $K \in \{0, 1\}^k$, and
 526 a submitted CMTDs ciphertext must be a valid verification/decryption ciphertext. Invalid
 527 encryption queries are rejected and not recorded, invalid plaintext-computing queries
 528 return $\perp$, invalid verification queries return 0, and invalid committing-game outputs make
 529 the game return 0. Such rejected inputs contribute no construction cost and create no
 530 construction transcripts. Throughout, an *accepted* encryption query is one that passes this
 531 validity check and any game-specific nonce or user check.

532 **Construction oracles.** Whenever a game uses a secret key K and permutation π , $\text{Enc}_K[\pi]$
 533 and $\text{Dec}_K[\pi]$ are the algorithms of Algorithms 2 and 3 with that permutation. The
 534 verification oracle is $\text{Ver}_K[\pi](U, A, C, T) = [\text{Dec}_K[\pi](U, A, C \parallel T) \neq \perp]$. For INT-RUP,
 535 $\text{PDec}_K[\pi](U, A, C, T)$ is the plaintext recovered by the decryption transcript before the
 536 final root-tag comparison; it returns neither the recomputed tag nor the verification bit.

537 4.2 The TW128 Random Oracle Model

538 The proofs of Sections 5 and 6 and Corollary 5 use an intermediate random oracle model.
 539 This model keeps the same construction-level games but replaces every duplex output by
 540 a lookup to a single random oracle through the [BDPVA11] bridge of Section 2.2. The
 541 flat sponge lift of Section B then converts the resulting bounds to the public-permutation
 542 form.

543 The model has a single random oracle $\text{RO}_{\text{flat}} : \{0, 1\}^* \rightarrow \{0, 1\}^\infty$ over the unpadded
 544 H -input domain of the [BDPVA08] simple-padded sponge interface, with lazy sampling:
 545 each distinct input receives an independent infinite bit string, and repeated queries return
 546 the same string. A direct adversary query is finite-output: on input (Y, ℓ) with finite ℓ ,
 547 the oracle returns the first ℓ bits of the lazy-sampled value for Y .

548 Construction calls do not query RO_{flat} on arbitrary strings. Every root or leaf duplex
 549 output is read from RO_{flat} at the bridged flattened transcript prefix for that duplex call,
 550 and the caller takes only the requested output projection. Direct adversary queries to
 551 RO_{flat} are the only random oracle queries counted as direct oracle queries; construction
 552 lookups are accounted for through the construction cost N of Section 4.3.

553 For a well-formed TW128 duplex transcript prefix X in the language of Definition 2, at
 554 one of the construction transcript lookup points formalized by Lemma 8 and Table 11, let
 555 $\text{flat}(X) = I[1344, 1344](\text{raw_flat}(X))$ be the bridged image of the native flattened input
 556 under the [BDPVA11, Theorem 3] preprocessing map of Section 2.2, and define

$$557 \quad \text{RF}(X) = \text{RO}_{\text{flat}}(\text{flat}(X)).$$

558 The construction interfaces Enc_K^{RO} , Dec_K^{RO} , $\text{PDec}_K^{\text{RO}}$, and Ver_K^{RO} are the TW128 algorithms
 559 of Section 3 with every root or leaf duplex output replaced by the corresponding requested
 560 projection of $\text{RF}(X)$; the projection length is the duplexing call's output parameter ℓ ,
 561 which may be zero. The bridge and the typed-restriction convention are formalized in
 562 Lemma 8; the direct-query key-hit predicate ignores the adversary's requested output
 563 length and is formalized as KeyedTWOOut in Lemma 8.

564 The random oracle games are obtained from the public-permutation games of Sec-
 565 tions 5.1, 6.1 and 6.2 by this mechanical substitution: direct primitive access (π, π^{-1})
 566 becomes finite-output access to RO_{flat} , real construction or deterministic challenger checks
 567 use the corresponding $^{\text{RO}}$ algorithm, and the ideal oracles ($\text{Rand}, \text{Replay}$) are unchanged.
 568 We use the same advantage names with subscript RO: $\text{Adv}_{\text{RO}}^{\text{Coll}}$, $\text{Adv}_{\text{RO}}^{\text{Pre}[a, \mu]}$, $\text{Adv}_{\text{RO}}^{\text{ePre}}$,
 569 $\text{Adv}_{\text{RO}}^{\text{CMTDs-}\ell}$, $\text{Adv}_{\text{RO}}^{\text{CMTs-}\ell}$, and $\text{Adv}_{\text{RO}}^{\text{INT-RUP}}$ are the win probabilities of the corresponding

Table 5: Resource caps and recurring loss terms. Uppercase resource symbols are the caps that appear in public-permutation theorems; lowercase symbols are realized counts in random oracle executions or source games.

Symbol	Meaning	Primary use / definition
N	Construction-internal KECCAK- p [1600, 12] calls induced by valid construction oracle queries and deterministic challenger checks.	Public-permutation lift and concrete work cap.
N_{prim}	Direct adversary queries to the public primitive interface.	Key-guess, root-collision, root-preimage terms; direct-query part of the lift.
$N_{\text{tot}} = N + N_{\text{prim}}$	Converted primitive-query cost used by the flat sponge replacement.	Argument of $\text{Lift}_c(N_{\text{tot}})$.
$q_{\text{RO}}(\mathcal{B}), Q_{\text{RO}}$	Direct finite-output queries to RO_{flat} in the intermediate random oracle model; construction lookups are excluded.	Random oracle bounds; after the lift $q_{\text{RO}} \leq N_{\text{prim}}$.
$q_v(\mathcal{A}), Q_v$	Valid verification queries that are not replayable hits.	Verification-tag terms in AE, mu-ind, and INT-RUP bounds.
$n_{\text{leaf}}(\mathcal{B}), N_{\text{leaf}}$	Distinct construction-generated final leaf transcripts.	Leaf tag collision and tag-guessing terms; always at most N .
D	Execution-uniform cap on users activated in the mu-ind game.	Multi-user key-collision and key-guess terms.
$\text{Lift}_c(N_{\text{tot}})$	[BDPVA08] flat sponge differentiating loss.	Defined in § 4.4.
$\text{KeyGuess}_k(Q)$	Ideal-system key-guessing term.	Defined in § 4.4.
$\text{RootColl}_\tau(Q, s)$	s -way root-tag collision term.	Defined in § 4.4.
$\text{RootPre}_\tau(Q)$	Root-tag preimage term.	Defined in § 4.4.
$\text{LeafColl}_\tau(Q)$	Known-key leaf-tag collision term.	Defined in § 4.4.

random oracle games, with CMTs challengers comparing deterministic encryptions via Enc^{RO} . The all-in-one AE advantage $\text{Adv}_{\text{RO}}^{\text{ae}}$ is the corresponding real-vs-ideal distinguishing advantage with direct RO_{flat} access in both worlds.

For mu-ind, $\text{Adv}_{\text{RO}}^{\text{mu-ind}}(\mathcal{B})$ denotes the Section 6.2 game with the same New interface and per-user nonce discipline: user d uses $(\text{Enc}_{K_d}^{\text{RO}}, \text{Ver}_{K_d}^{\text{RO}})$ when $b = 1$ and $(\text{Rand}, \text{Replay})$ when $b = 0$. Equivalently, in the two-game notation of Section 2.7,

$$\text{Adv}_{\text{RO}}^{\text{mu-ind}}(\mathcal{B}) = |\Pr[\mathcal{B}^{\text{Real}^{\text{mu}}, \text{RO}_{\text{flat}}} \Rightarrow 1] - \Pr[\mathcal{B}^{\text{Ideal}^{\text{mu}}, \text{RO}_{\text{flat}}} \Rightarrow 1]|,$$

where the superscripts name the entire multi-user construction interface in the corresponding world. Adversary queries to RO_{flat} are finite-output and counted in $q_{\text{RO}}(\mathcal{B})$ of Section 4.3. Construction-internal $\text{RF}(\cdot)$ lookups (those made by Enc_K^{RO} , Dec_K^{RO} , $\text{PDec}_K^{\text{RO}}$, Ver_K^{RO} , or a hash or committing challenger) are not.

4.3 Resources

Adversary resources are measured by fixed, execution-uniform caps on the number and size of oracle queries. An execution-uniform cap is one that holds on every execution path, including over adversary randomness, oracle randomness, and adaptive query choices. Proofs may also introduce local execution-uniform caps, such as q_e for the number of accepted encryption queries, solely to index a finite hybrid sequence. Such local caps are part of the adversary’s fixed query bounds, but they are not listed as headline resource parameters unless they affect the final bound.

Table 5 gives the compact notation summary used in later theorems. Lowercase resource quantities are realized counts in a random oracle experiment, while uppercase quantities are execution-uniform caps used in the public-permutation bounds. The detailed definitions below fix exactly which computations contribute to each count, and Section 4.4 defines the closed-form loss terms.

- 594 **N** : the construction interface cost, measured as the number of $\text{KECCAK-}p[1600, 12]$ calls
 595 induced after the validity convention has been applied. It counts at per-duplex-call
 596 granularity, summing each root initialization, associated data duplex block, root
 597 message duplex block, leaf initialization, leaf message duplex block, and aggregation
 598 duplex block.
- 599 *AE-style games.* In real games, N counts the valid construction computations
 600 actually run by the adversary's oracle queries. In ideal games, the same accounting
 601 applies to the corresponding real TW128 computation on the valid query, even
 602 when the ideal oracle answers by table lookup or without running the construction.
 603 Valid plaintext-computing computations and valid AE-style non-replay verification
 604 computations are counted whether they accept or reject; replay-table verification hits
 605 are transcript-consistency checks and are not charged to N or to the verification-tag
 606 terms.
- 607 *Hash games.* There is no adversary construction oracle. For the H_{TW128} collision
 608 game, N counts the s deterministic encryptions on the submitted inputs. For the
 609 standard preimage game it counts both challenger encryptions: the source encryption
 610 computing $T^* = H_{\text{TW128}}(X^*)$ and the deterministic encryption on the submitted
 611 input. For the everywhere-preimage game it counts the deterministic encryption on
 612 the submitted input. For the Sec and eSec games of Corollary 2, N and N_{prim} are
 613 those of the explicit induced two-way collision adversary in that corollary, whose
 614 source and candidate deterministic encryptions are counted exactly as in the $s = 2$
 615 collision game.
- 616 *Committing games.* For CMTDs- ℓ , N uses conservative committing-game accounting:
 617 after the syntax, domain, and message-length checks, it charges the deterministic
 618 decryption checks on the common ciphertext whether or not the particular WiC_ℓ
 619 distinctness check would reject before those checks. Likewise, for the CMTs- ℓ source
 620 games of Corollary 5, after syntax and domain checks N charges the deterministic
 621 encryptions that would be used to compare the s ciphertexts whether or not the
 622 WiC_ℓ distinctness check would reject earlier.
- 623 **N_{prim}** : the number of direct adversary queries to the primitive interface, the pair (π, π^{-1})
 624 in public permutation games.
- 625 **$N_{\text{tot}} = N + N_{\text{prim}}$** : the total primitive-query cost used by the flat sponge lift. The lift of
 626 Section B is applied at N_{tot} because after the operational duplex-trace accounting of
 627 Lemma 19, the construction-internal duplex calls counted by N and the adversary's
 628 direct primitive-interface queries counted by N_{prim} form a primitive-query sequence
 629 of cost at most $N + N_{\text{prim}}$.
- 630 **$q_v(\mathcal{A})$, Q_v** : the number of verification queries with valid inputs that are not replay-table
 631 hits, and a cap on it. Used in the verification terms of the direct mu-ind, AE, and
 632 INT-RUP bounds of Section 6. Every counted verification query contributes at least
 633 one verification/decryption construction call in the N accounting, so $q_v(\mathcal{A}) \leq N$.
- 634 **$q_{\text{RO}}(\mathcal{B})$, Q_{RO}** : the number of direct finite-output queries to RO_{flat} by a random oracle
 635 model adversary $\mathcal{B}$, and a cap on it. Repeated direct queries on the same Y are
 636 counted again. Construction-internal $\text{RF}(\cdot)$ lookups are not counted. After the
 637 derived-adversary construction, q_{RO} counts only simulator-induced RO_{flat} calls from
 638 the adversary's direct primitive-interface queries, not the construction calls already
 639 counted in N for the flat sponge lift.
- 640 **$n_{\text{leaf}}(\mathcal{B})$, N_{leaf}** : the number of distinct construction-generated final leaf transcripts in the
 641 execution of $\mathcal{B}$, and a cap on it. Distinct final leaf transcripts are generated by
 642 valid encryption computations, by valid plaintext-computing computations, by valid

AE-style verification computations (whether they accept or reject), and by hash and committing challenger construction computations that pass the early validity checks, including the preimage game's source encryption.

For a byte string X ,

$$\text{leaves}(X) = \max(0, \lceil \|X\|/\beta \rceil - 1)$$

counts the leaf chunks induced by a body byte string of length $\|X\|$. Thus n_{leaf} is bounded by the sum of $\text{leaves}(X)$ over the valid construction computations that create final leaf transcripts, where X is the plaintext body for encryption computations and the submitted ciphertext body for plaintext-computing, verification, and committing decryption checks. Each distinct final leaf transcript contains a final leaf duplex call counted in N , so $n_{\text{leaf}}(\mathcal{B}) \leq N$ unconditionally. The ratio n_{leaf}/N is largest when most processed bytes lie in leaf chunks, where each full leaf contributes $1 + \beta/\rho = 50$ duplex calls to N per final leaf transcript.

All theorems are stated for adversaries whose execution-uniform caps hold at the stated bounds.

4.4 Loss Terms

Five closed-form expressions recur across the security bounds: Lift_c throughout, with KeyGuess_k , RootColl_τ , RootPre_τ , and LeafColl_τ in some. We define them here, after the resource caps that parameterize them.

Flat sponge loss. The flat sponge differentiating loss is

$$\text{Lift}_c(N_{\text{tot}}) = N_{\text{tot}}(N_{\text{tot}} + 1)/2^{c+1}.$$

For $N_{\text{tot}} < 2^c$ it upper-bounds the [BDPVA08] sponge-differentiating advantage; Lemma 20 carries out the derivation from the [BDPVA08, Lemmas 4 and 5] product bounds through the Weierstrass product inequality. For $N_{\text{tot}} \geq 2^c$, $\text{Lift}_c(N_{\text{tot}}) \geq 1$ and the advantage bound is trivial without any appeal to indistinguishability, and already at $N_{\text{tot}} \geq 2^{c/2}$ the bound exceeds $1/2$ and provides no meaningful security.

Key-guessing term. The ideal system key-guessing term of [BDPVA11] is

$$\text{KeyGuess}_k(Q) = Q/2^k.$$

It accounts for direct RO_{flat} queries whose inputs satisfy the decoder KeyedTWOut of Table 11. Each such query is an adaptive test of the decoded candidate key against the hidden key K , independent of the query's requested output length. Lemma 10 bounds the probability that any of Q direct queries tests the right key by $Q/2^k$.

Root-collision term. The s -way root tag multi-collision term is

$$\text{RootColl}_\tau(Q, s) = \binom{Q+s}{s} / 2^{\tau(s-1)},$$

with the abbreviation $\text{RootColl}_{\tau_{\text{root}}}(Q, s) = \text{RootColl}_\tau(Q, s)$ at $\tau = \tau_{\text{root}}$. The $+s$ is candidate-sequence slack for the s final root transcript inputs generated by the challenger after the adversary outputs its tuples. These lookups are construction-internal and are not added to $q_{\text{RO}}(\mathcal{B})$; after the flat sponge lift of Section B, the public-permutation bounds instantiate Q as N_{prim} .

682 **Root-preimage term.** The root tag preimage term is

$$683 \quad \text{RootPre}_\tau(Q) = (Q + 1)/2^\tau,$$

684 with the abbreviation $\text{RootPre}_{\tau_{\text{root}}}(Q) = \text{RootPre}_\tau(Q)$ at $\tau = \tau_{\text{root}}$. It is the preimage
 685 analog of RootColl_τ : a single query-independent target tag replaces the s -way internal
 686 collision, one challenger lookup replaces the s final-root lookups, and the exponent drops
 687 from $\tau(s - 1)$ to τ . As in the root-collision term, the flat sponge lift gives $Q = N_{\text{prim}}$.

688 **Leaf-collision term.** The known-key leaf tag collision term is

$$689 \quad \text{LeafColl}_\tau(Q) = \binom{Q}{2}/2^\tau.$$

690 It is the pairwise birthday bound on τ -bit leaf tag projections used by the collision-
 691 resistance theorem (Theorem 4). In this game the keys are adversary-supplied, so direct
 692 primitive queries can pre-read candidate leaf tag points; after the flat sponge lift this gives
 693 $Q = N_{\text{prim}} + N_{\text{leaf}}$. The hidden-key AE-style bounds instead carry the linear same-slot
 694 term $n_{\text{leaf}}/2^{\tau_{\text{leaf}}}$ of Lemma 11, written inline.

695 5 The Authentication Tag as a Hash Output

696 Externally, a TW128 encryption returns a ciphertext body and one authentication tag.
 697 Internally, that tag is the final root-duplex output: leaf tags summarize leaf chunks, the
 698 root aggregates those summaries with the context and root ciphertext chunk, and the
 699 wrapper H_{TW128} returns the tag alone. This section proves that the visible tag is a hash
 700 output for the full opening (K, U, A, M) , not merely an AEAD verification value.

701 The hash notions are the standard ones for H_{TW128} : collision, preimage, second-preimage,
 702 and everywhere preimage resistance. The committing statements are consequences of that
 703 hash security: a successful compact opening must force distinct openings to agree on the
 704 same τ_{root} -bit tag, the root-collision event bounded below.

705 The proof uses three ingredients. Transcript parsing separates final root inputs, except
 706 for the explicitly charged leaf-tag collision event. Lemma 3 orders every relevant final-root
 707 input into a chronological candidate sequence, and Lemma 4 applies the adaptive collision
 708 and preimage bound of Section A to that sequence. The random oracle theorems are
 709 short instantiations; the flat sponge lift gives the public-permutation bounds; the [BH22]
 710 committing notions reuse the same root-collision bound.

711 5.1 Tag Hash and Committing Games

712 The public-permutation games in this section use the primitive, validity, and construction-
 713 oracle conventions of Section 4.1. The random oracle versions are obtained by the
 714 mechanical substitution of Section 4.2.

715 **Hash games.** The hash games for the wrapper H_{TW128} (Definition 1) have no challenger-
 716 sampled key. In the s -way collision game, all inputs are adversary-supplied: the adversary
 717 outputs s inputs, and the challenger returns 1 iff they are pairwise distinct, in the TW128
 718 domain, and share a root tag:

719 $\pi \leftarrow \$ \text{Perm}(\{0, 1\}^{1600}), \quad (K_i, U_i, A_i, M_i)_{i=1}^s \leftarrow \$ \mathcal{A}^{\pi, \pi^{-1}},$
 720 return 0 unless the inputs are pairwise distinct and in the TW128 domain,
 721 return $[H_{\text{TW128}}(K_1, U_1, A_1, M_1) = \dots = H_{\text{TW128}}(K_s, U_s, A_s, M_s)],$

with advantage $\text{Adv}_{\text{TW128}}^{\text{Coll}}(\mathcal{A}) = \Pr[\text{the game returns 1}]$; no nonce uniqueness is imposed. The [RS04] standard and everywhere second-preimage notions are used on fixed input slices; in Corollary 2 they are reduced explicitly to this two-way collision game. For fixed byte lengths a, μ , the standard preimage game samples $X^* \leftarrow \$ \mathcal{X}_{a, \mu}$ independently of the primitive, computes $T^* = \text{H}_{\text{TW128}}(X^*)$, gives T^* to $\mathcal{A}^{\pi, \pi^{-1}}$, and returns 1 iff $\mathcal{A}$ outputs a valid input X in the TW128 domain with $\text{H}_{\text{TW128}}(X) = T^*$; the advantage is $\text{Adv}_{\text{TW128}}^{\text{Pre}[a, \mu]}(\mathcal{A})$. The everywhere preimage game fixes a target tag $T^* \in \{0, 1\}^{\tau_{\text{root}}}$ before the primitive sampling and returns 1 iff $\mathcal{A}(T^*)$ outputs an input X in the domain with $\text{H}_{\text{TW128}}(X) = T^*$, the advantage $\text{Adv}_{\text{TW128}}^{\text{ePre}}$ taken as the worst case over query-independent targets.

Committing games. The [BH22] committing games are instantiated as in Section 2.5, with no challenger-sampled key. The s -way CMTDs- ℓ (decryption) game has the adversary output a ciphertext $C \parallel T$ and s tuples (K_i, U_i, A_i, M_i) with pairwise distinct WiC_ℓ -images, and returns 1 iff $\text{Dec}_{K_i}[\pi](U_i, A_i, C \parallel T) = M_i$ for every i , with the eager length rejection justified by Algorithm 3 ($\text{Dec}_K(U, A, C \parallel T)$ returns $\perp$ or a message of length $\|C\|$); its advantage is $\text{Adv}_{\text{TW128}}^{\text{CMTDs-}\ell}(\mathcal{A})$. The CMTs- ℓ (encryption) game has the adversary output the s tuples alone and returns 1 iff the encryptions $\text{Enc}_{K_i}[\pi](U_i, A_i, M_i)$ are all equal; its advantage is $\text{Adv}_{\text{TW128}}^{\text{CMTs-}\ell}(\mathcal{A})$.

5.2 Separating Final Tag Inputs

The tag value is read from a final root transcript input. The next two facts rule out structural collisions at that input, except for the explicitly charged possibility that two leaf tags collide before the root aggregates them.

Proposition 1 (Opening Triple Separation). *If two TW128 root computations have distinct (K, U, A) triples, then their bridged final root transcript inputs under $\text{flat}(\cdot)$ are distinct, regardless of whether each computation is an encryption or a decryption check, and whether or not they act on a common ciphertext.*

Proof. Consider two root computations i and j with contexts $(K_i, U_i, A_i) \neq (K_j, U_j, A_j)$, reaching final root transcripts R_i and R_j . Suppose for contradiction that $\text{flat}(R_i) = \text{flat}(R_j)$. Since the bridge $\text{flat}(\cdot)$ is injective on well-formed transcript prefixes (Lemma 8), the native flattened inputs of the two final-root prefixes are equal. Then Lemma 6 recovers the full duplex-call sequence and reads off the pair $(\sigma, \sigma_{\text{ph}})$ for each call. The seven 4-bit phase suffixes are pairwise distinct, so the $\sigma_{\text{init}}, \sigma_{\text{ad}}^{\pm}, \sigma_{\text{msg}}^{\pm}$, and (when present) $\sigma_{\text{agg}}^{\pm}$ calls in the recovered sequence are unambiguously identified. The aggregation phase is present exactly when $n \geq 1$, and the two recovered sequences are equal, so they either both carry an aggregation phase or both omit it; either way the recovery of (K, U, A) below uses only the σ_{init} and AD calls, which precede any message or aggregation block.

Equality of the recovered sequences forces equality of the σ_{init} call, whose σ is $p_{\text{root}} \parallel K \parallel U$ with fixed-width K and U , hence $K_i = K_j$ and $U_i = U_j$. It also forces equality of the recovered AD subsequences. By the AD bullet of Lemma 6 the associated data phase contributes a σ_{ad}^+ -tagged block exactly when the associated data is nonempty, so the recovered sequences agree on the presence of the AD phase. If exactly one of A_i, A_j were empty the recovered sequences would differ in the presence of a σ_{ad}^+ call, contradicting their equality; hence either both are empty, giving $A_i = A_j = \varepsilon$, or both are nonempty, in which case the phase-recovery sub-claim of Lemma 6 inverts the ρ -byte-block partition map on the AD byte string and equal AD σ -block sequences imply $A_i = A_j$. In all cases $(K_i, U_i, A_i) = (K_j, U_j, A_j)$, contradicting the hypothesis of distinct triples. $\square$

Triple Separation handles distinct contexts; when inputs share a context but differ in

their messages, distinctness of the final root transcripts needs the absence of a leaf tag collision, recorded next.

Proposition 2 (Final-Root Input Separation). *Let $\text{bad}_{\text{leaf}}^*$ be the event that two distinct construction-generated final leaf transcripts receive the same τ_{leaf} -bit RF projection. On $\neg\text{bad}_{\text{leaf}}^*$, if two TW128 construction computations reach final root transcripts with $\text{flat}(R) = \text{flat}(R')$, then they share the same context (K, U, A) and the same full ciphertext C . Equivalently, on $\neg\text{bad}_{\text{leaf}}^*$ computations with distinct (K, U, A, C) reach distinct bridged final root transcripts.*

Proof. By bridge injectivity (Lemma 8), $\text{flat}(R) = \text{flat}(R')$ gives equal native flattened final-root inputs. Lemma 6 then recovers the same root initialization $p_{\text{root}} \parallel K \parallel U$ —hence the same K and U —the same associated data blocks—hence the same A —the same root ciphertext chunk, and the same leaf count and ordered leaf tag sequence. Under $\neg\text{bad}_{\text{leaf}}^*$, Leaf-Transcript Recovery (Lemma 7) identifies the equal leaf tag sequences with the same final leaf transcripts, hence the same leaf ciphertext bodies; with the common root ciphertext chunk this gives the same full C . $\square$

5.3 Candidate Bound for Tag Inputs

After separation, the probabilistic question is local: how likely is a collision or target hit among the final root inputs whose tag values matter to a game? The candidate machinery rests on one engine, stated here and proved in full in Section A.5: adaptively chosen distinct random oracle inputs whose values are still unread when chosen behave, for collision and target-hitting purposes, as fresh uniform draws.

Lemma 2 (Adaptive Candidate Collision and Preimage). *Consider any interaction with a lazy random oracle $\text{RO}_{\text{flat}} : \{0, 1\}^* \rightarrow \{0, 1\}^\infty$ that builds a sequence of distinct candidate inputs $Y_1, \dots, Y_m$. The process may make arbitrary non-candidate oracle queries and may choose each candidate adaptively from the full prior transcript and private state sampled before the candidate’s unread oracle value. Whenever a new candidate Y_j is appended, two premises hold: predictability—both the decision to append and the string Y_j are determined by the state preceding the append step—and unrevealed value—no oracle lookup before the append step has requested a positive number of bits of $\text{RO}_{\text{flat}}(Y_j)$; zero-length lookups at Y_j are permitted. Fix a projection length τ , and suppose the sequence has at most L candidates ($m \leq L$).*

1. Collision. *For every $s \geq 2$, the probability $\Pr[\text{coll}_{s,\tau}]$ that some s distinct candidates have equal τ -bit projections,*

$$\lfloor \text{RO}_{\text{flat}}(Y_{i_1}) \rfloor_\tau = \dots = \lfloor \text{RO}_{\text{flat}}(Y_{i_s}) \rfloor_\tau \quad \text{for some } 1 \leq i_1 < \dots < i_s \leq m,$$

$$\text{satisfies } \Pr[\text{coll}_{s,\tau}] \leq \binom{L}{s} \cdot 2^{-\tau(s-1)}.$$

2. Preimage. *For every target $t \in \{0, 1\}^\tau$ chosen independently of RO_{flat} , the probability $\Pr[\text{pre}_{\tau,t}]$ that some candidate’s τ -bit projection equals t , i.e. $\lfloor \text{RO}_{\text{flat}}(Y_j) \rfloor_\tau = t$ for some $1 \leq j \leq m$, satisfies $\Pr[\text{pre}_{\tau,t}] \leq L \cdot 2^{-\tau}$.*

The preimage case has a second-preimage corollary: for a designated candidate position j_0 , the probability $\Pr[\text{sec}_{\tau,j_0}]$ that some candidate $j \neq j_0$ has $\lfloor \text{RO}_{\text{flat}}(Y_j) \rfloor_\tau = \lfloor \text{RO}_{\text{flat}}(Y_{j_0}) \rfloor_\tau$, i.e. collides with the designated candidate on its τ -bit projection, satisfies $\Pr[\text{sec}_{\tau,j_0}] \leq (L-1) \cdot 2^{-\tau}$.

Proof sketch. The premises license lazy sampling at the append step: predictability fixes the candidate string before its value is drawn, and the unrevealed-value premise states that no bit of that value has been exposed, so $\text{RO}_{\text{flat}}(Y_j)$ is uniform and independent of

the entire prior transcript. Conditioning position by position, each fixed position set then contributes at most $2^{-\tau(s-1)}$ (each fixed position at most $2^{-\tau}$), and a union bound over the $\binom{L}{s}$ position sets (the L positions) gives the bounds. The full argument, including the second-preimage corollary, appears in Section A.5. $\square$

The remaining work is structural: exhibiting, in each game, a candidate sequence that satisfies these premises and contains the final root transcripts the win event constrains.

Lemma 3 (Root Tag Candidate Sequence). *Consider one of the hash or committing random oracle games of Section 4.2. The adversary $\mathcal{B}$ makes direct RO_{flat} queries and has no construction oracle. The only construction-internal positive-length lookups at root tag inputs are the root tag samplings of $h \geq 1$ deterministic challenger encryption or decryption checks at game-fixed points. Apart from the standard preimage game’s designated sampled source check, all check inputs and keys are adversary supplied. In that preimage game the sampled source input, including its key, is private challenger state sampled before the source tag is revealed, and the source computation counts as one of the h checks. Let the i -th check reach a final root transcript R_i whose root tag is read by a single RO_{flat} lookup at $\text{flat}(R_i)$. Call an RO_{flat} input a root tag input if it decodes under KeyedTWOut to a root tag output-point class— $\mathcal{R}_{\text{msg}}^+$ (the final root message block, which carries the root tag only on the no-leaf path) or $\mathcal{R}_{\text{tag}}$ (the aggregated root tag); multi-chunk computations also reach $\mathcal{R}_{\text{msg}}^+$ with zero requested output before aggregation, so this predicate over-approximates the root tag candidate set, which only loosens the count. Every execution then admits a chronological candidate sequence in which each root tag input is appended exactly once, immediately before the first event that is either a direct RO_{flat} query at it or a positive-length RO_{flat} lookup reading it, and inputs never reached by such an event are never appended. This sequence is such that*

1. the candidates are pairwise distinct, and if the challenger reaches its checks, each $\text{flat}(R_i)$ is in the sequence no later than the sampling of the i -th root tag answer;
2. it satisfies the predictability and unrevealed-value premises of Lemma 2; and
3. its length is at most $q_{\text{RO}}(\mathcal{B}) + h$.

Proof. A direct query to a keyed transcript is one of $\mathcal{B}$ ’s visible RO_{flat} queries counted in $q_{\text{RO}}(\mathcal{B})$, not a separate construction query. The only construction computations in the game are the h deterministic challenger checks named in the statement.

Distinctness and coverage. Each root tag input is appended at its first triggering event and never again, so the candidates are pairwise distinct by construction, and the rule is well defined however direct queries and challenger checks interleave. In particular, in the standard preimage game the source check’s root tag sampling may be the trigger for $\text{flat}(R_1)$; a later direct query at the same address is not a first trigger and appends nothing. Each $\text{flat}(R_i)$ is appended no later than the sampling of the i -th root tag answer, since that sampling is a positive-length lookup at $\text{flat}(R_i)$, giving (1).

Predictability. The decoded root-tag-input predicate depends on the input string alone through the exact decoder of Lemma 8, which ignores the requested output length; by Output-Point Separation (Corollary 8) each input decodes to at most one output-point class. The append rule also uses the already fixed next event: a direct query triggers at the address, while a construction lookup triggers only when it requests positive output length. The string itself is fixed in the state preceding the trigger: a direct query’s input is chosen by $\mathcal{B}$ before its lookup, and a check’s final root transcript is determined by the check’s input and the prior oracle answers—including any private challenger state sampled before the candidate’s unread value, which Lemma 2 expressly allows.

Unrevealed value. The trigger is the first direct query or positive-length lookup at the address, so no earlier direct query at it exists, and it suffices that no earlier construction

lookup has read a positive number of its bits. An earlier positive-length construction lookup is either a stream output point, a leaf-tag point $\mathcal{L}_{\text{tag}}(j)$, or a root-tag sampling. The first two are different output-point classes and hence, by Corollary 8, have different addresses from root tag inputs; the last would itself have been the first trigger at this address. Zero-requested-output lookups, including the $\mathcal{R}_{\text{msg}}^+$ lookup a multi-chunk computation makes before aggregation, are permitted by the unrevealed-value premise of Lemma 2, including when such a zero-length lookup lands at another check's future no-leaf final-root address. Together with predictability this gives (2).

Length. Appends are triggered only by direct queries—at most $q_{\text{RO}}(\mathcal{B})$ of them—or by the h checks' root tag samplings; zero-length construction lookups never trigger an append. Hence the sequence has length at most $q_{\text{RO}}(\mathcal{B}) + h$, giving (3). $\square$

Lemma 4 (Root Tag Hash and Target-Hit Bound). *Consider one of the hash or committing TW128 random oracle games (Section 4.2) for an adversary $\mathcal{B}$, satisfying the hypotheses of Lemma 3, with win event W . By the Root Tag Candidate Sequence lemma (Lemma 3) the root tag inputs form a chronological candidate sequence meeting the premises of Lemma 2 with $\tau = \tau_{\text{root}}$. Writing $q_{\text{RO}} = q_{\text{RO}}(\mathcal{B})$:*

1. Root-collision case. *If W forces s checks to reach pairwise distinct bridged final root transcripts carrying a common root tag, then, with $h = s$ checks, $\Pr[W] \leq \text{RootColl}_{\tau_{\text{root}}}(q_{\text{RO}}, s)$;*
2. Fixed-target case. *If W forces a single check ($h = 1$) to carry a root tag equal to a target fixed independently of RO_{flat} , then $\Pr[W] \leq \text{RootPre}_{\tau_{\text{root}}}(q_{\text{RO}})$;*
3. Sampled-source case. *If the game first runs a designated source check, reveals its root tag, and W forces one later check to reach a distinct bridged final root transcript carrying that designated tag, then, with $h = 2$ checks, $\Pr[W] \leq \text{RootPre}_{\tau_{\text{root}}}(q_{\text{RO}})$.*

Proof. In each case the candidate sequence has the stated length and meets the premises of Lemma 2. (1) The s distinct candidates carrying a common τ_{root} -projection are an s -way collision; the collision case with $L = q_{\text{RO}} + s$ gives $\binom{q_{\text{RO}}+s}{s} \cdot 2^{-\tau_{\text{root}}(s-1)} = \text{RootColl}_{\tau_{\text{root}}}(q_{\text{RO}}, s)$. (2) The candidate hitting the fixed target is a preimage; the preimage case with $L = q_{\text{RO}} + 1$ gives $(q_{\text{RO}} + 1) \cdot 2^{-\tau_{\text{root}}} = \text{RootPre}_{\tau_{\text{root}}}(q_{\text{RO}})$. (3) Let j_{src} be the position of the designated source check. The later check is required to be a distinct bridged final root transcript, so it is a candidate position $j \neq j_{\text{src}}$ with the same τ_{root} -bit projection. This is an application of the second-preimage corollary, not the fixed-target case, because the designated source tag is sampled from RO_{flat} rather than fixed independently of it. The candidate sequence has length at most $L = q_{\text{RO}} + 2$, and the second-preimage corollary of Lemma 2 gives $(L - 1)2^{-\tau_{\text{root}}} = (q_{\text{RO}} + 1)2^{-\tau_{\text{root}}} = \text{RootPre}_{\tau_{\text{root}}}(q_{\text{RO}})$. $\square$

5.4 Random Oracle Tag-Hash Bounds

Each random oracle bound now matches a hash game to a case of Lemma 4. Collision and fixed-slice preimage resistance also carry a leaf-collision term, because distinct openings may have different ciphertext bodies before reaching the root.

Theorem 1 (Random Oracle Collision Resistance of H_{TW128}). *For every $s \geq 2$ and every s -way collision adversary $\mathcal{B}$ in the TW128 random oracle model,*

$$\text{Adv}_{\text{RO}}^{\text{Coll}}(\mathcal{B}) \leq \text{LeafColl}_{\tau_{\text{leaf}}}(q_{\text{RO}}(\mathcal{B}) + n_{\text{leaf}}(\mathcal{B})) + \text{RootColl}_{\tau_{\text{root}}}(q_{\text{RO}}(\mathcal{B}), s).$$

Proof. Run the random oracle collision game. Let $\text{bad}_{\text{leaf}}^*$ be the event that two distinct construction-generated final leaf transcripts receive the same τ_{leaf} -bit RF projection; by the general count of Lemma 11, $\Pr[\text{bad}_{\text{leaf}}^*] \leq \text{LeafColl}_{\tau_{\text{leaf}}}(q_{\text{RO}}(\mathcal{B}) + n_{\text{leaf}}(\mathcal{B}))$, so it remains

908 to bound the win under $\neg \text{bad}_{\text{leaf}}^*$. After rejecting unless the s output inputs are pairwise
 909 distinct and in the TW128 domain, the challenger performs s deterministic encryptions
 910 $\text{Enc}_{K_i}^{\text{RO}}(U_i, A_i, M_i)$, reaching $R_1, \dots, R_s$, all carrying a common root tag T on a win. Pair-
 911 wise distinct tuples (K_i, U_i, A_i, M_i) have pairwise distinct (K_i, U_i, A_i, C_i) : two tuples
 912 differing in their context already differ in (K, U, A) , while two sharing a context but differ-
 913 ing in the message yield distinct ciphertexts, since for a fixed context encryption maps the
 914 message to the ciphertext bijectively. So on $\neg \text{bad}_{\text{leaf}}^*$ Final-Root Input Separation (Proposi-
 915 tion 2) makes $\text{flat}(R_1), \dots, \text{flat}(R_s)$ pairwise distinct. The Root Tag Hash and Target-Hit
 916 Bound (Lemma 4, item 1) bounds the win under $\neg \text{bad}_{\text{leaf}}^*$ by $\text{RootColl}_{\tau_{\text{root}}}(q_{\text{RO}}(\mathcal{B}), s)$; adding
 917 $\Pr[\text{bad}_{\text{leaf}}^*]$ gives the theorem. $\square$

918 **Theorem 2** (Random Oracle Preimage Resistance of H_{TW128}). *Fix byte lengths $0 \leq a \leq$
 919 A_{max} and $0 \leq \mu \leq M_{\text{max}}$, and let $m_{a,\mu} = k + \nu + 8a + 8\mu$. For every Pre adversary $\mathcal{B}$ against
 920 H_{TW128} on the fixed input slice $\mathcal{X}_{a,\mu}$ in the TW128 random oracle model (Section 4.2),*

$$\begin{aligned} \text{Adv}_{\text{RO}}^{\text{Pre}[a,\mu]}(\mathcal{B}) &\leq \text{LeafColl}_{\tau_{\text{leaf}}}(q_{\text{RO}}(\mathcal{B}) + n_{\text{leaf}}(\mathcal{B})) \\ &\quad + \text{RootPre}_{\tau_{\text{root}}}(q_{\text{RO}}(\mathcal{B})) + 2^{\tau_{\text{root}} - m_{a,\mu}}. \end{aligned}$$

921 *Proof.* Run the random oracle Pre game. The challenger samples $X^* \leftarrow \$ \mathcal{X}_{a,\mu}$, writes
 922 $X^* = (K^*, U^*, A^*, M^*)$, computes $T^* = \text{H}_{\text{TW128}}(X^*)$, gives T^* to $\mathcal{B}$, and later checks
 923 the output $X = (K, U, A, M)$. Let **same** be the event $X = X^*$. For a fixed random
 924 oracle table and fixed adversary coins, the adversary's output is a function of the τ_{root} -bit
 925 challenge tag T^* , so over the uniform source X^* it can equal X^* with probability at most
 926 $2^{\tau_{\text{root}}}/2^{m_{a,\mu}} = 2^{\tau_{\text{root}} - m_{a,\mu}}$. Hence $\Pr[\text{same}] \leq 2^{\tau_{\text{root}} - m_{a,\mu}}$.
 927

928 It remains to bound winning executions with $X \neq X^*$. Let $\text{bad}_{\text{leaf}}^*$ be the event that
 929 two distinct construction-generated final leaf transcripts in the source and output checks
 930 receive the same τ_{leaf} -bit RF projection. By the general count of Lemma 11—which charges
 931 every direct query and every construction-generated final leaf transcript regardless of key
 932 provenance, covering the challenger-sampled source key—

$$\Pr[\text{bad}_{\text{leaf}}^*] \leq \text{LeafColl}_{\tau_{\text{leaf}}}(q_{\text{RO}}(\mathcal{B}) + n_{\text{leaf}}(\mathcal{B})).$$

933 On $\neg \text{bad}_{\text{leaf}}^*$, the distinct tuples X^* and X have distinct final root transcripts: distinct
 934 contexts are separated by Proposition 1, while equal contexts and distinct messages
 935 produce distinct ciphertexts, and then Proposition 2 applies. A win with $X \neq X^*$
 936 therefore forces the later output check to hit the designated source tag T^* at a distinct
 937 bridged final root transcript. The sampled-source case, item 3, of Lemma 4 bounds this
 938 probability by $\text{RootPre}_{\tau_{\text{root}}}(q_{\text{RO}}(\mathcal{B}))$. Adding the source-guess and leaf-collision events gives
 939 the theorem. $\square$

940 **Theorem 3** (Random Oracle Everywhere Preimage Resistance of H_{TW128}). *For every*
 941 *query-independent target tag $T^* \in \{0, 1\}^{\tau_{\text{root}}}$ and every ePre adversary $\mathcal{B}$ in the TW128*
 942 *random oracle model (Section 4.2),*

$$\text{Adv}_{\text{RO}}^{\text{ePre}}(\mathcal{B}) \leq \text{RootPre}_{\tau_{\text{root}}}(q_{\text{RO}}(\mathcal{B})) = \frac{q_{\text{RO}}(\mathcal{B}) + 1}{2^{\tau_{\text{root}}}}.$$

943 *Proof.* Run the random oracle preimage game for the query-independent target T^* , which
 944 is independent of RO_{flat} . After rejecting unless the output input $X = (K, U, A, M)$ is in the
 945 TW128 domain, the challenger performs one encryption $\text{Enc}_K^{\text{RO}}(U, A, M)$, reaching a final
 946 root transcript R . A win means $\text{H}_{\text{TW128}}(X) = T^*$, i.e. the root tag $[\text{RO}_{\text{flat}}(\text{flat}(R))]_{\tau_{\text{root}}}$ at
 947 R equals the fixed target T^* . The fixed-target case, item 2, of Lemma 4 gives $\text{Adv}_{\text{RO}}^{\text{ePre}}(\mathcal{B}) \leq$
 948 $\text{RootPre}_{\tau_{\text{root}}}(q_{\text{RO}}(\mathcal{B}))$. $\square$

5.5 Public Permutation Tag-Hash Bounds

These are the public-permutation hash claims for the authentication tag. The flat sponge lift transfers the preceding random oracle bounds to the public permutation model. For these one-world hash games, the lift contributes $\text{Lift}_c(N_{\text{tot}})$, and the derived random oracle adversary has $q_{\text{RO}}(\mathcal{B}) \leq N_{\text{prim}}$; challenger construction computations—including the preimage game’s source encryption—are construction-internal work counted in N .

Theorem 4 (Collision Resistance of H_{TW128}). *For every $s \geq 2$ and every adversary $\mathcal{A}$ against the s -way multi-collision resistance of H_{TW128} in the sense of [BH22], with the resources of Section 4.3,*

$$\text{Adv}_{\text{TW128}}^{\text{Coll}}(\mathcal{A}) \leq \text{Lift}_c(N_{\text{tot}}) + \text{LeafColl}_{\tau_{\text{leaf}}}(N_{\text{prim}} + N_{\text{leaf}}) + \text{RootColl}_{\tau_{\text{root}}}(N_{\text{prim}}, s).$$

Proof. By the Lift Application corollary (Corollary 9) for the collision game, the derived random oracle adversary $\mathcal{B}$ has $q_{\text{RO}}(\mathcal{B}) \leq N_{\text{prim}}$ and $n_{\text{leaf}}(\mathcal{B}) \leq N_{\text{leaf}}$. The random oracle bound of Theorem 1 is nondecreasing in both $q_{\text{RO}}(\mathcal{B})$ and $n_{\text{leaf}}(\mathcal{B})$, so the monotone form of Corollary 9 gives the stated bound. The s colliding inputs may differ in their messages, hence in their ciphertext bodies; the LeafColl term accounts for the only way two distinct inputs sharing a context (K, U, A) can reach the same final root transcript—a collision among the leaf tags it absorbs—while distinct contexts already reach distinct final root transcripts by Opening Triple Separation (Proposition 1). $\square$

Corollary 1 (Two-Way Collision Resistance of H_{TW128}). *Specializing Theorem 4 to $s = 2$ gives the classical two-way collision resistance of [RS04]: for every collision adversary $\mathcal{A}$ against H_{TW128} with the resources of Section 4.3,*

$$\text{Adv}_{\text{TW128}}^{\text{Coll}}(\mathcal{A}) \leq \text{Lift}_c(N_{\text{tot}}) + \text{LeafColl}_{\tau_{\text{leaf}}}(N_{\text{prim}} + N_{\text{leaf}}) + \text{RootColl}_{\tau_{\text{root}}}(N_{\text{prim}}, 2),$$

with $\text{RootColl}_{\tau_{\text{root}}}(N_{\text{prim}}, 2) = \binom{N_{\text{prim}}+2}{2} / 2^{\tau_{\text{root}}}$.

Proof. The $s = 2$ case of Theorem 4; [BH22] note that $s = 2$ recovers the classical notion of collision resistance, which under the keyed-by-the-permutation convention of Section 2.5 is the Coll notion of [RS04]. $\square$

Corollary 2 (Second-Preimage Resistance of H_{TW128}). *Fix byte lengths $0 \leq a \leq A_{\text{max}}$ and $0 \leq \mu \leq M_{\text{max}}$. For every adversary against H_{TW128} , with the resources of Section 4.3, in either the standard second-preimage (Sec) or everywhere second-preimage (eSec) sense of [RS04] on the fixed input slice $\mathcal{X}_{a,\mu}$, the corresponding advantage is bounded, independently of (a, μ) , by the two-way collision bound of Corollary 1:*

$$\text{Lift}_c(N_{\text{tot}}) + \text{LeafColl}_{\tau_{\text{leaf}}}(N_{\text{prim}} + N_{\text{leaf}}) + \text{RootColl}_{\tau_{\text{root}}}(N_{\text{prim}}, 2).$$

Proof. We use [RS04] to identify the standard and everywhere second-preimage notions with the fixed-slice notions $\text{Sec}[m_{a,\mu}]$ and $\text{eSec}[m_{a,\mu}]$ under the canonical encoding of $\mathcal{X}_{a,\mu}$; the quantitative reduction is explicit.

For a standard Sec adversary $\mathcal{A}$, define a two-way collision adversary $\mathcal{C}$ as follows. It samples $X_0 \leftarrow \$ \mathcal{X}_{a,\mu}$, runs $\mathcal{A}^{\pi, \pi^{-1}}(X_0)$, receives X_1 , and outputs (X_0, X_1) . Whenever $\mathcal{A}$ wins, X_1 is a valid input, $X_1 \neq X_0$, and $H_{\text{TW128}}(X_1) = H_{\text{TW128}}(X_0)$, so $\mathcal{C}$ wins the two-way collision game with exactly the same probability.

For eSec, fix a source $X_0 \in \mathcal{X}_{a,\mu}$ and define $\mathcal{C}_{X_0}$ to run the eSec adversary on that source and output (X_0, X_1) . The same pointwise implication holds for every X_0 ; taking the maximum over sources gives the eSec bound.

The induced collision adversary makes exactly the direct primitive queries made by $\mathcal{A}$. Its two deterministic challenger checks are the source and candidate checks, counted exactly as in the $s = 2$ collision game. Applying Corollary 1 to the induced collision adversary gives the displayed bound. $\square$

997 **Theorem 5** (Preimage Resistance of H_{TW128}). *Fix byte lengths $0 \leq a \leq A_{\max}$ and*
 998 *$0 \leq \mu \leq M_{\max}$, and let $m_{a,\mu} = k + \nu + 8a + 8\mu$. For every adversary $\mathcal{A}$ against H_{TW128} ,*
 999 *with the resources of Section 4.3, in the standard preimage (Pre) sense of [RS04] on the*
 1000 *fixed input slice $\mathcal{X}_{a,\mu}$,*

$$1001 \quad \text{Adv}_{TW128}^{\text{Pre}[a,\mu]}(\mathcal{A}) \leq \text{Lift}_c(N_{\text{tot}}) + \text{LeafColl}_{\tau_{\text{leaf}}}(N_{\text{prim}} + N_{\text{leaf}}) \\ + \text{RootPre}_{\tau_{\text{root}}}(N_{\text{prim}}) + 2^{\tau_{\text{root}} - m_{a,\mu}}.$$

1002 *Proof.* By the Lift Application corollary (Corollary 9) for the preimage game, the derived
 1003 random oracle adversary $\mathcal{B}$ has $q_{\text{RO}}(\mathcal{B}) \leq N_{\text{prim}}$ and $n_{\text{leaf}}(\mathcal{B}) \leq N_{\text{leaf}}$; per Section 4.3,
 1004 these caps cover both challenger encryptions' final leaf transcripts, the source encryption's
 1005 included; the encryptions' duplex calls are counted in N and enter through $\text{Lift}_c(N_{\text{tot}})$. The
 1006 random oracle bound of Theorem 2 is nondecreasing in both resources, so the monotone
 1007 form of Corollary 9 gives the stated bound. $\square$

1008 **Theorem 6** (Everywhere Preimage Resistance of H_{TW128}). *For every target tag $T^* \in$*
 1009 *$\{0,1\}^{\tau_{\text{root}}}$ fixed in advance and every ePre adversary $\mathcal{A}$ against H_{TW128} with the resources*
 1010 *of Section 4.3,*

$$1011 \quad \text{Adv}_{TW128}^{\text{ePre}}(\mathcal{A}) \leq \text{Lift}_c(N_{\text{tot}}) + \text{RootPre}_{\tau_{\text{root}}}(N_{\text{prim}}) = \text{Lift}_c(N_{\text{tot}}) + \frac{N_{\text{prim}} + 1}{2^{\tau_{\text{root}}}}.$$

1012 *Proof.* By the Lift Application corollary (Corollary 9) for the everywhere-preimage game—
 1013 whose target T^* is fixed before the primitive sampling—the derived random oracle adversary
 1014 $\mathcal{B}$ has $q_{\text{RO}}(\mathcal{B}) \leq N_{\text{prim}}$, and the random oracle bound of Theorem 3 is nondecreasing in
 1015 $q_{\text{RO}}(\mathcal{B})$, so the monotone form of Corollary 9 gives the stated bound. Compared with the
 1016 s -way root-collision term $\text{RootColl}_{\tau_{\text{root}}}(\cdot, s)$ of Section 4.4, the single fixed target replaces
 1017 the exponent $\tau_{\text{root}}(s - 1)$ by τ_{root} . $\square$

1018 **Corollary 3** (Tag Hash Security and Compact Commitment). *The $TW128$ authentication*
 1019 *tag, represented by the wrapper H_{TW128} , is a secure hash of the full encryption context*
 1020 *(K, U, A, M) in the sense of [RS04]: collision-resistant (Theorem 4), preimage-resistant on*
 1021 *each fixed input slice (Theorem 5), second-preimage-resistant (Corollary 2), and everywhere*
 1022 *preimage-resistant (Theorem 6). Under the implemented parameters and work cap $N_{\text{tot}} \leq$*
 1023 *2^{63} , these advantages are at most 2^{-128} (Corollary 7). These hash bounds make the single*
 1024 *τ_{root} -bit tag a binding compact commitment to (K, U, A, M) , with standard and everywhere*
 1025 *preimage resistance, native to the mode.*

1026 *Proof.* Collision resistance is Theorem 4; standard preimage resistance on fixed input slices
 1027 is Theorem 5; second-preimage resistance follows from collision resistance by Corollary 2;
 1028 and everywhere preimage resistance is Theorem 6. These bounds quantify the adversary's
 1029 success at the same τ_{root} -bit projection of the final root transcript, and Corollary 7 evaluates
 1030 them at the implemented parameters. Binding is the compact-commitment property
 1031 supplied by collision and second-preimage resistance. Standard preimage resistance is the
 1032 property that a tag produced from a uniformly sampled fixed-length opening does not
 1033 reveal an efficiently findable opening; everywhere preimage resistance is the corresponding
 1034 fixed-target property for tags chosen independently of the primitive. $\square$

1035 5.6 Committing Consequences of Tag Hash Security

1036 The committing consequences use the same tag-collision analysis under the opening
 1037 constraints of the [BH22] games. CMTDs-4 is slightly tighter than general collision
 1038 resistance: all openings share one ciphertext body, so distinct openings separate through
 1039 their opening triples and no leaf-collision term is needed.

Theorem 7 (Random Oracle CMTDs-4). *For every $s \geq 2$ and every s -way CMTDs-4 adversary $\mathcal{B}$ in the TW128 random oracle model (Section 4.2),*

$$\mathbf{Adv}_{\text{RO}}^{\text{CMTDs-4}}(\mathcal{B}) \leq \text{RootColl}_{\tau_{\text{root}}}(q_{\text{RO}}(\mathcal{B}), s) = \frac{\binom{q_{\text{RO}}(\mathcal{B})+s}{s}}{2^{\tau_{\text{root}}(s-1)}}.$$

Proof. Run the random oracle CMTDs-4 game. After rejecting on any failed syntax, message-length, or pairwise-full-input check, the challenger performs s deterministic decryption checks $\text{Dec}_{K_i}^{\text{RO}}(U_i, A_i, C \parallel T)$ on the one common ciphertext $C \parallel T$, reaching $R_1, \dots, R_s$. On a winning execution the tuples (K_i, U_i, A_i, M_i) are pairwise distinct, and since all checks decrypt the common ciphertext, determinism of Dec makes the triples (K_i, U_i, A_i) pairwise distinct—equal triples would return equal messages, hence equal tuples. By Opening Triple Separation (Proposition 1) the candidates $\text{flat}(R_1), \dots, \text{flat}(R_s)$ are pairwise distinct (leaf tag collisions cannot merge them), and every accepting check carries the common root tag T . The Root Tag Hash and Target-Hit Bound (Lemma 4, item 1) gives $\mathbf{Adv}_{\text{RO}}^{\text{CMTDs-4}}(\mathcal{B}) \leq \text{RootColl}_{\tau_{\text{root}}}(q_{\text{RO}}(\mathcal{B}), s)$. $\square$

Corollary 4 (CMTDs-4 Committing Security). *For every $s \geq 2$ and every s -way CMTDs-4 adversary $\mathcal{A}$ against TW128 with the resources of Section 4.3,*

$$\mathbf{Adv}_{\text{TW128}}^{\text{CMTDs-4}}(\mathcal{A}) \leq \text{Lift}_c(N_{\text{tot}}) + \text{RootColl}_{\tau_{\text{root}}}(N_{\text{prim}}, s) = \text{Lift}_c(N_{\text{tot}}) + \frac{\binom{N_{\text{prim}}+s}{s}}{2^{\tau_{\text{root}}(s-1)}}.$$

Proof. A CMTDs-4 winner gives one ciphertext $C \parallel T$ and s distinct inputs (K_i, U_i, A_i, M_i) such that $\text{Dec}_{K_i}(U_i, A_i, C \parallel T) = M_i$ for every i . By TW128 tidiness (Lemma 1), each opened input re-encrypts to the same ciphertext: $\text{Enc}_{K_i}(U_i, A_i, M_i) = C \parallel T$. In particular the opened inputs share the root tag T . For the displayed no-LeafColl bound, use the dedicated root-collision derivation of Theorem 7, lifted by Corollary 9: in this all-bodies-equal case, the single shared body C forces the triples (K_i, U_i, A_i) pairwise distinct; by Opening Triple Separation (Proposition 1), a leaf tag collision cannot merge two openings and no LeafColl term arises. $\square$

Remaining committing notions. The hierarchy and s -way extension of the [BH22] committing notions are given in [BH22, §3, Figs. 4–5, and App. A]. We use those relations for the D-notions and a direct source-game argument for the CMTs- ℓ notions to preserve the resource split of Section 4.3.

Corollary 5 (Committing-Notion Root-Collision Bound). *For every $s \geq 2$, every*

$$X \in \{\text{CMTDs-1}, \text{CMTDs-3}, \text{CMTDs-4}, \text{CMTs-1}, \text{CMTs-3}, \text{CMTs-4}\},$$

and every s -way X -adversary $\mathcal{A}$ against TW128 with the resources of Section 4.3,

$$\mathbf{Adv}_{\text{TW128}}^X(\mathcal{A}) \leq \text{Lift}_c(N_{\text{tot}}) + \text{RootColl}_{\tau_{\text{root}}}(N_{\text{prim}}, s) = \text{Lift}_c(N_{\text{tot}}) + \frac{\binom{N_{\text{prim}}+s}{s}}{2^{\tau_{\text{root}}(s-1)}}.$$

Proof. CMTDs-4 is bounded directly by Corollary 4. For the remaining D-notions, the s -way hierarchy of [BH22, §3, Fig. 5 and App. A] gives, for $\ell \in \{1, 3\}$, $\mathbf{Adv}_{\text{TW128}}^{\text{CMTDs-}\ell}(\mathcal{A}) \leq \mathbf{Adv}_{\text{TW128}}^{\text{CMTDs-4}}(\mathcal{A})$ for the same output distribution. By the conservative committing-game accounting of Section 4.3, the CMTDs- ℓ and CMTDs-4 resource caps charge the same deterministic decryption checks after syntax, domain, and message-length validation, independently of their distinctness gates.

For CMTs- ℓ , the challenger performs s deterministic encryption checks, which supply the candidate final-root transcripts (Lemma 3 builds the candidate sequence for encryption and decryption checks alike). On a winning execution all s encryptions are equal, hence of

equal length and with one common root tag T . Pairwise WiC_1 -distinctness gives pairwise distinct keys, pairwise WiC_3 -distinctness gives pairwise distinct (K, U, A) triples, and pairwise WiC_4 -distinctness gives pairwise distinct full inputs; in the last case, equal triples and equal ciphertexts would decrypt deterministically to equal messages, contradicting full-input distinctness. Thus each CMTs- ℓ winner reaches pairwise distinct opening triples. Opening Triple Separation (Proposition 1) makes the s final-root candidates distinct while their τ_{root} -bit projections all equal T . The CMTs- ℓ encryption checks are the construction-internal work counted in N by Section 4.3.

By the Lift Application corollary (Corollary 9) for the CMTs- ℓ game, the derived random oracle adversary $\mathcal{B}$ has $q_{\text{RO}}(\mathcal{B}) \leq N_{\text{prim}}$. The Root Tag Hash and Target-Hit Bound (Lemma 4, item 1) then bounds its win by $\text{RootColl}_{\tau_{\text{root}}}(q_{\text{RO}}(\mathcal{B}), s)$ in the random oracle model, which the lift transfers to the public permutation setting at N_{prim} ; the encryption checks are construction-internal work counted in N , not in N_{prim} . $\square$

6 Baseline Authenticated Encryption Bounds

Section 5 proves the tag-centric properties of TW128. Here we discharge the AEAD-facing obligations: multi-user indistinguishability, its single-user all-in-one AE corollary, and INT-RUP integrity. These bounds use the same hidden-key, leaf-collision, and tag-guessing accounting as the tag analysis, but their role is narrower: they show that the mode remains a conventional nonce-respecting AEAD.

We first prove mu-ind in the flat random oracle model, lift the result to the public permutation model using Section B, and derive all-in-one AE as the $D = 1$ case. We then prove INT-RUP in the separated-interface model of [ABL⁺14], where plaintext-computing queries increase the construction work counts but do not change the non-replay verification count. All concrete instantiations are summarized in Section 7.

6.1 AEAD Public Permutation Games

The public-permutation games in this section use the primitive, validity, and construction-oracle conventions of Section 4.1. The random oracle versions are obtained by the mechanical substitution of Section 4.2.

All-in-one AE. The all-in-one authenticated encryption game [RS06] samples $K \leftarrow \$\{0, 1\}^k$ and compares the real pair $(\text{Enc}_K[\pi], \text{Ver}_K[\pi])$ with the ideal pair $(\text{Rand}, \text{Replay})$. On each accepted encryption query (U, A, M) , Rand samples a uniformly random byte string of length $\|M\| + \tau_{\text{root}}/8$, parses it as $C \parallel T$ with $T \in \{0, 1\}^{\tau_{\text{root}}}$, records (U, A, C, T) , and returns $C \parallel T$; Replay accepts exactly recorded tuples. Encryption queries are nonce-respecting, and

$$\text{Adv}_{\text{TW128}}^{\text{ae}}(\mathcal{A}) = \left| \Pr[\mathcal{A}^{\text{Enc}_K[\pi], \text{Ver}_K[\pi], \pi, \pi^{-1}} \Rightarrow 1] - \Pr[\mathcal{A}^{\text{Rand}, \text{Replay}, \pi, \pi^{-1}} \Rightarrow 1] \right|.$$

This all-in-one AE advantage is the single-user case of the multi-user real-vs-ideal notion of Section 2.7; Corollary 6 derives it from Theorem 9.

RUP integrity. The INT-RUP game samples $K \leftarrow \$\{0, 1\}^k$ and $\pi \leftarrow \$\text{Perm}(\{0, 1\}^{1600})$, exposes $\text{Enc}_K[\pi]$, $\text{PDec}_K[\pi]$, $\text{Ver}_K[\pi]$, and (π, π^{-1}) , and maintains the replay table W of encryption outputs. Accepted encryption queries are nonce-respecting: an encryption query is rejected if its nonce has appeared in a previous accepted encryption query. Plaintext-computing and verification queries may use arbitrary nonces. On an accepted encryption query (U, A, M) the game returns $\text{Enc}_K[\pi](U, A, M)$, parsed and recorded as (U, A, C, T) in W . On a valid plaintext-computing query (U, A, C, T) it returns $\text{PDec}_K[\pi](U, A, C, T)$,

and on an invalid query it returns $\perp$. On a valid verification query (U, A, C, T) it returns $\text{Ver}_K[\pi](U, A, C, T)$; if this bit is 1 and $(U, A, C, T) \notin W$ when the query is asked, the game records a forgery. Invalid verification queries return 0. The game returns 1 iff a forgery is recorded, and

$$\text{Adv}_{\text{TW128}}^{\text{INT-RUP}}(\mathcal{A}) = \Pr[\text{G}_{\text{TW128}}^{\text{INT-RUP}}(\mathcal{A}) \Rightarrow 1].$$

6.2 Multi-User Setting

This subsection instantiates the mu-ind game of [BT16] (Section 2.7) for TW128 in the public permutation model, under the primitive convention of Section 4.1. The challenger samples a challenge bit $b \leftarrow \{0, 1\}$ and a permutation $\pi \leftarrow \text{Perm}(\{0, 1\}^{1600})$, and exposes (π, π^{-1}) as the primitive interface in both worlds. The adversary $\mathcal{A}$ creates users on demand via **New**, and the game maintains an active-user counter v . We write D for an execution-uniform cap on the number of **New** calls. Equivalently, the game may sample independent keys $K_1, \dots, K_D$ at the start of the experiment and let the d -th **New** call activate K_d ; inactive indices are never valid users.

The mu-ind construction oracles inherit the validity convention of Section 4.1. The encryption oracle on input (d, U, A, M) rejects, without recording, if (U, A, M) is outside the TW128 encryption domain, if $d \notin \{1, \dots, v\}$, or if $(d, U) \in V$; otherwise it sets $C = \text{Enc}_{K_d}[\pi](U, A, M)$ when $b = 1$, and samples a uniformly random byte string C of length $\|M\| + \tau_{\text{root}}/8$ when $b = 0$. The game records (d, U) in V and (d, U, A, C) in W , and returns C . The verification oracle on input (d, U, A, C) rejects if $d \notin \{1, \dots, v\}$ or if (U, A, C) is not a valid TW128 verification/decryption input; if $(d, U, A, C) \in W$ it returns true; if $b = 0$ it returns false; otherwise it returns $[\text{Dec}_{K_d}[\pi](U, A, C) \neq \perp]$. Nonce uniqueness is enforced per user by the set V .

After querying these oracles, $\mathcal{A}$ outputs $b' \in \{0, 1\}$. The mu-ind advantage of $\mathcal{A}$ against TW128 is

$$\text{Adv}_{\text{TW128}}^{\text{mu-ind}}(\mathcal{A}) = |2 \Pr[b' = b] - 1|.$$

A valid verification query is *non-replay* if its tuple is not in W when the query is asked; replay hits are answered by the recorded transcript and are not counted in the verification-query resource below.

Resource caps split per potential user $d \leq D$: $N^{(d)}$, $q_v^{(d)}$, $n_{\text{leaf}}^{(d)}$ are the per-user counts under Section 4.3, with inactive users contributing zero. The global totals $N = \sum_{d=1}^D N^{(d)}$, $q_v = \sum_{d=1}^D q_v^{(d)}$, $n_{\text{leaf}} = \sum_{d=1}^D n_{\text{leaf}}^{(d)}$ appear in the final bound: N through the lift term $\text{Lift}_c(N_{\text{tot}})$, and n_{leaf} and q_v through the leaf-collision and verification-tag terms. N_{prim} remains a single shared count over the primitive interface.

The mu-ind treatment applies only to indistinguishability. The committing security games already give the adversary control of all s keys, so the s -way CMTDs- ℓ statements of Section 5.1 are themselves the multi-key committing statement and need no separate mu-cmt notion.

All random oracle adversaries in this section are flat-RO adversaries (Section 4.2) with the displayed resources, and the flat sponge lift transfers each random oracle bound using the derived-adversary cap $q_{\text{RO}}(\mathcal{B}) \leq N_{\text{prim}}$.

Theorem 8 (Random Oracle Multi-User Indistinguishability). *For every flat-RO mu-ind adversary $\mathcal{B}$ with execution-uniform user cap D and resource counts of Section 6.2,*

$$\text{Adv}_{\text{RO}}^{\text{mu-ind}}(\mathcal{B}) \leq \frac{D(D-1)}{2^{k+1}} + D \cdot \text{KeyGuess}_k(q_{\text{RO}}(\mathcal{B})) + \frac{n_{\text{leaf}}(\mathcal{B})}{2^{\tau_{\text{leaf}}}} + \frac{q_v(\mathcal{B})}{2^{\tau_{\text{root}}}}.$$

Here q_v is the non-replay verification-query count of Section 4.3.

Proof. Use the pre-sampled-key formulation of Section 6.2: sample keys $K_1, \dots, K_D$ at the start of the experiment, and let the d -th New call activate K_d , unused keys being ignored. Let coll denote the event that two of the pre-sampled keys $K_1, \dots, K_D$ collide; $\Pr[\text{coll}] \leq \binom{D}{2}/2^k = D(D-1)/2^{k+1}$ by union bound. In the random oracle model, use a standard hybrid over the D users: H_d answers users $1, \dots, d$ with (Rand, Replay) and users $d+1, \dots, D$ with the real construction interface, so H_0 is the real mu-ind game and H_D is the ideal one. Let H_d^* be H_d with a fixed output bit on coll . Since coll has the same probability in all hybrids,

$$\text{Adv}_{\text{RO}}^{\text{mu-ind}}(\mathcal{B}) \leq \Pr[\text{coll}] + |\Pr[\mathcal{B}^{H_0^*} \Rightarrow 1] - \Pr[\mathcal{B}^{H_D^*} \Rightarrow 1]|.$$

By the triangle inequality, it remains to bound each adjacent stopped hybrid. For each d , Lemma 17 gives

$$|\Pr[\mathcal{B}^{H_{d-1}^*} \Rightarrow 1] - \Pr[\mathcal{B}^{H_d^*} \Rightarrow 1]| \leq \text{KeyGuess}_k(q_{\text{RO}}(\mathcal{B})) + \frac{n_{\text{leaf}}^{(d)}}{2^{\tau_{\text{leaf}}}} + \frac{q_v^{(d)}}{2^{\tau_{\text{root}}}}.$$

Lemma 17 is the only construction-specific step in the baseline mu-ind proof. It couples the adjacent user- d hybrids through the all-reject continuation of Definition 5. The coupling agrees until a direct random-oracle query tests the hidden key K_d or a non-replay verification comparison succeeds. The resulting first-occurrence decomposition has exactly the three terms displayed above: the foreign-key test is bounded by $\text{KeyGuess}_k(q_{\text{RO}}(\mathcal{B}))$; a same-slot collision between final leaf tags for user d is bounded by $n_{\text{leaf}}^{(d)}/2^{\tau_{\text{leaf}}}$ via Lemma 11; and a successful verification-first root comparison is bounded by $q_v^{(d)}/2^{\tau_{\text{root}}}$ via Lemma 15. Encryption transcript marginal uniformity and environment neutrality (Lemmas 12 and 13) are the facts that make this coupling independent of the other users.

Summing across $d = 1, \dots, D$ gives $D \text{KeyGuess}_k(q_{\text{RO}}(\mathcal{B}))$ for the key-test contribution. The leaf tag contribution sums exactly:

$$\sum_d \frac{n_{\text{leaf}}^{(d)}}{2^{\tau_{\text{leaf}}}} = \frac{n_{\text{leaf}}(\mathcal{B})}{2^{\tau_{\text{leaf}}}},$$

since $\sum_d n_{\text{leaf}}^{(d)} = n_{\text{leaf}}(\mathcal{B})$. The verification-tag contribution satisfies

$$\sum_d q_v^{(d)}/2^{\tau_{\text{root}}} = q_v(\mathcal{B})/2^{\tau_{\text{root}}}.$$

Combining these estimates with the coll penalty gives the stated RO bound. $\square$

Theorem 9 (Multi-User Indistinguishability). *For every mu-ind adversary $\mathcal{A}$ against TW128 with execution-uniform user cap D and the global resources of Section 6.2,*

$$\text{Adv}_{\text{TW128}}^{\text{mu-ind}}(\mathcal{A}) \leq \text{Lift}_c(N_{\text{tot}}) + \text{Lift}_c(N_{\text{prim}}) + \frac{D(D-1)}{2^{k+1}} + D \cdot \text{KeyGuess}_k(N_{\text{prim}}) + \frac{N_{\text{leaf}}}{2^{\tau_{\text{leaf}}}} + \frac{Q_v}{2^{\tau_{\text{root}}}}.$$

Proof. By the Lift Application corollary (Corollary 9) for the mu-ind game—a real-vs-ideal game, so the ideal-side proximity term applies—there is a random oracle mu-ind adversary $\mathcal{B} = \mathcal{B}_{\text{derived}}(\mathcal{A})$ with $q_{\text{RO}}(\mathcal{B}) \leq N_{\text{prim}}$, $n_{\text{leaf}}(\mathcal{B}) \leq N_{\text{leaf}}$, $q_v(\mathcal{B}) \leq Q_v$, and the per-user construction-side caps of Section 6.2 such that

$$\text{Adv}_{\text{TW128}}^{\text{mu-ind}}(\mathcal{A}) \leq \text{Lift}_c(N_{\text{tot}}) + \text{Adv}_{\text{RO}}^{\text{mu-ind}}(\mathcal{B}) + \text{Lift}_c(N_{\text{prim}}).$$

Theorem 8 bounds the random oracle term. Substituting $q_{\text{RO}}(\mathcal{B}) \leq N_{\text{prim}}$, $n_{\text{leaf}}(\mathcal{B}) \leq N_{\text{leaf}}$, and $q_v(\mathcal{B}) \leq Q_v$ gives the stated bound. $\square$

1207 **Corollary 6** (All-in-One AE Security). *For every nonce-respecting AE adversary $\mathcal{A}$*
 1208 *against TW128 with the resources of Section 4.3,*

$$1209 \quad \text{Adv}_{TW128}^{\text{ae}}(\mathcal{A}) \leq \text{Lift}_c(N_{\text{tot}}) + \text{Lift}_c(N_{\text{prim}}) + \text{KeyGuess}_k(N_{\text{prim}}) + \frac{N_{\text{leaf}}}{2^{\tau_{\text{leaf}}}} + \frac{Q_v}{2^{\tau_{\text{root}}}}.$$

1210 *Proof.* Build a mu-ind adversary $\mathcal{A}^\mu$ that makes one New query, runs $\mathcal{A}$, and forwards
 1211 each encryption and verification query of $\mathcal{A}$ to user 1. The primitive interface is forwarded
 1212 unchanged. The real and ideal views of $\mathcal{A}$ in the all-in-one AE experiment are exactly
 1213 the real and ideal views of $\mathcal{A}^\mu$ in the mu-ind experiment with $D = 1$, and the resource
 1214 counts are unchanged. On a replayed encryption tuple the two real verification oracles
 1215 agree: the mu-ind oracle short-circuits through its replay table, while the AE oracle
 1216 re-runs decryption, which accepts by TW128 decryption correctness (Section 3.5); all other
 1217 queries are answered by identical code. Applying Theorem 9 with $D = 1$ eliminates the
 1218 key-collision term and gives the stated bound. $\square$

1219 The INT-RUP bound is AEAD-facing, but it is closer to the tag-hash analysis than to
 1220 the mu-ind baseline: plaintext-computing queries expose unverified stream plaintext, and
 1221 freshness must still reduce to hidden key tests, leaf-tag collisions, and root-tag guessing.
 1222 The separated interface changes the accounting by adding plaintext-computing work to N
 1223 and n_{leaf} , while leaving the non-replay verification count q_v unchanged.

1224 **Theorem 10** (Random Oracle INT-RUP Integrity). *For every flat-RO INT-RUP adversary*
 1225 *$\mathcal{B}$ against TW128 with nonce-respecting encryption queries and the resources of Section 4.3,*

$$1226 \quad \text{Adv}_{\text{RO}}^{\text{INT-RUP}}(\mathcal{B}) \leq \text{KeyGuess}_k(q_{\text{RO}}(\mathcal{B})) + \frac{n_{\text{leaf}}(\mathcal{B})}{2^{\tau_{\text{leaf}}}} + \frac{q_v(\mathcal{B})}{2^{\tau_{\text{root}}}}.$$

1227 *Plaintext-computing queries contribute to N and n_{leaf} , but not to q_v .*

1228 *Proof.* The proof uses the same hidden-tag accounting as the adjacent-hybrid bound, with
 1229 the plaintext-computing interface included in the construction transcript. Work in the
 1230 flat-RO INT-RUP game of Section 4.2. Let **forge** be the event that a valid non-replay
 1231 verification query accepts. Let **bad_{key}** be the event that a direct RO_{flat} query decodes under
 1232 **KeyedTWO** to the hidden key K and a counted output-point class, and let **bad_{leaf}** be the
 1233 event that some construction-generated final leaf transcript receives the same τ_{leaf} -bit leaf
 1234 tag projection as a distinct final leaf transcript that shares its key, nonce, and leaf index
 1235 and is reached by an encryption computation (the same-slot event of Lemma 11). Final leaf
 1236 transcripts generated by encryption, plaintext-computing, and verification computations
 1237 are all included in n_{leaf} .

1238 The key-test term is unchanged by the plaintext-computing oracle. By Lemma 14,
 1239 stated for the INT-RUP extension, the hidden key remains uniform before **bad_{key}** outside
 1240 the candidate keys already tested by direct queries, and Lemma 10 gives

$$1241 \quad \Pr[\text{bad}_{\text{key}}] \leq \text{KeyGuess}_k(q_{\text{RO}}(\mathcal{B})).$$

1242 For the leaf term, no direct query has pre-read a hidden-key final leaf tag point before
 1243 **bad_{key}**: such a query would decode under **KeyedTWO** to K and the corresponding $\mathcal{L}_{\text{tag}}(j)$
 1244 class. Coupling the real INT-RUP execution to the all-reject continuation used in Lemma 16
 1245 preserves the construction computations and leaf-tag table accesses up to the first accepting
 1246 non-replay submission. The same-slot case of Lemma 11, applied to the enlarged set of
 1247 construction computations, gives

$$1248 \quad \Pr[\text{bad}_{\text{leaf}} \text{ before } \text{bad}_{\text{key}} \cup \text{forge}] \leq \frac{n_{\text{leaf}}(\mathcal{B})}{2^{\tau_{\text{leaf}}}}.$$

It remains to bound accepting verification before $\text{bad}_{\text{key}} \cup \text{bad}_{\text{leaf}}$. The INT-RUP root-guessing lemma (Lemma 16) gives

$$\Pr[\text{forge before } \text{bad}_{\text{key}} \cup \text{bad}_{\text{leaf}}] \leq q_v(\mathcal{B})/2^{\tau_{\text{root}}}.$$

Combining the three bounds through the first-occurrence decomposition of $\Pr[\text{forge}]$ proves the theorem. $\square$

Theorem 11 (INT-RUP Integrity). *For every INT-RUP adversary $\mathcal{A}$ against TW128 with nonce-respecting encryption queries and the resources of Section 4.3,*

$$\text{Adv}_{\text{TW128}}^{\text{INT-RUP}}(\mathcal{A}) \leq \text{Lift}_c(N_{\text{tot}}) + \text{KeyGuess}_k(N_{\text{prim}}) + \frac{N_{\text{leaf}}}{2^{\tau_{\text{leaf}}}} + \frac{Q_v}{2^{\tau_{\text{root}}}}.$$

Proof. By the Lift Application corollary (Corollary 9) for the INT-RUP game, there is a random oracle INT-RUP adversary $\mathcal{B} = \mathcal{B}_{\text{derived}}(\mathcal{A})$ with $q_{\text{RO}}(\mathcal{B}) \leq N_{\text{prim}}$, $n_{\text{leaf}}(\mathcal{B}) \leq N_{\text{leaf}}$, and $q_v(\mathcal{B}) \leq Q_v$ such that

$$\text{Adv}_{\text{TW128}}^{\text{INT-RUP}}(\mathcal{A}) \leq \text{Lift}_c(N_{\text{tot}}) + \text{Adv}_{\text{RO}}^{\text{INT-RUP}}(\mathcal{B}).$$

Theorem 10 bounds the random oracle term, and substituting the derived-adversary resource caps gives the stated bound. $\square$

7 Concrete Bounds and Work Cap

The preceding sections prove two families of bounds: the tag-centric hash and committing bounds of Section 5, and the AEAD-facing bounds of Section 6. This section instantiates all of them at the implemented TW128 parameters and records the common work cap used for the paper’s concrete security claim.

7.1 Concrete Security Claim

Corollary 7 (Concrete TW128 Security). *Under the implemented parameters $k = c = \nu = \tau_{\text{root}} = \tau_{\text{leaf}} = 256$, every adversary with work cap $N_{\text{tot}} \leq 2^{63}$ —and, in the multi-user case, user cap $D \leq 2^{63}$ —has advantage at most 2^{-128} in each of TW128’s notions: multi-user indistinguishability (Theorem 9), all-in-one AE (Corollary 6), INT-RUP integrity (Theorem 11), the hash security of H_{TW128} — s -way collision resistance for every $s \geq 2$ in the sense of [BH22], with the [RS04] notions of standard and everywhere second-preimage resistance inherited from collision resistance on every fixed input slice, standard preimage resistance on every fixed input slice, and everywhere preimage resistance (Theorems 4 to 6 and Corollary 2)—and the [BH22] CMTDs-4 committing bound it yields (Corollary 4).*

Proof. The resource consequences of Section 4.3 give $N_{\text{prim}} \leq N_{\text{tot}}$, $Q_v \leq N_{\text{tot}}$, $N_{\text{leaf}} \leq N_{\text{tot}}$, and $N_{\text{prim}} + N_{\text{leaf}} \leq N_{\text{tot}}$. The user cap D remains independent, since created-but-idle users need not contribute to N .

At $N_{\text{tot}} \leq 2^{63}$, each flat-sponge term—at N_{tot} or at N_{prim} —and each birthday-scale root or leaf term at $s = 2$ is at most $2^{-131} + 2^{-192}$, while each linear root-preimage, key-guessing, verification, or leaf tag term with no factor D is at most a 2^{-193} -scale term. The fixed-slice Pre degradation is at most 2^{-256} , since $m_{a,\mu} \geq k + \nu = 512$. Thus the largest non-multi-user bound is the $s = 2$ collision bound, which is below $4 \cdot 2^{-131} = 2^{-129}$; larger s only decreases the root-collision term. The all-in-one AE bound carries its two flat-sponge terms as its only birthday-scale terms and lies below 2^{-129} , and the INT-RUP bound carries $\text{Lift}_c(N_{\text{tot}})$ alone. In the multi-user case, using $D \leq 2^{63}$,

$$\text{Adv}_{\text{TW128}}^{\text{mu-ind}}(\mathcal{A}) < (2^{-130} + 2^{-193}) + 2^{-130} + 2^{-192} + 2^{-131} < 2^{-128}.$$

Table 6: Concrete TW128 security at the work cap $N_{\text{tot}} \leq 2^{63}$ (with $D \leq 2^{63}$ for mu-ind), under $k = c = \nu = \tau_{\text{root}} = \tau_{\text{leaf}} = 256$ and the resource consequences $N_{\text{prim}}, Q_v, N_{\text{leaf}} \leq N_{\text{tot}}$ and $N_{\text{prim}} + N_{\text{leaf}} \leq N_{\text{tot}}$ of Section 4.3. The Coll and CMTDs-4 rows hold for every $s \geq 2$; larger s only shrinks the root-collision term.

Notion	Dominant scale	At the cap
All-in-one AE	Lift_c	$\leq 2^{-128}$
Multi-user ind.	$D \text{KeyGuess}_k(N_{\text{prim}})$	$\leq 2^{-128}$
INT-RUP	Lift_c	$\leq 2^{-128}$
H_{TW128} collision (Coll)	$\text{Lift}_c, \text{LeafColl}, \text{RootColl}$	$\leq 2^{-128}$
H_{TW128} second-preimage (Sec/eSec)	inherited from Coll	$\leq 2^{-128}$
H_{TW128} preimage (Pre)	$\text{Lift}_c, \text{LeafColl}$	$\leq 2^{-128}$
H_{TW128} everywhere preimage (ePre)	Lift_c	$\leq 2^{-128}$
CMTDs-4 [BH22]	Lift_c and RootColl	$\leq 2^{-128}$

the two flat-sponge terms sum to at most $2 \cdot (2^{-131} + 2^{-194}) = 2^{-130} + 2^{-193}$, the key-guessing term satisfies $DN_{\text{prim}}/2^{256} \leq 2^{126}/2^{256} = 2^{-130}$, the remaining linear terms satisfy $(Q_v + N_{\text{leaf}})/2^{256} \leq 2^{64}/2^{256} = 2^{-192}$, and $D(D-1)/2^{257} < 2^{-131}$; the total is below $5 \cdot 2^{-131} + 2^{-192} + 2^{-193} < 2^{-128}$. These estimates instantiate every closed form in Section 7.2; Section 7.3 discusses the margin they leave. $\square$

7.2 Instantiated Bounds

The implemented parameters of Section 3.2 fix

$$k = c = \nu = \tau_{\text{root}} = \tau_{\text{leaf}} = 256.$$

Substituting into Corollaries 2, 4 and 6 and Theorems 4 to 6, 9 and 11 gives the following closed-form bounds.

All-in-one AE.

$$\text{Adv}_{\text{TW128}}^{\text{ae}}(\mathcal{A}) \leq \frac{N_{\text{tot}}(N_{\text{tot}} + 1)}{2^{257}} + \frac{N_{\text{prim}}(N_{\text{prim}} + 1)}{2^{257}} + \frac{N_{\text{prim}} + Q_v + N_{\text{leaf}}}{2^{256}}.$$

Multi-user indistinguishability.

$$\text{Adv}_{\text{TW128}}^{\text{mu-ind}}(\mathcal{A}) \leq \frac{N_{\text{tot}}(N_{\text{tot}} + 1)}{2^{257}} + \frac{N_{\text{prim}}(N_{\text{prim}} + 1)}{2^{257}} + \frac{DN_{\text{prim}} + Q_v + N_{\text{leaf}}}{2^{256}} + \frac{D(D-1)}{2^{257}}.$$

INT-RUP integrity.

$$\text{Adv}_{\text{TW128}}^{\text{INT-RUP}}(\mathcal{A}) \leq \frac{N_{\text{tot}}(N_{\text{tot}} + 1)}{2^{257}} + \frac{N_{\text{prim}} + Q_v + N_{\text{leaf}}}{2^{256}}.$$

s -way collision resistance of H_{TW128} . For every $s \geq 2$,

$$\text{Adv}_{\text{TW128}}^{\text{Coll}}(\mathcal{A}) \leq \frac{N_{\text{tot}}(N_{\text{tot}} + 1)}{2^{257}} + \frac{\binom{N_{\text{prim}} + N_{\text{leaf}}}{2}}{2^{256}} + \frac{\binom{N_{\text{prim}} + s}{s}}{2^{256(s-1)}}.$$

For $s = 2$, this is the ordinary two-input collision bound:

$$\begin{aligned} \text{Adv}_{\text{TW128}}^{\text{Coll}}(\mathcal{A}) &\leq \frac{N_{\text{tot}}(N_{\text{tot}} + 1)}{2^{257}} + \frac{(N_{\text{prim}} + N_{\text{leaf}})(N_{\text{prim}} + N_{\text{leaf}} - 1)}{2^{257}} \\ &\quad + \frac{(N_{\text{prim}} + 1)(N_{\text{prim}} + 2)}{2^{257}}. \end{aligned}$$

1307 **Second-preimage resistance of H_{TW128} .** By Corollary 2, the standard and everywhere
 1308 [RS04] second-preimage advantages of H_{TW128} are bounded by the same $s = 2$ collision
 1309 expression above.

1310 **Preimage resistance of H_{TW128} .** For fixed byte lengths a, μ , let $m_{a,\mu} = k + \nu + 8a + 8\mu$.
 1311 By Theorem 5,

$$1312 \quad \text{Adv}_{TW128}^{\text{Pre}[a,\mu]}(\mathcal{A}) \leq \frac{N_{\text{tot}}(N_{\text{tot}} + 1)}{2^{257}} + \frac{(N_{\text{prim}} + N_{\text{leaf}})(N_{\text{prim}} + N_{\text{leaf}} - 1)}{2^{257}} \\ + \frac{N_{\text{prim}} + 1}{2^{256}} + 2^{256 - m_{a,\mu}}.$$

1313 Under the implemented parameters $k = \nu = 256$, the slice term is $2^{-256 - 8a - 8\mu}$.

1314 **s -way CMTDs-4 committing security.** For every $s \geq 2$, the all-bodies-equal special case
 1315 drops the known-key leaf tag collision term:

$$1316 \quad \text{Adv}_{TW128}^{\text{CMTDs-4}}(\mathcal{A}) \leq \frac{N_{\text{tot}}(N_{\text{tot}} + 1)}{2^{257}} + \frac{\binom{N_{\text{prim}} + s}{s}}{2^{256(s-1)}}.$$

1317 *Remark 1* (Two-Opening CMTDs-4 Specialization). Setting $s = 2$ recovers the ordinary
 1318 two-opening CMTDs-4 statement of [BH22]:

$$1319 \quad \text{Adv}_{TW128}^{\text{CMTDs-4}}(\mathcal{A}) \leq \frac{N_{\text{tot}}(N_{\text{tot}} + 1)}{2^{257}} + \frac{\binom{N_{\text{prim}} + 2}{2}}{2^{256}} = \frac{N_{\text{tot}}(N_{\text{tot}} + 1) + (N_{\text{prim}} + 1)(N_{\text{prim}} + 2)}{2^{257}}.$$

Everywhere preimage resistance of H_{TW128} .

$$1320 \quad \text{Adv}_{TW128}^{\text{ePre}}(\mathcal{A}) \leq \frac{N_{\text{tot}}(N_{\text{tot}} + 1)}{2^{257}} + \frac{N_{\text{prim}} + 1}{2^{256}}.$$

1321 7.3 Interpreting the Work Cap

1322 The bounds of Section 7.2 aggregate the loss terms of Section 4.4. The cap $N_{\text{tot}} \leq 2^{63}$ is
 1323 a simultaneous work point for all notions, not a separate parameter choice per theorem.
 1324 Table 6 records the dominant terms and aggregate cap bounds; Corollary 7 supplies the
 1325 individual term sizes at the cap.

1326 **Term tightness.** The loss terms of Section 4.4 are mostly generic search or birthday
 1327 terms, and their leading scales are tight in the usual random oracle sense. The key-guessing
 1328 term is matched at constants by direct random guesses for a k -bit key. In the multi-user
 1329 theorem’s free-parameter setting, the term $D \text{KeyGuess}_k(N_{\text{prim}})$ is also tight at leading
 1330 scale: using the per-user nonce discipline of Section 2.7, which permits one nonce value
 1331 across users, an adversary obtains one known nonempty encryption from each targeted
 1332 user and compares each direct candidate-key query against all collected root-initialization
 1333 keystream blocks at once. Each guess then matches some targeted user with probability
 1334 $D/2^k$. The concrete joint cap charges the target encryptions to N , lowering the feasible
 1335 scale as discussed below.

1336 The root terms are the standard random oracle search terms: an s -way birthday search
 1337 matches the leading scale of $\text{RootColl}_\tau(Q, s)$, and fixed-target preimage search matches the
 1338 per-query $2^{-\tau}$ rate of $\text{RootPre}_\tau(Q)$. The hidden-key same-slot leaf term $n_{\text{leaf}}/2^{\tau_{\text{leaf}}}$ is also
 1339 tight at constants for nonce-respecting verification: after encrypting a two-chunk message
 1340 under nonce U , an adversary can submit non-replay verification queries that replace the
 1341 leaf ciphertext chunk by fresh strings of the same length while keeping the root tag. Each

query recomputes the hidden leaf tag at the same slot $(K, U, 1)$, and acceptance occurs when the fresh leaf tag equals the original hidden leaf tag, with probability $2^{-\tau_{\text{leaf}}}$ per verification-side final leaf transcript.

The flat sponge lift term $\text{Lift}_c(N_{\text{tot}}) = N_{\text{tot}}(N_{\text{tot}} + 1)/2^{c+1}$ dominates the BDPV differentiating bound $f_P(N_{\text{tot}})$ of [BDPVA08, Lemma 5] via Lemma 20. That differentiating bound is matched at leading order by the standard inner-collision distinguisher; the remaining conservatism is in composing indifferenciability into the keyed games, not in an obviously improvable simulator bound. Real-vs-ideal games pay the same differentiating bound once more, at N_{prim} , to replace the ideal-side simulator by the real permutation (Lemma 21); since $N_{\text{prim}} \leq N_{\text{tot}}$, that term never dominates the N_{tot} term.

Dominant terms and margin. At $N_{\text{tot}} \leq 2^{63}$, the hash bounds other than ePre and the CMTDs-4 bound have flat-sponge and root or leaf birthday terms on the same 2^{-131} scale. The ePre and INT-RUP bounds retain the flat sponge loss as their only birthday-scale term, and the single-user AE bound carries its two flat-sponge terms, at N_{tot} and N_{prim} , as its only birthday-scale terms. Thus the flat sponge loss is not always the largest summand: under the loose per-term cap, the $s = 2$ root-collision term $\binom{N_{\text{prim}}+2}{2}/2^{256}$ is on the same scale as $\text{Lift}_c(N_{\text{tot}})$.

The multi-user bound is the exception. Its hybrid-induced term $D \text{KeyGuess}_k(N_{\text{prim}})$ can reach 2^{-130} at the extreme cap $D = N_{\text{prim}} = 2^{63}$, and the displayed estimate then has value $5 \cdot 2^{-131} + 2^{-194}$ because that corner forces $N = 0$ and hence $Q_v = N_{\text{leaf}} = 0$. The corner is a proof artifact rather than an attack point: an adversary without construction computations receives identical primitive answers in both worlds (Section 4.1). At feasible resource splits, the multi-user key-guessing witness needs one accepted nonempty encryption per targeted user; with t targeted users and $q = N_{\text{prim}}$ direct guesses, the joint cap imposes $2t + q \leq 2^{63}$, so the feasible key-guessing scale is about 2^{-133} , reached near $t = 2^{61}$ and $q = 2^{62}$ (Section 4.4). The remaining distance to the proof bound is hybrid slack from charging created-but-idle users plus the two flat-sponge composition terms.

Expressed in bytes, $N_{\text{tot}} \leq 2^{63}$ corresponds to at most $\rho \cdot 2^{63} = 167 \cdot 2^{63} < 2^{71}$ bytes of plaintext, associated data, ciphertext, and aggregation through the duplex (Section 3.2). The margin widens away from the multi-user corner: at $D \leq 2^{32}$, the D -indexed terms drop below 2^{-160} and the multi-user bound falls to its two flat-sponge terms, below $2^{-130} + 2^{-159}$.

Reading the work cap. Corollary 7 is a success-probability statement at a fixed cap, not a statement that security is exhausted there. The closed forms of Section 7.2 remain explicit functions of the resources beyond 2^{63} , and their dominant terms are birthday-type in N_{tot} on the 256-bit capacity, so the bounds degrade quadratically rather than abruptly: at $N_{\text{tot}} = 2^{100}$, for example, the $s = 2$ collision bound is still below 2^{-55} . Once N_{tot} reaches $2^{c/2} = 2^{128}$, the flat sponge loss exceeds $1/2$ and no longer gives meaningful security (Section 4.4); the displayed closed form itself becomes a trivial advantage bound only when it reaches 1, near $N_{\text{tot}} = 2^{(c+1)/2}$. The cap $N_{\text{tot}} \leq 2^{63}$ is therefore a data-scale choice: a single work bound at which every notion's advantage lies below the 2^{-128} target simultaneously, not the largest workload the mode tolerates.

The residual conservatism in the cap comes from the indifferenciability route and, for mu-ind, the hybrid accounting above. A direct keyed-duplex analysis would be expected to replace the quadratic Lift_c terms by finer construction-primitive cross terms, but it would also need to handle the adversarial-initialization hash and committing games used here. Section 9.5 discusses that route and the gaps it would need to close.

Cross-scheme bound-level comparisons against prior sponge AEADs and the committing AEAD line of [BH22] are deferred to Sections 9.1 to 9.3.

8 Performance Evaluation

TW128 fixes the root/leaf transcript but leaves leaf scheduling to the implementation. This section evaluates that separation: the same ciphertext and tag are produced by scalar and vectorized implementations, while complete leaf chunks can be batched to recover practical long-message throughput. The comparison baseline is hardware-accelerated AES-128-GCM on the same hosts, so the measurements should be read as an implementation sanity check and same host reference point, not as a claim that TW128 is faster than AES-GCM.

8.1 Optimized Implementations

We analyze a Go implementation of TW128¹ with optimized assembly paths for the performance-critical permutation and absorb/squeeze loops. We compare three vectorized assembly paths: `amd64 AVX-512`, `amd64 AVX2`, and `arm64 NEON` using the Armv8.2 SHA-3 extension. The framing, domain separation, padding, tree bookkeeping, and dispatch remain in shared Go code. A pure Go path uses a scalar permutation and a scalar eight-instance loop, and serves both as a reference path for cross-checking the assembly cores and as a fallback on other architectures.

The tree structure of TW128 exposes two distinct permutation workloads. The root duplex—which absorbs the associated data, processes chunk 0, and absorbs the leaf tag aggregation transcript—is inherently sequential, since each call consumes the previous call’s output. It is driven by a *single-duplex* ($\times 1$) core, except for its chunk-0 message phase. The leaf duplexes operate on disjoint state and disjoint inputs (Section 3), so complete leaf chunks are batched and run through an *eight-way* ($\times 8$) core. Leftover runs of two to seven complete chunks use narrower remainder kernels, while a single leftover complete chunk, a partial trailing chunk, and the count-aggregation tail fall back to the single-duplex core.

Chunk 0 itself runs on the batched kernels rather than serially, a technique we refer to as *chunk 0 fusion*. The root duplex’s chunk-0 message phase has the same kernel-visible schedule as a leaf chunk and differs only in its initial state. The implementation therefore seeds lane 0 of the first batched call with the root state, runs chunk 0 alongside the first leaf chunks, then writes lane 0’s output state back to the root duplex before the root absorbs the leaf tags from that same call. This avoids a serial pass over chunk 0 and is the reason the cycles-per-byte curve drops as soon as complete leaves are present (Section 8.3). Section C gives the lane-major layout, remainder-kernel strategies, lane 0 writeback details, and instruction-level kernel structure.

Architecture-specific cores. On `amd64` the optimized implementation selects an AVX-512 or AVX2 core once at startup from the AVX512VL CPUID feature bit. On `arm64` the optimized implementation uses NEON with the Armv8.2 SHA-3 extension (FEAT_SHA3, available on Apple Silicon and Neoverse V1/V2 and N2). Both architecture-specific paths provide a single-duplex core and batched leaf kernels; the assembly stays below the mode boundary, so framing, domain separation, padding, and tree bookkeeping remain shared with the reference path.

Data-independent execution. The scalar reference and all three vectorized implementations avoid secret-dependent branches and secret-dependent memory or vector indices for the key, nonce, associated-data contents, and plaintext or ciphertext contents. The permutation and the XOR-based absorb/squeeze loops are straight-line with respect to data values. The only indexed table is the round-constant array, addressed by the public round number; dispatch, batched loads, stores, gathers, scatters, and inactive-lane handling

¹The implementation is available at <https://github.com/codahale/treewrap>.

are controlled by public lengths, chunk counts, and fixed chunk strides. Section C records the architecture-specific addressing details.

8.2 Benchmark Methodology

We measure on two hosts, one per architecture. The **amd64** host is a Google Compute Engine **c4-standard-4** instance with an Intel Xeon Platinum 8581C (Emerald Rapids, 2.30 GHz base, AVX-512 including AVX512VL, AVX512_VBMI2, GFNI, and VAES), two physical cores with simultaneous multithreading enabled, 48 KiB L1d and 32 KiB L1i per core, 2 MiB L2 per core, and a 260 MiB shared L3, running Debian 12 (kernel 6.1) under KVM. The guest does not expose a **cpufreq** interface, so the frequency governor and turbo state are not under our control; we mitigate the resulting variance statistically (below). The **arm64** host is an Apple M4 Pro (10 performance cores and 4 efficiency cores; per performance core, 128 KiB L1i, 64 KiB L1d, and a 16 MiB shared L2) running macOS 26.5.1 (Darwin 25.5), with **FEAT_SHA3** present. macOS exposes no userspace frequency control.

Measurements are collected by a standalone harness (**cmd/cpb**) that locks its goroutine to an OS thread and, for each algorithm, operation, and message length, first calibrates an iteration count that fills at least 100 ms, then records 100 samples. It reports the median cycles per byte and median wall-clock throughput together with the interquartile range as a dispersion measure (Table 9). Both metrics are read from the same timed loop, so the cycles-per-byte and throughput figures for a given measurement come from one interleaved run rather than two separately scheduled ones.

Each timed call uses a fixed all-zero key and nonce, empty associated data, and preallocated source and destination buffers. Open measurements decrypt and verify a ciphertext prepared before timing; seal and open reset the destination slice length inside the timed loop but do not allocate per call. The reported figures are therefore warm-cache, steady-state rates for message processing under a fixed AEAD object. The single-threaded measurement leaves the SMT sibling of the **amd64** host idle. All reported figures are single-threaded: they exercise the SIMD axis of TW128’s parallelism, and multicore scaling remains unmeasured. We do not report multicore scaling separately: once leaf chunks are assigned to workers, the remaining scaling limit is scheduling, memory bandwidth, and ordered transport of 32-byte leaf tags back to the root, rather than a mode-level cryptographic effect. We report a sweep of seven message lengths from 1 B to 16 MiB.

The cycle counters differ by architecture, and this affects how the cycles-per-byte figures may be read. On **amd64** the harness reads **RDTSC**; on this part the time-stamp counter is invariant and ticks at the 2.30 GHz reference frequency, so the reported figures are *reference* cycles per byte and are unaffected by turbo (and convert directly to wall-clock time). On **arm64** the harness reads **CNTVCT_ELO** and scales it by the ratio of the peak core frequency to **CNTFRQ_ELO**; since Apple exposes no frequency API, the peak is auto-detected by timing a dependent integer-add chain and taking the top observed P-state, yielding *core* cycles per byte. Consequently the cycles-per-byte figures are comparable within an architecture and against the same-host AES-128-GCM baseline, but not directly across architectures; absolute throughput (Table 8) is the cross-platform metric.

The baseline is Go’s AES-128-GCM implementation (**crypto/aes** with **crypto/cipher**). It uses AES-NI and PCLMULQDQ on **amd64**, and the AES and PMULL instructions on **arm64**. The AES block cipher, GCM wrapper, and TW128 AEAD object are constructed once before timing, so the measurements exclude key schedule and GHASH key-power precomputation. Each timed TW128 call still performs the root duplex initialization inside the call. Both directions are measured for each scheme; the cycles-per-byte table reports both, and the throughput table reports the encryption direction.

Table 7: Median cycles per byte (100 samples). The **amd64** figures are RDTSC reference cycles at the 2.30 GHz invariant time-stamp counter; the **arm64** figures are core cycles at the auto-detected peak performance-core frequency. Cycle counts are not directly comparable across architectures (Section 8.2).

	Message length						
	1 B	64 B	8 KiB	32 KiB	64 KiB	1 MiB	16 MiB
<i>Intel Xeon 8581C (Emerald Rapids), AVX-512</i>							
TW128 encrypt	571	9.04	1.81	0.73	0.39	0.38	0.38
TW128 decrypt	605	9.55	1.82	0.73	0.39	0.38	0.38
AES-128-GCM seal	119	2.74	0.30	0.29	0.29	0.29	0.29
AES-128-GCM open	99	2.17	0.29	0.28	0.28	0.28	0.29
<i>Intel Xeon 8581C (Emerald Rapids), AVX2</i>							
TW128 encrypt	647	10.18	2.21	1.15	1.10	1.09	1.10
TW128 decrypt	682	10.70	2.21	1.06	1.10	1.08	1.06
<i>Apple M4 Pro, NEON + FEAT_SHA3</i>							
TW128 encrypt	620	9.88	1.97	0.92	0.89	0.89	0.91
TW128 decrypt	671	10.65	2.04	0.90	0.87	0.87	0.89
AES-128-GCM seal	113	2.84	0.44	0.43	0.43	0.44	0.46
AES-128-GCM open	123	2.56	0.42	0.41	0.41	0.42	0.43

8.3 Throughput

Table 7 reports cycles per byte across the message-length sweep and Table 8 the corresponding wall-clock throughput in the bulk regime. The main pattern in the TW128 rows is the descent produced by the tree-parallel leaf core. A message of at most one chunk has no complete leaf chunk for chunk 0 to share a kernel with, so it runs entirely on the single-duplex core, and the 8 KiB column sits at that core’s rate. Past one chunk, chunk 0 fusion (Section 8.1) keeps every complete chunk on the batched kernels, and the descent tracks lane occupancy.

On the **amd64** AVX-512 path, TW128 encryption falls $1.81 \rightarrow 0.73 \rightarrow 0.39$ cycles per byte across the 8, 32, and 64 KiB columns and then holds at 0.38–0.39 through 16 MiB. The AVX2 path has the same shape at lower throughput: it runs the eight instances as two groups of four rather than in a single 8-wide register, leaving the AVX-512 core about $2.9\times$ faster in bulk. The **arm64** curve flattens earlier because the eight-way core processes its batch as four two-instance pairs; a 32 KiB message, chunk 0 and three leaves as two full pairs, is already close to the long-message rate. In wall-clock terms TW128 reaches 6.00 GB/s on the **amd64** AVX-512 host and 4.93 GB/s on the M4 Pro for 16 MiB messages.

Dispersion. Each cell of Tables 7 and 8 is the median of 100 samples; Table 9 summarizes their spread as the interquartile range (IQR) relative to the median. The medians are stable: across the sweep the relative IQR of the TW128 cycles-per-byte measurements is at most 0.7% on the **amd64** AVX-512 path, 0.9% on AVX2, and 1.9% on the M4 Pro, with per-path medians of 0.3%, 0.4%, and 1.2%. The full min–max range is wider at isolated points—up to 4.9%, 12.0%, and 16.7% on the three paths—because turbo, governor, and scheduler states are not under our control (Section 8.2); these are tail excursions that the IQR excludes. We therefore report medians throughout and treat the IQR as the dispersion measure.

Table 8: Measured wall-clock throughput (GB/s, GB = 10^9 bytes) from the same `cmd/cpb` run as Table 7, encryption direction. The `amd64` AES-128-GCM row is the AVX-512 host and is unaffected by the TW128 core selection.

	8 KiB	32 KiB	64 KiB	1 MiB	16 MiB
<i>Intel Xeon 8581C (Emerald Rapids)</i>					
TW128 (AVX-512)	1.27	3.17	5.93	6.10	6.00
TW128 (AVX2)	1.04	2.00	2.10	2.10	2.08
AES-128-GCM	7.75	8.00	8.05	8.00	7.87
<i>Apple M4 Pro</i>					
TW128 (NEON+SHA3)	2.27	4.84	5.04	5.01	4.93
AES-128-GCM	10.04	10.27	10.33	10.07	9.71

Table 9: Cycles-per-byte measurement dispersion for TW128 across the seven-length sweep (seal and open, 100 samples per point), as the interquartile range relative to the median. The final column is the widest single-point min–max range; it is excluded from the IQR as a tail effect.

Path	Relative IQR		Max
	median	max	range
Xeon 8581C, AVX-512	0.3%	0.7%	4.9%
Xeon 8581C, AVX2	0.4%	0.9%	12.0%
Apple M4 Pro	1.2%	1.9%	16.7%

8.4 Latency

For short messages in this benchmark, the cost is set by the serial root duplex and is independent of the parallel leaf machinery. With empty associated data, a message of at most $\rho = 167$ bytes occupies a single rate block and never enters leaf mode, so it costs exactly two permutations: one to initialize the root duplex and one to close the message block and extract the tag. The 1 B and 64 B columns of Table 7 bear this out: per-message latency is nearly flat across them at roughly 571 reference cycles on `amd64` (571 at 1 B versus $9.04 \times 64 \approx 578$ at 64 B) and 620 core cycles on `arm64` (620 versus $9.88 \times 64 \approx 633$), i.e. about 285 and 310 cycles per KECCAK- $p[1600, 12]$ respectively. Because such messages bypass the leaf core entirely, leaf-level parallelism imposes no latency penalty on them.

The latency/throughput trade-off appears instead in the intermediate regime, which chunk 0 fusion has narrowed to roughly the one-to-eight-chunk band. The worst per-byte cost in the sweep is the single-chunk regime at 8 KiB (1.8–2.2 cycles per byte across the three paths), where the message is the chunk-0 phase alone on the single-duplex core. Once the message has complete leaf chunks for chunk 0 to share a kernel with, the cost falls quickly: by 32 KiB TW128 is within a factor of two of its AVX-512 floor and close to its `arm64` floor, and the floor itself is reached once the eight-way core fills at 64 KiB. The root duplex also bounds the tail of a long message: the final tag cannot be produced until chunk 0 has been processed and every leaf tag has been absorbed into the aggregation transcript. That absorption is cheap—each rate block of the root absorbs just over five 32-byte leaf tags, so a 16 MiB message (≈ 2000 leaves) adds on the order of 400 root permutations against roughly 10^5 permutations of leaf work, under 0.5% overhead—so the serial root does not materially cap bulk throughput.

8.5 AES-128-GCM Baseline

We compare against the one baseline measured under the same harness on the same hosts, the Go standard library’s hardware-accelerated AES-128-GCM (Tables 7 and 8). The harness times TW128 and AES-128-GCM back-to-back at each length and reads both metrics from the same loop, so the cycles-per-byte and throughput comparisons hold under identical conditions and agree. In the bulk regime AES-128-GCM is faster on both architectures, as expected for a primitive with dedicated AES and carryless-multiply instructions: at 16 MiB it runs at 0.29 cycles per byte on `amd64` and 0.46 on `arm64`, against 0.38 and 0.91 for TW128—about $1.3\times$ and $2.0\times$ faster. The wall-clock throughputs show the same margins: TW128 reaches 6.00 and 4.93 GB/s against AES-128-GCM’s 7.87 and 9.71 GB/s. The wider `arm64` gap reflects the strength of Apple’s AES and PMULL units. TW128 attains these rates with a permutation that has no dedicated hardware support, relying only on the SHA-3 extension on `arm64` and on general SIMD on `amd64`.

AES-128-GCM is also faster for short messages in this steady-state measurement: at 1 B it costs 119 and 113 cycles on the two hosts against TW128’s 571 and 620. This is the expected result when AES-GCM key setup is amortized outside the timed call. In the sub-bulk regime AES-128-GCM is several times faster at one chunk (0.30 versus 1.81 cycles per byte at 8 KiB on `amd64`), but the gap narrows once the fused kernels engage: about $2.5\times$ at 32 KiB and the bulk ratio from 64 KiB on. TW128’s intended value relative to AES-128-GCM lies in the tag-centric hash and committing properties analyzed in Section 5, with the baseline AEAD bounds of Section 6, rather than in raw speed.

9 Discussion

9.1 Comparison Table

Table 10 situates TW128 against representative authenticated encryption schemes along the axes that distinguish it: the underlying primitive, whether the mode is parallelizable, the key, nonce, and tag sizes, associated data support, nonce-misuse resistance, and the published committing result or attack most relevant to the scheme. The rows are representative rather than exhaustive: sponge/duplex relatives, a Farfalle-based parallel design, AES baselines, a misuse-resistant AES baseline, and a generic committing transform. The comparison is structural, not a total security ranking; TW128’s own concrete bounds and the regimes in which each term dominates are treated in Section 7.3, and its measured throughput against hardware-accelerated AES-128-GCM in Section 8.5. The committing column should be read within the named notions only, and a negative entry there does not assert a confidentiality or integrity break of the underlying AEAD. It records TW128’s 128-bit CMTDs-4 bound derived from tag hash security, Ascon’s roughly 64-bit committing bound [KSW23], published low-cost attacks where known [KSW23, CR22, ADG⁺22], and the hash-collision level inherited by CTX. The table does not separately track compact (tag-only) commitment, since that property has not been systematically analyzed for the comparison schemes.

9.2 Comparison with Prior Sponge AEADs

TW128 belongs to the single-permutation duplex AEAD family introduced in [BDPVA11]. Representative lightweight members of that family, Ascon [DEMS21] and Xoodyak [DHP⁺20], target 128-bit security on small permutations (320 and 384 bits) with a strictly serial duplex: each call depends on the previous one, so there is no mode-level parallelism to exploit. TW128 instead pays for a larger permutation, KECCAK- p [1600, 12], and a 256-bit capacity—the same round count as KangarooTwelve [BDP⁺16]—in exchange for a tree that processes leaf chunks in parallel and for the headroom that meets the 2^{-128}

Table 10: Structural comparison of TW128 with representative AEADs. Sizes are in bytes; AES-GCM-SIV also admits a 32-byte key, and Kravatte-SAE’s key and nonce lengths are parameters (keys of up to 40 bytes, arbitrary-length nonces). “Misuse” is nonce-misuse resistance in the sense of [RS06]; “Commit” records a published committing result or attack in the relevant committing notion, with “broken” denoting a published low-cost committing attack, “hash CR” denoting the collision-resistance level of the hash used by the transform, and “open” marking a scheme with no published committing analysis. A dash means the entry depends on the base scheme. These entries are notion-specific summaries, not broad AE security claims. TW128’s concrete bounds appear in Section 7.3.

Scheme	Primitive	Par.	K/U/T	AD	Misuse	Commit
TW128	KECCAK- p [1600, 12]	yes	32/32/32	yes	no	CMTDs-4
Ascon-128a [DEMS21]	Ascon- p (320 b)	no	16/16/16	yes	no	≈ 64 b
Xoodoo [DHP ⁺ 20]	Xoodoo (384 b)	no	16/16/16	yes	no	2^{17} atk.
Kravatte-SAE [BDH ⁺ 17]	KECCAK- p [1600, 6]	yes	$\leq 40/\text{var.}/16$	yes	no	open
AES-GCM [Dwo07]	AES	yes	16/12/16	yes	no	broken
AES-GCM-SIV [GLL17]	AES	yes	16/12/16	yes	yes	broken
nAE + CTX [CR22]	any nAE	—	—	yes	—	hash CR

target at the stated work cap. The cost of this choice is visible in Section 8: on short, single-leaf inputs TW128 is slower per byte than a compact serial duplex would be, and its advantage appears only once a message spans enough leaves to fill the vector units.

TW128 reuses KangarooTwelve’s final-node/leaf split but not its Sakura coding [BDPVA14]. KangarooTwelve uses Sakura to make the final node parseable as a tree-hash transcript [BDP⁺16]. In TW128, the analogous chaining values are typed leaf tags: each leaf duplex is initialized with its leaf prefix and index, and the root absorbs leaf tags in index order followed by the leaf count (Sections 3 and 3.3). Sakura’s structural role is therefore handled by mode initialization and domain separation rather than by a tree coding. The transcript separation lemmas make this explicit (Lemma 6 and Corollary 8); the hash collision bound sees the leaf encoding only through the explicit leaf tag collision term of Theorem 4.

Keyak [BDP⁺15], whose Motorist mode is already parallelizable, is the closest prior duplex AEAD in spirit; Section 1.2 contrasts its wire-visible, encryptor-and-decryptor-shared degree of parallelism with TW128’s run-time leaf choice. Strobe [Ham17] is a serial sponge framework aimed at whole protocols rather than at a standalone AEAD, and like Ascon and Xoodoo it does not parallelize.

Farfalle [BDH⁺17] leaves the duplex setting altogether: it is a parallel pseudorandom function whose compression layer sums masked-and-permuted input blocks into an accumulator and whose expansion layer generates output from a rolling state, with authenticated encryption supplied by the Farfalle-SAE and Farfalle-SIV modes on top (Section 1.2). Its Kravatte instance runs KECCAK- p [1600, 6], and the parallelism available to an implementation is arbitrary, as in TW128. The two designs price their permutations for different adversary models. Farfalle’s six-round compression layer is analyzed in a PRF setting where inputs are randomized by secret masks and an adversary does not see both the input and output of the same permutation call [BDH⁺17]. TW128’s tag-hash claims are adversarial-initialization claims: in the hash and committing games, the adversary may choose the key as part of the input being hashed. TW128 therefore keeps the strict duplex object, uses the hash-grade round count discussed in Section 9.4, and takes its parallelism from the tree topology rather than from a parallel PRF compression layer.

Committing security is not unique to TW128 in this family: a dedicated analysis of the sponge-based NIST lightweight finalists proves Ascon committing while breaking others [KSW23], and some constructions, such as Strobe and the Farfalle modes, have not been examined in that setting at all. What distinguishes TW128 is that the authentication

tag is analyzed directly as a hash output for the full input. Collision, second-preimage, preimage, and everywhere preimage bounds for that tag then yield the compact commitment statements, including the 128-bit full-input CMTDs-4 bound, as mode-native consequences without leaving the single-permutation duplex setting.

9.3 Comparison with Committing AEAD Constructions

The dominant route to committing AEAD is a generic transform over a non-committing base scheme. [BH22] give the UtC, RtC, and HtE transforms, which add key- or context-commitment to an arbitrary AEAD, and [CR22] give CTX, which makes any nonce-based AE scheme commit to its full context at the cost of a single hash over a short string and with no ciphertext expansion. These transforms are inexpensive and broadly applicable; CTX in particular is nearly free. It is nevertheless a committing-AE transform, not a tag-hash transform: it binds the transformed ciphertext, but does not make the tag a standalone hash output for the full input. TW128 differs not by undercutting their cost but by integration. The same permutation provides confidentiality, integrity, and the tag hash output; no separate commitment primitive, key-derivation step, or auxiliary hash is invoked. The multi-input full-commitment bound follows from the same root tag collision bound used for hash security (Section 5), and Theorem 4 applies even when the ciphertext bodies used as openings differ, up to the additional leaf tag birthday term. This places TW128 alongside the dedicated committing primitives—the compactly committing AEAD of [GLR17] and the encryption of [DGRW18]—which likewise build commitment in rather than bolt it on, but which were designed for message franking rather than as general nonce-based AEADs. The contrast with transformed conventional schemes is sharpened by the attacks of Section 1.2: base schemes such as GCM and OCB are not committing on their own [CR22], and a transform is the only way to make them so.

Closest to TW128 here is the recent work of [DHM⁺24], which builds nonce-based AE directly on the SHA-3 functions SHAKE and TurboSHAKE—the latter using the same round-reduced KECCAK- p permutation as TW128—and likewise achieves CMT-4 natively, from the collision resistance of the underlying function rather than from a transform. The two designs make different structural choices: [DHM⁺24] targets session-based communication, chaining a stream of messages through intermediate tags and generalizing the keyed duplex into a *duplex cipher*, whereas TW128 is a single two-level tree whose independent leaves let the implementation choose parallelism at runtime with constant memory (Sections 1.2 and 8.1). Because a session chain processes each message serially, TW128’s principal advantage is throughput on large messages (Section 8): its parallel leaves fill vector units a serial duplex leaves idle, at the cost of the session support [DHM⁺24] provides and TW128 does not. Both demonstrate that committing AE needs no add-on transform once it is built on a well-scrutinized KECCAK- p permutation.

9.4 Cryptanalysis of the Underlying Permutation

Every bound in this paper models KECCAK- p [1600, 12] as a uniformly random permutation (Section 4). That modeling choice is supported by cryptanalytic scrutiny rather than by proof, so this subsection summarizes the third-party record and the safety margin it indicates. The Keccak round function has been unchanged since 2008, and TW128 uses the permutation at the round count introduced by KangarooTwelve [BDP⁺16] and later exposed directly as TurboSHAKE [BDH⁺23], whose designers maintain a survey of third-party cryptanalysis. TurboSHAKE128 shares TW128’s rate and capacity ($r = 1344$, $c = 256$) as well as its permutation, so attacks on it and on round-reduced SHAKE128 transfer to TW128’s geometry directly.

In the unkeyed setting, the deepest collision attacks on any SHA-3 or SHAKE instance reach 6 of TW128’s 12 rounds: a SAT-aided attack on 6-round SHAKE128 with time $2^{123.5}$

[GLST22], and internal-differential attacks reaching 6 rounds of SHAKE256 and 5 rounds of SHA3-384 [ZHL24]. Preimage attacks reach 5 rounds [ZHL25]. Against the collision- and preimage-style attacks most relevant to the root tag hash claims of Section 5, the permutation therefore retains a margin of 6 of 12 rounds—the assessment under which the KangarooTwelve round count was chosen and later reaffirmed [BDP⁺16, BDH⁺23]. The structural distinguisher reaching the most rounds of the permutation itself, the SymSum property on 9 rounds reported in the survey of [BDH⁺23], has, like the zero-sum distinguishers before it, not been extended to an attack on any sponge or duplex mode.

The keyed setting is closer to TW128’s AE games: a hidden key, with adversary-controlled nonces and messages entering through the duplex rate. The attacks reaching the most rounds here are cube and conditional cube attacks. Dinur et al. break up to 9 rounds of Keccak-based stream ciphers and MACs faster than generic search, with keystream prediction on 8 rounds at time and data 2^{128} and on 9 rounds (of a 512-bit-key variant) at 2^{256} [DMP⁺15]. With MILP-optimized conditional cubes, nonce-respecting key recovery reaches 8 of Lake Keyak’s 12 rounds at time 2^{71} for 128-bit keys, 9 rounds at 2^{137} for 256-bit keys, and 9 of KMAC256’s 24 rounds at 2^{147} [SGSL18]. Lake Keyak is a keyed duplex over the same 1600-bit permutation whose initialization, like TW128’s root and leaf initializations, absorbs the key and nonce ahead of nonce-varied cube queries, so these results are the closest published analog to TW128’s keyed setting. They leave a margin of 3 to 4 of 12 rounds against attacks whose complexities start at 2^{71} .

In sum, the known attack frontier stands at 6 of 12 rounds for collisions and preimages and at 8 to 9 rounds for keyed-mode key recovery and structural distinguishers. TW128 uses KECCAK- p [1600, 12] at the same round count and with the same rate/capacity split as KangarooTwelve and TurboSHAKE128, so its margin assessment tracks theirs, and any future advance against the permutation applies to all three alike. What TW128 adds at the mode level and these analyses do not cover is the tree structure itself—in particular the family of leaf initializations sharing (K, U) and differing only in the leaf index—and we consider dedicated cryptanalysis of that structure a natural target for future work.

9.5 Limitations and Open Problems

Section 4 states the formal scope of the security claims; this subsection discusses the consequences of the excluded settings and the corresponding open problems.

Nonce misuse. As scoped in Section 4, TW128’s AE and mu-ind bounds require nonce-respecting encryption queries, and the mode is not misuse-resistant: a repeated nonce reuses leaf keystream and breaks confidentiality, as for any online duplex stream cipher. Schemes such as AES-GCM-SIV [GLL17], in the deterministic/misuse-resistant lineage of [RS06], tolerate nonce repetition at the price of a second pass. A misuse-resistant variant of TW128 is left open.

Unverified plaintext release. The INT-RUP theorem of Theorem 11 covers integrity when an adversary can obtain plaintext-computing oracle outputs before verification, in the release-of-unverified-plaintext setting of [ABL⁺14]. This is an integrity-only statement. It does not give plaintext awareness or privacy under release of unverified plaintext: TW128 is online and single-pass, and exposing unverified stream plaintext can reveal keystream information. Application behavior on unverified plaintext is likewise outside the provable model.

Conservative lift. Per-term tightness is itemized in Section 4.4; the flat-sponge terms are conservative because our proof follows the unkeyed sponge/hash line. This route matches the role of the authentication tag: TW128’s committing statements are consequences of

tag hash security, and sponge hash security is supported by a mature literature. The price is the indistinguishability lift of Section B: one $\text{Lift}_c(N_{\text{tot}})$ term on the real side, and in real-vs-ideal games a second proximity term at N_{prim} (Corollary 9).

A direct keyed-duplex treatment would have to close two gaps. First, existing keyed-duplex bounds are hidden-key AE bounds: the game samples the key, and the keyed initialization material is not part of the adversary’s output. The root tag claims here are adversarial-initialization statements. In the hash and committing games, the adversary may choose the key, nonce, associated data, message, or competing opening. Thus the proof must handle adversarially chosen initialization transcripts, not only hidden-key encryption and verification queries.

Second, the available keyed-duplex AE interface is not a direct substitute for a byte-string AEAD with arbitrary final-block length. The modern bounds synthesized by [Men23] use a duplex call phased as permutation, squeeze, then absorb or overwrite ([Men23, Sec. 3.4, Table 1]). In the authenticated-encryption application, this interface is instantiated as MonkeySpongeWrap [Men23, Alg. 8]. As written, however, the overwrite interface does not reconstruct the missing padding bits of a truncated final ciphertext block on decryption: encryption absorbs the padded final plaintext block after squeezing and then truncates the resulting ciphertext to the message length, while decryption receives only this truncated ciphertext, pads it independently, and uses the overwrite branch to set the rate. The decryption transcript can therefore diverge before tag recomputation.

TW128 instead stays with the BDPV11 duplex phasing: a call absorbs the actual input string, padded by the duplex rule, and then returns the requested output. At the stream layer this supports arbitrary length messages by splitting them into bounded duplex blocks, with the final short ciphertext block absorbed as it is rather than reconstructed through an overwrite branch (Section 3). A direct keyed-duplex theorem for TW128 would therefore need both this BDPV11 phasing and adversarial-initialization hash and committing games. We retain the indistinguishability lift because it covers these claims uniformly and already meets the 128-bit target (Section 7.3).

Beyond-birthday security and leakage. All bounds are birthday-type on the 256-bit capacity, delivering the 128-bit target but no more; a beyond-birthday analysis is open. The model is also single-stage and leakage-free: side-channel resistance is addressed only at the implementation level, through the data-independent execution properties of Section 8.1, and is not part of the provable-security claims.

10 Conclusion

TW128 starts from the premise that an AEAD tag should be the compact digest applications often treat it as: a hash output for the full encryption context, with committing security following from that hash story rather than added as an external layer. The construction adapts the KangarooTwelve tree-hashing pattern into a two-level tree of keyed duplexes over a single KECCAK- $p[1600, 12]$ permutation. The chunk decomposition is canonical, while the number of leaf duplexes evaluated in parallel is an implementation choice rather than a mode parameter. This gives scalar, SIMD, and multicore implementations the same ciphertexts and tags, with constant working memory and scalable parallelism.

The main security result is tag hash security for the wrapper that returns the final authentication tag on input (K, U, A, M) . In the public permutation model, via an intermediate random oracle and the flat sponge lift, we prove collision resistance, standard and everywhere second-preimage resistance, preimage resistance on fixed input slices, and everywhere preimage resistance. Those bounds yield the AEAD-facing committing statements, including full-input CMTDs-4. The same root structure gives INT-RUP integrity under release of unverified plaintext. As baseline AEAD properties, we prove concrete

multi-user indistinguishability and derive all-in-one AE as its single-user corollary. At the implemented parameters, all stated notions meet a 128-bit target.

The implementation results show the cost and payoff of putting that tag hash inside a tree. Short inputs pay for the larger permutation and the root/leaf structure, but long inputs expose independent leaves while preserving one canonical wire format. Vectorized `amd64` and `arm64` implementations reach bulk throughputs of 48.0 Gbit/s on the AVX-512 path, 39.4 Gbit/s on the `arm64` path, and 16.6 Gbit/s on the AVX2 path, with the AVX-512 and `arm64` paths at 51–76% of hardware-accelerated AES-128-GCM throughput.

The remaining questions are those left open in Section 9.5: nonce misuse would require a different construction, privacy under release of unverified plaintext is not provided by the INT-RUP theorem, and sharper bounds would need either a direct keyed-duplex treatment of the BDPV11-style phasing with adversarial initialization or a beyond-birthday argument. More broadly, TW128 supports the thesis that a single KECCAK- p permutation with substantial cryptanalytic history can provide authenticated encryption whose tag has hash security strong enough to drive commitment, while retaining scalable hashing-style performance.

Acknowledgments

A hearty thank-you to my wife, who remains reasonably mystified at my choice of hobbies.

References

- [ABL⁺14] Elena Andreeva, Andrey Bogdanov, Atul Luykx, Bart Mennink, Nicky Mouha, and Kan Yasuda. How to securely release unverified plaintext in authenticated encryption. In *Advances in Cryptology – ASIACRYPT 2014*, volume 8873 of *Lecture Notes in Computer Science*, pages 105–125. Springer, 2014.
- [ADG⁺22] Ange Albertini, Thai Duong, Shay Gueron, Stefan Kölbl, Atul Luykx, and Sophie Schmieg. How to abuse and fix authenticated encryption without key commitment. In *31st USENIX Security Symposium (USENIX Security 22)*, pages 3291–3308. USENIX Association, 2022.
- [BDH⁺17] Guido Bertoni, Joan Daemen, Seth Hoffert, Michaël Peeters, Gilles Van Assche, and Ronny Van Keer. Farfalle: Parallel permutation-based cryptography. *IACR Transactions on Symmetric Cryptology*, 2017(4):1–38, 2017.
- [BDH⁺23] Guido Bertoni, Joan Daemen, Seth Hoffert, Michaël Peeters, Gilles Van Assche, Ronny Van Keer, and Benoît Viguier. TurboSHAKE. *Cryptology ePrint Archive*, Paper 2023/342, 2023.
- [BDP⁺15] Guido Bertoni, Joan Daemen, Michaël Peeters, Gilles Van Assche, and Ronny Van Keer. CAESAR submission: Keyak v2. CAESAR Competition, Round 3 Candidate, 2015.
- [BDP⁺16] Guido Bertoni, Joan Daemen, Michaël Peeters, Gilles Van Assche, Ronny Van Keer, and Benoît Viguier. KangarooTwelve: Fast hashing based on Keccak-p. *Cryptology ePrint Archive*, Paper 2016/770, 2016.
- [BDPVA08] Guido Bertoni, Joan Daemen, Michaël Peeters, and Gilles Van Assche. On the indistinguishability of the sponge construction. In *Advances in Cryptology – EUROCRYPT 2008*, volume 4965 of *Lecture Notes in Computer Science*, pages 181–197. Springer, 2008.

- [BDPVA11] Guido Bertoni, Joan Daemen, Michaël Peeters, and Gilles Van Assche. Duplexing the sponge: Single-pass authenticated encryption and other applications. In *Selected Areas in Cryptography – SAC 2011*, volume 7118 of *Lecture Notes in Computer Science*, pages 320–337. Springer, 2011.
- [BDPVA14] Guido Bertoni, Joan Daemen, Michaël Peeters, and Gilles Van Assche. Sakura: A flexible coding for tree hashing. In *Applied Cryptography and Network Security – ACNS 2014*, volume 8479 of *Lecture Notes in Computer Science*, pages 217–234. Springer, 2014.
- [BH22] Mihir Bellare and Viet Tung Hoang. Efficient schemes for committing authenticated encryption. In *Advances in Cryptology – EUROCRYPT 2022*, volume 13276 of *Lecture Notes in Computer Science*, pages 845–875. Springer, 2022. Citations refer to the full version of July 23, 2024.
- [BT16] Mihir Bellare and Björn Tackmann. The multi-user security of authenticated encryption: AES-GCM in TLS 1.3. In *Advances in Cryptology – CRYPTO 2016*, volume 9814 of *Lecture Notes in Computer Science*, pages 247–276. Springer, 2016.
- [CR22] John Chan and Phillip Rogaway. On committing authenticated encryption. In *Computer Security – ESORICS 2022*, Lecture Notes in Computer Science. Springer, 2022.
- [DEMS21] Christoph Dobraunig, Maria Eichlseder, Florian Mendel, and Martin Schläffer. Ascon v1.2: Lightweight authenticated encryption and hashing. *Journal of Cryptology*, 34(3):33, 2021.
- [DGRW18] Yevgeniy Dodis, Paul Grubbs, Thomas Ristenpart, and Joanne Woodage. Fast message franking: From invisible salamanders to encryptment. In *Advances in Cryptology – CRYPTO 2018*, volume 10991 of *Lecture Notes in Computer Science*, pages 155–186. Springer, 2018.
- [DHM⁺24] Joan Daemen, Seth Hoffert, Silvia Mella, Gilles Van Assche, and Ronny Van Keer. Shaking up authenticated encryption. Cryptology ePrint Archive, Paper 2024/1618, 2024.
- [DHP⁺20] Joan Daemen, Seth Hoffert, Michaël Peeters, Gilles Van Assche, and Ronny Van Keer. Xoodyak, a lightweight cryptographic scheme. *IACR Transactions on Symmetric Cryptology*, 2020(S1):60–87, 2020.
- [DMP⁺15] Itai Dinur, Paweł Morawiecki, Josef Pieprzyk, Marian Srebrny, and Michał Straus. Cube attacks and cube-attack-like cryptanalysis on the round-reduced Keccak sponge function. In *Advances in Cryptology – EUROCRYPT 2015*, volume 9056 of *Lecture Notes in Computer Science*, pages 733–761. Springer, 2015.
- [Dwo07] Morris J. Dworkin. Recommendation for block cipher modes of operation: Galois/Counter Mode (GCM) and GMAC. Technical Report Special Publication 800-38D, National Institute of Standards and Technology, 2007.
- [GLL17] Shay Gueron, Adam Langley, and Yehuda Lindell. AES-GCM-SIV: Specification and analysis. Cryptology ePrint Archive, Paper 2017/168, 2017.
- [GLR17] Paul Grubbs, Jiahui Lu, and Thomas Ristenpart. Message franking via committing authenticated encryption. In *Advances in Cryptology – CRYPTO 2017*, volume 10403 of *Lecture Notes in Computer Science*, pages 66–97. Springer, 2017.

- [GLST22] Jian Guo, Guozhen Liu, Ling Song, and Yi Tu. Exploring SAT for cryptanalysis: (quantum) collision attacks against 6-round SHA-3. In *Advances in Cryptology – ASIACRYPT 2022*, volume 13793 of *Lecture Notes in Computer Science*, pages 645–674. Springer, 2022.
- [Ham17] Mike Hamburg. The STROBE protocol framework. Cryptology ePrint Archive, Paper 2017/003, 2017.
- [KSW23] Juliane Krämer, Patrick Struck, and Maximiliane Weishäupl. Committing AE from sponges: Security analysis of the NIST LWC finalists. Cryptology ePrint Archive, Paper 2023/1525, 2023.
- [LGR21] Julia Len, Paul Grubbs, and Thomas Ristenpart. Partitioning oracle attacks. In *30th USENIX Security Symposium (USENIX Security 21)*, pages 195–212. USENIX Association, 2021.
- [Men23] Bart Mennink. Understanding the duplex and its security. *IACR Transactions on Symmetric Cryptology*, 2023(2):1–46, 2023.
- [MLGR23] Sanketh Menda, Julia Len, Paul Grubbs, and Thomas Ristenpart. Context discovery and commitment attacks: How to break CCM, EAX, SIV, and more. In *Advances in Cryptology – EUROCRYPT 2023*, volume 14007 of *Lecture Notes in Computer Science*, pages 379–407. Springer, 2023.
- [Nat15] National Institute of Standards and Technology. SHA-3 standard: Permutation-based hash and extendable-output functions. Federal Information Processing Standards Publication FIPS PUB 202, U.S. Department of Commerce, August 2015.
- [RS04] Phillip Rogaway and Thomas Shrimpton. Cryptographic hash-function basics: Definitions, implications, and separations for preimage resistance, second-preimage resistance, and collision resistance. In *Fast Software Encryption – FSE 2004*, volume 3017 of *Lecture Notes in Computer Science*, pages 371–388. Springer, 2004. Citations refer to the full version of July 16, 2009.
- [RS06] Phillip Rogaway and Thomas Shrimpton. A provable-security treatment of the key-wrap problem. In *Advances in Cryptology – EUROCRYPT 2006*, volume 4004 of *Lecture Notes in Computer Science*, pages 373–390. Springer, 2006.
- [SGSL18] Ling Song, Jian Guo, Danping Shi, and San Ling. New MILP modeling: Improved conditional cube attacks on Keccak-based constructions. In *Advances in Cryptology – ASIACRYPT 2018*, volume 11273 of *Lecture Notes in Computer Science*, pages 65–95. Springer, 2018.
- [ZHL24] Zhongyi Zhang, Chengan Hou, and Meicheng Liu. Probabilistic linearization: Internal differential collisions in up to 6 rounds of SHA-3. In *Advances in Cryptology – CRYPTO 2024*, volume 14923 of *Lecture Notes in Computer Science*, pages 241–272. Springer, 2024.
- [ZHL25] Zhongyi Zhang, Chengan Hou, and Meicheng Liu. Preimage attacks on up to 5 rounds of SHA-3 using internal differentials. In *Advances in Cryptology – EUROCRYPT 2025*, volume 15601 of *Lecture Notes in Computer Science*, pages 333–363. Springer, 2025.

A Supporting Lemmas

The proof core separates structural decoding from probabilistic accounting. The well-formed transcript language of Definition 2 fixes which flattened queries are valid TW128 construction prefixes. Lemmas 5, 6 and 8 then show how such a query is flattened, parsed as a typed transcript, and assigned an output point; the exact keyed decoder of Definition 4 then decodes direct queries to candidate keys, when any. Lemma 10 turns those decoded keys into an adaptive key-test bound. The remaining interfaces expose the local events consumed by the main proofs: encryption-side freshness, verification hidden-key exposure and root tag guessing, leaf tag collisions, and final-root candidate sequencing. The tag hash proof applies the candidate bound, committing statements reuse the resulting root-collision bound, and the mu-ind and INT-RUP proofs derive the premises for their accounting lemmas.

How to read this appendix. Figure 2 records the proof dependency map. It groups local interfaces when the exact proof citations would obscure the spine. The Bridge & Decode node is Lemma 8 together with the exact keyed decoder of Definition 4. The Root/Output Separation node groups Output-Point Separation (Corollary 8), Opening Triple Separation (Proposition 1), and Final-Root Input Separation (Proposition 2); the use of Leaf-Transcript Recovery (Lemma 7) is folded into the final-root separation fact. The Key-Indexed Renaming node (Lemma 9) supplies the key-field disjointness and renaming used by hidden-key encryption and verification accounting. The Adaptive Candidate node (Lemma 2) is the generic probabilistic engine: it bounds the Leaf Tag Collision lemma (Lemma 11) and, through the Root Tag Candidate Sequence and Root Tag Hash and Target-Hit Bound (Lemmas 3 and 4) of Section 5, the tag hash proof; the committing consequences reuse the same root-collision bound under their opening constraints. The mu-ind proof reuses the same structural lemmas through the explicit hybrid game sequence of Definition 5, adding freshness, key-test, leaf-collision, and verification-first root guessing accounting.

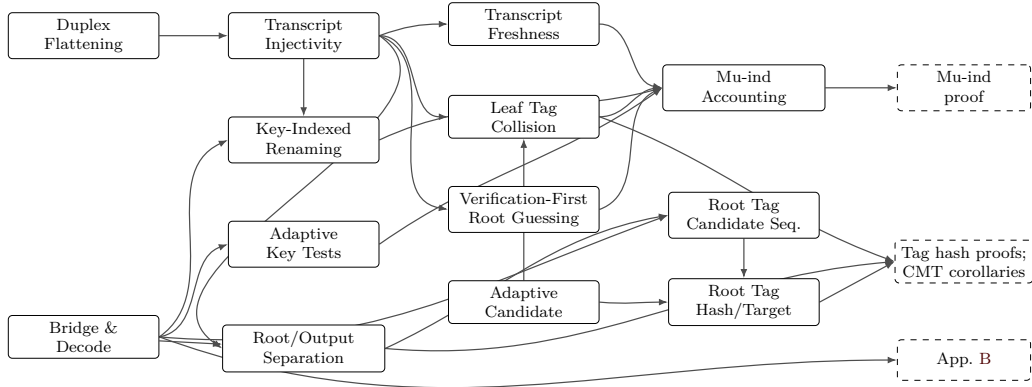


Figure 2: Proof dependency map for the supporting components in this appendix and their consumers. Solid boxes are lemmas or grouped local proof interfaces; dashed boxes are proof components that consume them.

A.1 Duplex Flattening

Lemma 5 (Duplex Flattening). *Every TW128 root or leaf duplex output is equal to a [BDPVA11] sponge output on the corresponding flattened duplex-call transcript, and the*

1921 *ciphertext-absorbing message phase admits a flattened duplex transcript that is well defined*
 1922 *for both encryption and decryption.*

1923 *Proof.* Let D be a duplex object using permutation π , rate r , and padding pad10^*1 . For a
 1924 sequence of duplex calls $Z_i = D.\text{duplexing}(\sigma_i, \ell_i)$ with each σ_i short enough to fit in one
 1925 duplex block after padding, [BDPVA11, Lemma 3] gives

$$1926 \quad Z_i = \text{Sponge}[\pi, \text{pad10}^*1, r](\sigma_0 \parallel \text{pad}_0 \parallel \sigma_1 \parallel \text{pad}_1 \parallel \cdots \parallel \sigma_i, \ell_i),$$

1927 where $\text{pad}_j = \text{pad10}^*1[r](|\sigma_j|)$, and Lemma 4 says that, for fixed pad10^*1 and r , the map
 1928 from the duplex input-string sequence to the flattened sponge input is injective.

1929 In TW128, each duplex call absorbs a byte-aligned block σ followed by a 4-bit phase
 1930 suffix $\sigma_{\text{ph}} \in \{\sigma_{\text{init}}, \sigma_{\text{ad}}^-, \dots, \sigma_{\text{agg}}^+\}$ (Section 3.3), so the duplex-call input is $\sigma \parallel \sigma_{\text{ph}}$ followed
 1931 by pad10^*1 . With $r = 1344$ and $\rho_{\text{max}}(\text{pad10}^*1, r) = r - 2 = 1342$ bits, the maximum
 1932 byte-aligned input is $\rho = 167$ bytes (Section 3.2), giving a worst-case duplex-call input of
 1933 $8 \cdot 167 + 4 = 1340 \leq 1342$ bits. The two initialization calls are shorter still: $|p_{\text{root}} \parallel K \parallel U| + 4 =$
 1934 $8 \cdot (6 + 32 + 32) + 4 = 564$ bits and $|p_{\text{leaf}} \parallel K \parallel U \parallel \langle j \rangle_8| + 4 = 8 \cdot (6 + 32 + 32 + 8) + 4 = 628$
 1935 bits. [BDPVA11, Lemmas 3 and 4] therefore apply independently to every TW128 duplex
 1936 object.

1937 The root duplex is initialized by absorbing $p_{\text{root}} \parallel K \parallel U$ under σ_{init} (Algorithm 2), then
 1938 absorbs the associated data blocks under $\sigma_{\text{ad}}^- / \sigma_{\text{ad}}^+$, the root ciphertext blocks in the message
 1939 phase under $\sigma_{\text{msg}}^- / \sigma_{\text{msg}}^+$, and the leaf tag aggregation blocks under $\sigma_{\text{agg}}^- / \sigma_{\text{agg}}^+$. By the
 1940 duplex-to-sponge equality, every root duplex output is the sponge function evaluated on
 1941 the root transcript prefix accumulated up to the relevant output point. The same holds for
 1942 each leaf duplex, initialized by $p_{\text{leaf}} \parallel K \parallel U \parallel \langle j \rangle_8$ and absorbing only its own ciphertext
 1943 blocks in the message phase. In decryption and verification, Algorithm 3 absorbs the
 1944 submitted ciphertext blocks before the root tag comparison, so the same flattened message
 1945 phase transcript is defined whether the comparison accepts or rejects. $\square$

1946 A.2 Well-formed Transcript Prefixes

1947 For a duplex-call prefix $((\sigma_0, \sigma_{\text{ph},0}), \dots, (\sigma_i, \sigma_{\text{ph},i}))$, write

$$1948 \quad \text{Flat}((\sigma_0, \sigma_{\text{ph},0}), \dots, (\sigma_i, \sigma_{\text{ph},i})) = \left\|_{j=0}^{i-1} (\sigma_j \parallel \sigma_{\text{ph},j} \parallel \text{pad}_j) \parallel \sigma_i \parallel \sigma_{\text{ph},i},\right.$$

1949 where $\text{pad}_j = \text{pad10}^*1[r](|\sigma_j \parallel \sigma_{\text{ph},j}|)$, the padding of the j -th full duplex-call input. The
 1950 current call's padding pad_i is not part of Flat; it is applied internally by the sponge
 1951 function. Equivalently, after the i -th call the duplex state is the sponge state after
 1952 absorbing $\text{Flat}(\cdot) \parallel \text{pad}_i$.

1953 **Definition 2** (Well-Formed TW128 Transcript Prefixes). A typed TW128 transcript prefix
 1954 is *well formed* if it is a prefix, ending at a construction transcript lookup point, of one of
 1955 the following two call languages.

- 1956 • A root transcript starts with $(p_{\text{root}} \parallel K \parallel U, \sigma_{\text{init}})$, with fixed-width fields $K \in \{0, 1\}^k$
 1957 and $U \in \{0, 1\}^u$. It then absorbs the associated data byte string (only when the
 1958 associated data is nonempty; the phase is otherwise elided, Section 3.4), the root
 1959 ciphertext byte string, and the aggregation byte string (only when $n \geq 1$; when
 1960 $n = 0$ the aggregation phase is elided, Section 3.4, and the root tag is emitted by the
 1961 final σ_{msg}^+ message call), in that order. Each present phase byte string is partitioned
 1962 by the block rule of Algorithm 1: the empty string gives one empty final block, and
 1963 a nonempty string gives zero or more ρ -byte non-final blocks followed by one final
 1964 block. The phases use respectively $\sigma_{\text{ad}}^- / \sigma_{\text{ad}}^+$, $\sigma_{\text{msg}}^- / \sigma_{\text{msg}}^+$, and $\sigma_{\text{agg}}^- / \sigma_{\text{agg}}^+$.

- A leaf- j transcript starts with $(p_{\text{leaf}} \parallel K \parallel U \parallel \langle j \rangle_8, \sigma_{\text{init}})$, where $1 \leq j < 2^{64}$ and the fields have fixed width. It then absorbs the leaf ciphertext byte string under $\sigma_{\text{msg}}^- / \sigma_{\text{msg}}^+$, using the same block rule.

The message byte string in this language is always the ciphertext byte string absorbed by the construction: encryption obtains it by XORing the plaintext with the current keystream, while decryption and verification use the submitted ciphertext. Thus a verification transcript is well formed for every in-domain ciphertext before the tag comparison is made; acceptance is not part of the transcript language.

The root aggregation byte string has the form $T_1 \parallel \dots \parallel T_n \parallel \langle n \rangle_8$, where each T_j has fixed byte length $\tau_{\text{leaf}}/8$. The trailing fixed-width field therefore gives a unique parse of the leaf tag sequence. The output-point class of a well-formed counted prefix is the class in Table 11 determined by its duplex instance and final-call suffix.

Remark 2 (Well-Formedness versus Realizability). The language of Definition 2 is deliberately larger than the set of transcript prefixes that TW128 computations realize. For example, it does not require an aggregation phase to follow a root message phase of exactly β bytes, although every real computation with $n \geq 1$ absorbs a full β -byte root chunk (Algorithm 2). The inclusion runs in the safe direction everywhere: every construction-generated prefix is well formed, so the structural lemmas of this appendix and the separation propositions of Section 5 apply to construction transcripts as restrictions of statements proved on the larger language, while the direct-query decoder of Definition 4 and the root tag candidate predicate of Lemma 3 accept the larger language and therefore only over-count adversary queries, which loosens no bound.

Name the tag-emitting transcript families:

$$\begin{aligned} \mathcal{R}_{\text{tag}} &= \text{root transcript prefix whose sponge output gives the final root tag,} \\ \mathcal{L}_{\text{tag}}(j) &= \text{leaf-}j \text{ transcript prefix whose sponge output gives the leaf tag.} \end{aligned}$$

The final root tag is emitted by the σ_{agg}^+ aggregation call (the $\mathcal{R}_{\text{tag}}$ family) when $n \geq 1$, and by the closing σ_{msg}^+ message call (an $\mathcal{R}_{\text{msg}}^+$ prefix, in the notation of Corollary 8) when $n = 0$ and the aggregation phase is elided. Since $n = \text{leaves}(C)$ is fixed by $\|C\|$, every computation on a given ciphertext emits its tag at the output point selected by $\|C\|$. Intermediate construction transcript lookup points (init, non-final AD, non-final message, and non-final aggregation) are also typed prefixes covered by Lemma 8; the exact keyed direct-query decoder names them in Table 11.

Definition 3 (Root Tag Output Point and Final Root Transcript Class). For any in-domain encryption or verification computation on ciphertext C , the *root tag output point* is the construction lookup that emits the root tag: $\mathcal{R}_{\text{tag}}$ when $\text{leaves}(C) \geq 1$, and the closing root message point $\mathcal{R}_{\text{msg}}^+$ when $\text{leaves}(C) = 0$ and aggregation is elided. The *final root transcript class* of such a computation is the set of construction computations that realize the same bridged final root input at this root tag output point. A class is *encryption-first* if its first construction generation is by an encryption query, and *verification-first* if its first construction generation is by a non-replay verification query.

A.3 Transcript Injectivity

Lemma 6 (Transcript Injectivity). *Well-formed TW128 flattened duplex-call transcript prefixes are injectively encoded and separated by duplex instance (root vs leaf), leaf index j , phase suffix, and duplex-call order. For final-root prefixes, equality of flattened inputs recovers the same root initialization, AD blocks, root ciphertext blocks, aggregation input, and leaf tag sequence.*

Proof. By Lemma 5, every well-formed output point can be written as a [BDPVA11] flattened sponge input for the corresponding duplex-call prefix, namely Flat as defined above. The deterministic current-call pad10^*1 is applied internally by the sponge function and is not part of the input to which [BDPVA11, Lemma 4] is applied. That lemma recovers the duplex-call sequence from the flattened sponge input; in TW128, each call input is $\sigma \parallel \sigma_{\text{ph}}$ with byte-aligned σ and fixed 4-bit suffix σ_{ph} , so recovering the call input recovers the pair $(\sigma, \sigma_{\text{ph}})$.

Definition 2 fixes the root and leaf initialization fields, the admissible phase order, and the final-call suffixes. The distinct prefixes p_{root} and p_{leaf} separate root from leaf transcripts; the fixed-width leaf field $\langle j \rangle_8$ separates leaves; and the seven suffix values of Table 2 separate phases and non-final/final calls. Thus equality of flattened inputs forces equality of the recovered instance type, key, nonce, leaf index when present, phase suffixes, and duplex-call order.

Phase-recovery sub-claim. For each present phase $P \in \{\text{ad}, \text{msg}, \text{agg}\}$ (the ad phase is present exactly when the associated data is nonempty; see Section 3.4), TW128 absorbs the phase’s input byte string s_P by partitioning into ρ -byte blocks (with the empty case yielding a single empty block, per Section 3.4) and feeding each block to the duplex with suffix σ_P^- (non-final) or σ_P^+ (final). Lemma 5 and [BDPVA11, Lemma 4] recover the corresponding $(\sigma, \sigma_{\text{ph}})$ sequence from the flattened input. Because the partition map is deterministic and uniquely invertible on its image (the empty case yielding the single empty block carrying σ_P^+), concatenating the recovered σ blocks reconstructs s_P exactly. The map from s_P to the phase’s $(\sigma, \sigma_{\text{ph}})$ sequence is therefore injective.

Applied phase by phase to a well-formed final-root transcript:

- AD. The associated data phase contributes a σ_{ad}^+ -tagged block exactly when the associated data is nonempty; an empty associated data elides the phase and contributes no block. When the phase is present the sub-claim recovers the AD byte string. Thus distinct associated data strings $A_i \neq A_j$ are separated either by the presence versus absence of the AD phase (when exactly one is empty) or, when both are nonempty, by their distinct AD σ -block sequences.
- MSG. Distinct absorbed root or leaf ciphertext byte strings produce distinct MSG σ -block sequences; the message language of Definition 2 is ciphertext-absorbing for encryption and decryption alike.
- AGG. The aggregation phase is present exactly when $n \geq 1$ (Section 3.4). When it is present, the sub-claim applied to the aggregation byte string $T_1 \parallel \dots \parallel T_n \parallel \langle n \rangle_8$ recovers that byte string. The final eight bytes give $\langle n \rangle_8$, and each leaf tag has fixed byte length $\tau_{\text{leaf}}/8$, so the byte string has a unique parse as a leaf tag sequence. Two root aggregation encodings are therefore equal exactly when their leaf tag sequences are equal. When $n = 0$ the aggregation phase is absent and the final-root transcript ends at the recovered σ_{msg}^+ root message call; the MSG bullet recovers the root ciphertext byte string, and the absence of any $\sigma_{\text{agg}}^-/\sigma_{\text{agg}}^+$ call distinguishes such a prefix from every $n \geq 1$ final-root prefix.

Therefore the recovered final-root transcript determines the root initialization, AD blocks, root ciphertext blocks, and, when $n \geq 1$, the aggregation byte string and leaf tag sequence. $\square$

Lemma 7 (Leaf-Transcript Recovery). *Let two construction-generated final-root prefixes have equal flattened inputs, and suppose that, for each position j of their common leaf tag sequence, the two computations’ position- j final leaf transcripts are either equal or receive distinct leaf tags. Then equality of flattened inputs recovers the same leaf transcript*

sequence, and therefore the same full absorbed ciphertext. Both computations' position- j final leaf transcripts are leaf- j transcripts under the common recovered (K, U) , so the hypothesis holds whenever no two distinct construction-generated final leaf transcripts with the same key, nonce, and leaf index receive the same leaf tag; when one of the two prefixes is reached by an encryption computation, it suffices that no construction-generated final leaf transcript receives the same leaf tag as a distinct encryption-reached final leaf transcript with the same key, nonce, and leaf index.

Proof. By Transcript Injectivity (Lemma 6), the two equal flattened final-root inputs recover the same root initialization, AD blocks, root ciphertext blocks, and, when $n \geq 1$, the same aggregation byte string and leaf tag sequence; the trailing $\langle n \rangle_8$ field fixes the common leaf count n . When $n = 0$ no leaf is present and the recovered root ciphertext blocks already constitute the full absorbed ciphertext. When $n \geq 1$, the aggregation byte string parses uniquely as the leaf tag sequence $T_1 \parallel \dots \parallel T_n$; the two leaf tag sequences are equal, so for each j the two computations' position- j final leaf transcripts—leaf- j transcripts carrying the common recovered (K, U) (Definition 2, Section 3.4)—receive the same leaf tag, and by hypothesis are equal, hence so are their absorbed leaf ciphertext bodies. Together with the common root ciphertext blocks this recovers the same full absorbed ciphertext. $\square$

A.4 Bridge and Typed Decoding

Lemma 8 (Bridge Injectivity). *The random oracle games used in Sections 5 and 6 and Corollary 5 are ordinary random oracle games on a single random oracle $\text{RO}_{\text{flat}} : \{0, 1\}^* \rightarrow \{0, 1\}^\infty$ over the unpadded [BDPVA08] H -input domain (Section 4.2), with TW128 root and leaf duplex outputs answered by the typed restriction*

$$\text{RF}(X) = \text{RO}_{\text{flat}}(\text{flat}(X)), \quad \text{flat}(X) = I[r, r_{\max}](\text{raw_flat}(X)),$$

for $r = r_{\max} = 1344$. Here $\text{raw_flat}(X)$ is the native flattened input $\text{Flat}(\cdot)$ of Section A.2 applied to the duplex-call sequence of X , and $I[r, r_{\max}]$ is the [BDPVA11] Theorem 3 preprocessing map. The induced map $X \mapsto \text{flat}(X)$ is injective on well-formed typed TW128 duplex transcript prefixes.

Proof. The bridge $I[r, r_{\max}]$ is described by the proof of [BDPVA11, Theorem 3] in three steps: (1) pad the input with $\text{pad10}^*1[r]$; (2) for each r -bit block, append $r_{\max} - r$ zero bits; (3) strip the trailing pad10^* from the result. The pad10^*1 padding in step (1) appends a 1, then $q = (-|\text{raw_flat}(X)| - 2) \bmod r$ zero bits, then a closing 1; the pad10^* -unpadding in step (3) strips the $r_{\max} - r$ trailing zero bits introduced by step (2) together with the closing 1 of step (1). For TW128 with $r = r_{\max}$, step (2) appends an empty string and step (3) strips zero trailing zeros and the closing 1, so the net effect of steps (1) and (3) is to map a native flattened input $W = \text{raw_flat}(X)$ to $W \parallel 1 \parallel 0^q$. Any string in this image has a unique inverse: q is the number of trailing zero bits, the preceding bit is the inserted 1, and deleting this suffix recovers W .

Injectivity of flat on well-formed typed TW128 prefixes then follows from Lemma 6: distinct well-formed typed prefixes have distinct $\text{raw_flat}(\cdot)$ images, and the bridge is injective on that image. Direct adversary queries on inputs outside the bridged TW128 well-formed transcript language still count in $q_{\text{RO}}(\mathcal{B})$ (Section 4.3) but cannot trigger bad_{key} . $\square$

The bridge injectivity established here underlies Opening Triple Separation (Proposition 1, Section 5): distinct (K, U, A) triples yield distinct bridged final root transcripts.

Corollary 8 (Output-Point Separation). *Every well-formed TW128 typed transcript prefix at a counted construction transcript lookup point belongs to exactly one of the following*

2106 *output-point classes, identified by the recovered duplex instance and the 4-bit suffix of its*
 2107 *final call:*

- 2108 • $\mathcal{R}_{\text{init}}(\text{root duplex}, \text{suffix } \sigma_{\text{init}})$,
- 2109 • $\mathcal{L}_{\text{init}}(j)$ (leaf- j duplex, suffix σ_{init}),
- 2110 • $\mathcal{R}_{\text{ad}}^- / \mathcal{R}_{\text{ad}}^+$ (root duplex, suffix σ_{ad}^- or σ_{ad}^+),
- 2111 • $\mathcal{R}_{\text{msg}}^- / \mathcal{R}_{\text{msg}}^+$ (root duplex, suffix σ_{msg}^- or σ_{msg}^+),
- 2112 • $\mathcal{L}_{\text{msg}}^-(j)$ (leaf- j duplex, suffix σ_{msg}^-),
- 2113 • $\mathcal{L}_{\text{tag}}(j)$ (leaf- j duplex, suffix σ_{msg}^+),
- 2114 • $\mathcal{R}_{\text{agg}}(\text{root duplex}, \text{suffix } \sigma_{\text{agg}}^-)$,
- 2115 • $\mathcal{R}_{\text{tag}}(\text{root duplex}, \text{suffix } \sigma_{\text{agg}}^+)$.

2116 *For any two such prefixes whose bridged flattened inputs are equal, the separation of*
 2117 *Lemma 6 recovers the same duplex instance, leaf index j when present, (K, U) pair, output-*
 2118 *point class, and duplex-call sequence. Equivalently, any two distinct counted transcript*
 2119 *prefixes already differ in one of these recovered fields, and hence in their flattened inputs.*

2120 *Proof.* Lemma 8 recovers equality of native flattened inputs from equality of bridged inputs,
 2121 after which Lemma 6 recovers the duplex-call sequence and its per-call $(\sigma, \sigma_{\text{ph}})$ pairs. The
 2122 first call's σ is $p_{\text{root}} \parallel K \parallel U$ for root instances and $p_{\text{leaf}} \parallel K \parallel U \parallel \langle j \rangle_8$ for leaf instances,
 2123 with fixed-width fields, so it fixes the instance type (by p_{root} vs p_{leaf}), the (K, U) pair, and
 2124 the leaf index j ; within each instance type the phase suffixes are pairwise distinct, so the
 2125 recovered instance type together with the final-call suffix fixes the output-point class, and
 2126 each prefix therefore lies in exactly one class. Conversely, two distinct well-formed prefixes
 2127 in the same class on the same instance differ in some recovered call's $(\sigma, \sigma_{\text{ph}})$ pair or in the
 2128 sequence length, hence in their flattened inputs. $\square$

2129 **Definition 4** (Exact Keyed Decoder). For a direct query to RO_{flat} with input Y and finite
 2130 length ℓ , define $\text{KeyedTWOut}(Y)$ as a partial map from $\{0, 1\}^*$ to $\{0, 1\}^k \times \mathcal{C}_{\text{out}}$, where
 2131 $\mathcal{C}_{\text{out}}$ is the finite set of output-point classes of Table 11. The map ignores ℓ . It returns
 2132 $(K, \mathcal{C})$ iff there exists a typed TW128 duplex-call prefix X such that $Y = \text{flat}(X)$, the raw
 2133 flattened prefix of X parses as a well-formed TW128 transcript prefix in the class $\mathcal{C}$ listed in
 2134 Table 11, and the first call of X contains the candidate key K in the fixed-width root field
 2135 $p_{\text{root}} \parallel K \parallel U$ or leaf field $p_{\text{leaf}} \parallel K \parallel U \parallel \langle j \rangle_8$. If no such prefix exists, $\text{KeyedTWOut}(Y) = \perp$.
 2136 In particular, inputs with the right final suffix but a malformed phase order, noncanonical
 2137 block partition, malformed aggregation string, or out-of-range leaf index are outside the
 2138 well-formed language and decode to $\perp$. A non- $\perp$ decoding always fixes the candidate
 2139 key K and nonce U in the fixed-width initialization field. It fixes the associated data A
 2140 only for root prefixes whose recovered phase order determines the complete AD string: a
 2141 final AD prefix, or any later root message or aggregation prefix, with absence of AD calls
 2142 determining $A = \varepsilon$. We use the key projection $(K, \mathcal{C})$ throughout.

2143 By Lemma 8 and Corollary 8, KeyedTWOut is well defined: a bridged input cannot
 2144 parse as two distinct counted well-formed prefixes, and an accepted input fixes a unique
 2145 candidate key and output-point class. Inputs outside this well-formed transcript language
 2146 decode to $\perp$.

2147 The table is a transcript-lookup table, not an observation-length table. A class is
 2148 counted even when the construction's requested output length at that call is zero, including
 2149 root initialization, non-final AD and aggregation calls, a final AD call whose following root
 2150 chunk requests no keystream bytes, and the final root message call. A direct query triggers

Table 11: Counted keyed transcript classes for KeyedTWOOut.

Class $\mathcal{C}$	Instance	Final call / role
$\mathcal{R}_{\text{init}}$	root	σ_{init} , root initialization
$\mathcal{R}_{\text{ad}}^-$	root	σ_{ad}^- , non-final AD block
$\mathcal{R}_{\text{ad}}^+$	root	σ_{ad}^+ , final AD block
$\mathcal{R}_{\text{msg}}^-$	root	σ_{msg}^- , non-final root message block
$\mathcal{R}_{\text{msg}}^+$	root	σ_{msg}^+ , final root message block
$\mathcal{R}_{\text{agg}}^-$	root	σ_{agg}^- , non-final aggregation block
$\mathcal{R}_{\text{tag}}$	root	σ_{agg}^+ , final aggregation / root tag
$\mathcal{L}_{\text{init}}(j)$	leaf j	σ_{init} , leaf initialization
$\mathcal{L}_{\text{msg}}^-(j)$	leaf j	σ_{msg}^- , non-final leaf message block
$\mathcal{L}_{\text{tag}}(j)$	leaf j	σ_{msg}^+ , final leaf message block / leaf tag

the same decoder regardless of ℓ : short finite-output queries and zero-length queries ($Y, 0$) still count as direct queries to the address Y . Leaves have no AD or aggregation classes; a leaf transcript enters the table only through leaf initialization, non-final leaf message blocks, or the final leaf tag point. When the associated data is empty the root transcript likewise has no $\mathcal{R}_{\text{ad}}^-$ or $\mathcal{R}_{\text{ad}}^+$ class (Section 3.4); the $\mathcal{R}_{\text{init}}$ class then supplies the first root keystream block, exactly as $\mathcal{L}_{\text{init}}(j)$ supplies the first leaf keystream block, and the decoder still ignores ℓ . Symmetrically, when $n = 0$ the root transcript has no $\mathcal{R}_{\text{agg}}$ or $\mathcal{R}_{\text{tag}}$ class (Section 3.4); the final root message call $\mathcal{R}_{\text{msg}}^+$ then supplies the root tag, exactly as $\mathcal{L}_{\text{tag}}(j)$ supplies a leaf tag, and the decoder again ignores ℓ . An $n \geq 1$ computation also reaches an $\mathcal{R}_{\text{msg}}^+$ point when closing chunk 0, but there the construction requests zero output and continues into the aggregation phase, so the $\mathcal{R}_{\text{msg}}^+$ value is exposed as a tag only on the $n = 0$ path; since $n = \text{leaves}(C)$ is fixed by $\|C\|$, the two uses never coincide on a single ciphertext. We call $\mathcal{R}_{\text{init}}$, $\mathcal{R}_{\text{ad}}^+$, $\mathcal{R}_{\text{msg}}^-$, $\mathcal{L}_{\text{init}}(j)$, and $\mathcal{L}_{\text{msg}}^-(j)$ the *stream output points*: by the schedule just described, these are the only classes at which a construction computation requests keystream bytes, while tag values are exposed only at $\mathcal{L}_{\text{tag}}(j)$ and $\mathcal{R}_{\text{tag}}$ points and, on the $n = 0$ path, at $\mathcal{R}_{\text{msg}}^+$.

The hidden-key encryption, verification, and mu-ind accounting all reindex transcript addresses by the key field. We record that reindexing once.

Lemma 9 (Key-Indexed Transcript Renaming). *For a key $K \in \{0, 1\}^k$, let $\mathcal{Y}_K$ be the set of bridged inputs $\text{flat}(X)$ of well-formed TW128 transcript prefixes X whose first-call key field is K —the counted hidden-key transcript addresses of Table 11 for key K . Then:*

1. Disjointness. *For distinct keys $K_0 \neq K_1$, the sets $\mathcal{Y}_{K_0}$ and $\mathcal{Y}_{K_1}$ are disjoint.*
2. Renaming. *The map $\Phi_{K_0 \rightarrow K_1}$ that replaces the first-call key field K_0 by K_1 and leaves every other recovered field unchanged—nonce, leaf index, phase sequence, AD blocks, ciphertext blocks, aggregated leaf tag bytes, and root/leaf instance—induces through flat a bijection $\text{flat} \circ \Phi_{K_0 \rightarrow K_1} \circ \text{flat}^{-1} : \mathcal{Y}_{K_0} \rightarrow \mathcal{Y}_{K_1}$ preserving output-point classes and equality relations among addresses.*
3. Direct-query avoidance. *A direct query input $Y \in \mathcal{Y}_K$ has $\text{KeyedTWOOut}(Y) = (K, \mathcal{C})$ for a counted class $\mathcal{C}$; hence if no prior direct query decodes to K then $\mathcal{Y}_K$ is disjoint from all prior direct-query inputs.*

Proof. By bridge injectivity (Lemma 8) and Transcript Injectivity (Lemma 6), a bridged input $\text{flat}(X)$ determines the native flattened prefix and hence the first duplex call $p_{\text{root}} \| K \| U$ or $p_{\text{leaf}} \| K \| U \| \langle j \rangle_8$, in particular its fixed-width key field K . (1) If $\text{flat}(X) = \text{flat}(X')$ with key fields K_0 and K_1 then $K_0 = K_1$, so distinct keys give disjoint address sets. (2) $\Phi_{K_0 \rightarrow K_1}$ rewrites only the key field, leaving the recovered duplex-call sequence otherwise intact,

so it is a bijection on well-formed prefixes preserving the output-point class (Corollary 8) and the equality pattern among prefixes; flat transports it to the stated bijection on addresses. (3) By Lemma 8 a $Y = \text{flat}(X) \in \mathcal{Y}_K$ decodes under KeyedTWOut to $(K, \mathcal{C})$; the contrapositive gives the avoidance claim. $\square$

A.5 Adaptive Candidate Bounds

The Adaptive Candidate Collision and Preimage lemma is stated as Lemma 2 in Section 5, where a proof sketch also appears; this subsection supplies the full proof.

Proof of Lemma 2. Both parts expose the interaction in chronological order and sample each candidate's RO_{flat} value lazily at its append step, which the premises license: when Y_j is appended its string is fixed by the prior state (predictability) and no bit of $\text{RO}_{\text{flat}}(Y_j)$ has yet been revealed (unrevealed value), so $\text{RO}_{\text{flat}}(Y_j)$ may be drawn then, uniform and independent of everything so far.

Collision. For a fixed position set $I = \{i_1 < \dots < i_s\} \subseteq \{1, \dots, L\}$, let E_I be the event that these candidate positions exist and their τ -bit projections are all equal. Condition on any positive-probability prefix immediately before position i_r is appended, for $r \geq 2$, and on the values sampled at the earlier positions in I . By the above, $\text{RO}_{\text{flat}}(Y_{i_r})$ is then uniform and independent of the conditioning, so its τ -bit projection equals the already fixed projection of Y_{i_1} with probability $2^{-\tau}$. Multiplying over $r = 2, \dots, s$ gives $\Pr[E_I] \leq 2^{-\tau(s-1)}$; if some position in I is never appended then E_I is false and the bound still holds. A union bound over the $\binom{L}{s}$ position sets gives the claim.

Preimage. For a fixed position $j \in \{1, \dots, L\}$, let E_j be the event that position j exists and $\lfloor \text{RO}_{\text{flat}}(Y_j) \rfloor_\tau = t$. Condition on any positive-probability prefix immediately before Y_j is appended; the target t is a constant. By the above, $\text{RO}_{\text{flat}}(Y_j)$ is uniform and independent of the conditioning, so its τ -bit projection equals t with probability $2^{-\tau}$. If position j is never appended, E_j is false. A union bound over the L positions gives the claim.

Second-preimage corollary. If the designated position j_0 is not appended, the event is false. Otherwise split the other candidates according to whether they occur before or after j_0 . For earlier candidates, condition on any positive-probability prefix immediately before Y_{j_0} is appended, including the already sampled projections of candidates $1, \dots, j_0 - 1$. The value $\text{RO}_{\text{flat}}(Y_{j_0})$ is then fresh and uniform, so its τ -bit projection hits one of those at most $j_0 - 1$ earlier projections with probability at most $(j_0 - 1)2^{-\tau}$. For a later candidate Y_j , $j > j_0$, condition on the full prefix immediately before it is appended, including the fixed projection $t_0 = \lfloor \text{RO}_{\text{flat}}(Y_{j_0}) \rfloor_\tau$. The value $\text{RO}_{\text{flat}}(Y_j)$ is fresh and uniform at its append step, so it hits t_0 with probability $2^{-\tau}$. A union bound over the at most $L - j_0$ later candidates gives an additional $(L - j_0)2^{-\tau}$. Adding the two sides gives $\Pr[\text{sec}_{\tau, j_0}] \leq (L - 1) \cdot 2^{-\tau}$; the designated candidate is excluded, so its trivially equal projection does not enter the count. $\square$

A.6 Adaptive Key Tests

Lemma 10 (Adaptive Key-Test Bound). *Consider a TW128 random oracle model game with a hidden key K sampled uniformly from $\{0, 1\}^k$, direct adversary access to RO_{flat} , and an immediate abort event bad_{key} that fires on a direct query (Y, ℓ) exactly when Y decodes under KeyedTWOut to the candidate key K and a counted output-point class of Table 11. Suppose that, before the abort, the following invariant holds for every direct query position i : after conditioning on the full visible execution prefix and on the set S_i of distinct decoded candidate keys from earlier direct queries, the hidden K is uniform on $\{0, 1\}^k \setminus S_i$. Then*

$$\Pr[\text{bad}_{\text{key}}] \leq \text{KeyGuess}_k(q_{\text{RO}}) = q_{\text{RO}}/2^k,$$

2233 where q_{RO} is the execution-uniform count of direct RO_{flat} queries.

2234 *Proof.* If $q_{\text{RO}} \geq 2^k$ the right-hand side $q_{\text{RO}}/2^k$ is at least one, so the claim is trivial. Assume
 2235 $q_{\text{RO}} < 2^k$.

2236 Order the direct RO_{flat} queries. By the definition of **KeyedTWOout** and Lemma 8, each
 2237 query input either decodes to $\perp$ or decodes to a unique candidate key and counted output-
 2238 point class. Queries that decode to $\perp$, and repeated queries whose decoded candidate key
 2239 already lies in the current set S_i , do not change the set of possible first successful tests.
 2240 Removing them can only shorten the sequence, so it suffices to analyze the subsequence of
 2241 distinct fresh decoded candidate keys $K'_1, \dots, K'_m$, where $m \leq q_{\text{RO}}$.

2242 Let F_i be the event that the first i fresh decoded candidate keys all miss the hidden K .
 2243 The invariant says that, conditioned on the visible prefix before the $(i+1)$ -st fresh test
 2244 and on F_i , the hidden K is uniform over the $2^k - i$ values not yet tested. The next fresh
 2245 candidate key K'_{i+1} is one of those remaining values and is determined by the adversary's
 2246 next direct query, so

$$2247 \Pr[K = K'_{i+1} \mid F_i] = \frac{1}{2^k - i}, \quad \Pr[F_{i+1} \mid F_i] = 1 - \frac{1}{2^k - i}.$$

2248 Multiplying the miss probabilities gives

$$2249 \Pr[F_m] = \prod_{i=0}^{m-1} \left(1 - \frac{1}{2^k - i}\right) = \prod_{i=0}^{m-1} \frac{2^k - i - 1}{2^k - i} = \frac{2^k - m}{2^k}.$$

2250 Thus the probability that some fresh decoded candidate key equals the hidden K is
 2251 $m/2^k \leq q_{\text{RO}}/2^k$. This event is exactly the first bad_{key} abort, because **KeyedTWOout** returns
 2252 either $\perp$ or a unique candidate key. $\square$

2253 A.7 Leaf Tag Collision Bound

2254 **Lemma 11** (Leaf Tag Collision Bound). *Consider any TW128 random oracle model game*
 2255 *with an adversary $\mathcal{B}$ that has direct access to RO_{flat} and construction computations that*
 2256 *may generate final leaf transcripts—by encryption, plaintext-computing, or verification*
 2257 *computations, whether they accept or reject. Let bad_{leaf} be the event that two distinct*
 2258 *construction-generated final leaf transcript prefixes $X \neq X'$ in leaf tag output-point classes*
 2259 *receive the same τ_{leaf} -bit projection from $\text{RF}(\cdot)$, and let $n_{\text{leaf}}(\mathcal{B})$ be the execution-uniform*
 2260 *count of distinct construction-generated final leaf transcripts (Section 4.3). The first bound*
 2261 *applies the collision case ($s = 2$, $\tau = \tau_{\text{leaf}}$) of Lemma 2 to a chronological leaf tag candidate*
 2262 *sequence; the second is a same-slot freshness argument in an all-reject game.*

- 2263 1. General count. *Unconditionally—in particular when some or all keys are adversary-*
 2264 *supplied, so direct queries may themselves land on leaf tag output points—*

$$2265 \Pr[\text{bad}_{\text{leaf}}] \leq \text{LeafColl}_{\tau_{\text{leaf}}}(q_{\text{RO}}(\mathcal{B}) + n_{\text{leaf}}(\mathcal{B})) = \binom{q_{\text{RO}}(\mathcal{B}) + n_{\text{leaf}}(\mathcal{B})}{2} / 2^{\tau_{\text{leaf}}}.$$

- 2266 2. Hidden key, same slot. *Suppose the game has a hidden key, nonce-respecting en-*
 2267 *ryption queries, and all-reject verification: every non-replay verification submission*
 2268 *performs its construction computation and returns reject to the adversary. (The*
 2269 *game G_d^{rj} of Definition 5 and the INT-RUP continuation in the proof of Lemma 16*
 2270 *have this shape.) Say two final leaf transcripts share a slot if they recover the same*
 2271 *key, nonce, and leaf index (K, U, j) (Corollary 8). Let bad_{leaf} be the event that some*
 2272 *construction-generated final leaf transcript receives the same τ_{leaf} -bit projection as*
 2273 *a distinct final leaf transcript that shares its slot and is reached by an encryption*
 2274 *computation, and let hit_{leaf} be the event that a direct adversary query hits the bridged*

2275 *input of a construction-generated hidden-key final leaf transcript before that transcript*
 2276 *is first created. Then, in such a game,*

$$2277 \quad \Pr[\text{bad}_{\text{leaf}} \text{ before hit}_{\text{leaf}}] \leq \frac{n_{\text{leaf}}(\mathcal{B})}{2^{\tau_{\text{leaf}}}}.$$

2278 *Proof.* Build a chronological leaf tag candidate sequence: a candidate is the bridged input
 2279 of a leaf tag output point ($\mathcal{L}_{\text{tag}}(j)$), appended the first time it is reached—by a direct query
 2280 in $\mathcal{B}$'s oracle phase or by a construction computation generating a final leaf transcript—and
 2281 not appended again. By Lemma 8, Corollary 8, and Lemma 6, two reaches of the same
 2282 bridged input come from the same final leaf transcript, so repeated evaluations add no
 2283 candidate—the lazy RO_{flat} value is reused—and distinct final leaf transcripts give distinct
 2284 candidates. No candidate's RO_{flat} value is read before its append step, so the sequence meets
 2285 the predictability and unrevealed-value premises of Lemma 2, whose collision case with
 2286 $s = 2$ and $\tau = \tau_{\text{leaf}}$ bounds the probability that two candidates share a τ_{leaf} -bit projection
 2287 by $\binom{L}{2} \cdot 2^{-\tau_{\text{leaf}}}$ for a length- L sequence. Every bad_{leaf} pair is a pair of construction-generated
 2288 candidates.

2289 (1) *General count.* Direct queries may decode to a leaf tag class $\mathcal{L}_{\text{tag}}(j)$ and are
 2290 appended as candidates; together with the $n_{\text{leaf}}(\mathcal{B})$ construction-generated transcripts the
 2291 sequence has length at most $q_{\text{RO}}(\mathcal{B}) + n_{\text{leaf}}(\mathcal{B})$, so the case with $L = q_{\text{RO}}(\mathcal{B}) + n_{\text{leaf}}(\mathcal{B})$ gives
 2292 $\Pr[\text{bad}_{\text{leaf}}] \leq \binom{q_{\text{RO}}(\mathcal{B}) + n_{\text{leaf}}(\mathcal{B})}{2} \cdot 2^{-\tau_{\text{leaf}}} = \text{LeafColl}_{\tau_{\text{leaf}}}(q_{\text{RO}}(\mathcal{B}) + n_{\text{leaf}}(\mathcal{B}))$. The count charges
 2293 every direct query and every construction-generated final leaf transcript regardless of key
 2294 provenance, so no assumption on the keys is used.

2295 (2) *Hidden key, same slot.* Every bad_{leaf} pair is a pair of construction-generated
 2296 hidden-key transcripts sharing a slot, with at least one member reached by an encryption
 2297 computation. Nonce-respecting encryption uses each nonce in at most one encryption
 2298 query, and within one encryption computation the leaf index separates final leaf transcripts,
 2299 so each slot contains at most one encryption-reached final leaf transcript. That transcript
 2300 may have first been created by an earlier verification or plaintext-computing computation,
 2301 when the encryption's absorbed leaf body reproduces an earlier absorbed body. Charge
 2302 each possible bad_{leaf} pair to its member that is not reached by an encryption computation.
 2303 Each charged transcript has one slot, and that slot has at most one encryption-reached
 2304 counterpart, so each charged transcript receives at most one charge. Thus at most $n_{\text{leaf}}(\mathcal{B})$
 2305 charged pairs can occur.

2306 Each pair's collision is decided at the first creation of its second-created member. At
 2307 that moment the earlier member's τ_{leaf} -bit projection is a function of the prior execution,
 2308 while the later member's projection is a fresh uniform lazy sample: by Transcript Injectivity
 2309 (Lemma 6) two construction reaches of the same bridged input come from the same final
 2310 leaf transcript, so no construction computation has read that bridged input before the
 2311 creation, and a direct query reading it before the creation would decode under KeyedTWOut
 2312 to the hidden key and its $\mathcal{L}_{\text{tag}}(j)$ class (Lemma 8)—precisely hit_{leaf} . Hence before hit_{leaf} ,
 2313 conditioned on the entire prior execution, each pair collides on its τ_{leaf} -bit projections
 2314 with probability $2^{-\tau_{\text{leaf}}}$, and a union bound over the at most $n_{\text{leaf}}(\mathcal{B})$ pairs gives the claim.
 2315 No assumption on what the visible interface exposes is needed: a projection revealed
 2316 after its sampling cannot alter a collision already decided. By Leaf-Transcript Recovery
 2317 (Lemma 7), on $\neg \text{bad}_{\text{leaf}}$ equal flattened final-root inputs of an encryption computation and
 2318 any construction computation identify the same leaf transcript sequence. $\square$

2319 A.8 Encryption Transcript Marginal Uniformity

2320 **Lemma 12** (Encryption Transcript Marginal Uniformity). *For any fixed nonce-respecting*
 2321 *encryption query in the TW128 random oracle experiment, let $\text{hit}_{\text{fresh}}$ be the event that,*
 2322 *before this query reaches some counted transcript point, a prior direct adversary query*

2323 to RO_{flat} has already hit the bridged input of that hidden-key point. On $\neg\text{hit}_{\text{fresh}}$, every
 2324 transcript output of the query that contributes to a returned ciphertext block, to an internal
 2325 leaf tag, or to the final root tag is either fresh when this query first generates its bridged
 2326 input, or reuses the lazy-table value sampled uniformly when that input was first generated
 2327 by an earlier construction computation. The counted output points first generated by the
 2328 fixed query are pairwise distinct.

2329 *Proof.* Work on an execution outside $\text{hit}_{\text{fresh}}$ for the fixed query. This rules out prior
 2330 adversary reads of the hidden-key construction inputs considered below. A previous
 2331 construction computation may nevertheless have generated one of these inputs, for example
 2332 through a non-replay verification query using the same nonce. Such a lookup samples the
 2333 lazy-table value uniformly when it first occurs; if the fixed encryption query later reaches
 2334 the same bridged input, it reuses that same value. Thus it remains only to show that the
 2335 counted output points newly generated by the fixed encryption query are distinct from one
 2336 another and from previous nonce-respecting encryption queries. Conditional information
 2337 leaked by earlier verification comparisons of a final root tag is accounted for separately by
 2338 Lemma 15, not by this lemma.

2339 **First root keystream block.** The first root keystream block, when present, is produced
 2340 by the root duplex after the initialization call $p_{\text{root}} \parallel K \parallel U$ and any AD absorption (elided
 2341 when $A = \varepsilon$, in which case the σ_{init} call itself emits the first root keystream block). Since the
 2342 nonce-respecting adversary never repeats a nonce, no previous encryption query can have
 2343 used the same root initialization transcript. Within the current query, Lemma 6 separates
 2344 initialization, AD, message, and aggregation calls by their suffixes and by the duplex-call
 2345 encoding, so the first root keystream transcript is distinct from the other counted output
 2346 points generated by this query. If the requested keystream length is zero, no ciphertext
 2347 block is returned at that point and there is nothing to prove for that output.

2348 **First leaf keystream block.** For each leaf chunk $j \geq 1$, the first leaf keystream block is
 2349 produced after the initialization call $p_{\text{leaf}} \parallel K \parallel U \parallel \langle j \rangle_8$. The prefix separates leaves from
 2350 the root, the nonce separates this encryption query from all previous encryption queries,
 2351 and $\langle j \rangle_8$ separates leaves within this query. Hence every first leaf keystream transcript
 2352 newly generated by the query is distinct.

2353 **Later message outputs.** Argue inductively within each root or leaf message stream.
 2354 Suppose the current message-output transcript is either first generated by this query or
 2355 reuses a value generated by an earlier construction computation. If its output is used as a
 2356 keystream block, write its RO_{flat} value as Z_i . Since the adaptive plaintext block M_i is a
 2357 deterministic function of the prior view,

$$2358 \quad C_i = M_i \oplus Z_i$$

2359 is now fixed. TW128 then absorbs C_i with the appropriate message phase suffix (σ_{msg}^- or
 2360 σ_{msg}^+) before requesting the next keystream block. Once C_i is fixed, the next transcript is
 2361 fixed; Lemma 6's injectivity and domain separation imply it has not appeared earlier unless
 2362 the same duplex-call prefix has appeared earlier. Nonce uniqueness excludes this across
 2363 encryption queries. By Corollary 8, the next message-keystream output point can coincide
 2364 with a within-query earlier output point only if both belong to the same output-point
 2365 class on the same duplex instance (root or leaf- j with matching j) and have the same
 2366 duplex-call sequence; the current message-stream prefix is strictly longer than every earlier
 2367 prefix in that stream, so this is excluded. Across distinct instances within the same query
 2368 (root vs leaf, or two leaves with distinct j), the prefixes already differ in the first call's σ by
 2369 $p_{\text{root}}/p_{\text{leaf}}$ or $\langle j \rangle_8$ respectively. Each later message-output transcript that emits keystream

is therefore distinct from the other points generated by this query, unless it is a reuse of a point generated by an earlier construction computation. The terminal message-output transcript of a leaf emits the leaf tag rather than a following keystream block; the same separation argument applies to that transcript, so each internal leaf tag newly generated by the query is sampled at a distinct point. Empty inputs still induce a duplex call, but with zero keystream output they contribute no ciphertext block.

Final root tag transcript. The final root tag is sampled at the root tag output point selected by $\|C\|$: the $\mathcal{R}_{\text{tag}}$ point after the root duplex absorbs the aggregation transcript built from the leaf tags and $\langle n \rangle_8$ when $n \geq 1$, and the $\mathcal{R}_{\text{msg}}^+$ point closing the root message phase when $n = 0$ (Section 3.4). When $n \geq 1$, the leaf tags are themselves RO_{flat} outputs on final leaf transcripts; condition on their sampled values, on the sampled ciphertext, and on the adversary-visible prior view and RO_{flat} table state, so that the root aggregation transcript is fixed. When $n = 0$ there is no aggregation transcript; condition on the sampled ciphertext and the adversary-visible prior view alone. By Corollary 8, the selected root tag output-point class is separated from every other output-point class in the same query — including root initialization (suffix σ_{init}), AD-phase outputs, the remaining message phase outputs, leaf-side output points (separated by p_{leaf}), and, in the $n \geq 1$ case, earlier $\mathcal{R}_{\text{agg}}$ -prefix output points (separated by the σ_{agg}^- vs σ_{agg}^+ suffix). Even if two encryption queries realize the same aggregation transcript (or, for $n = 0$, the same root ciphertext), the full root transcript also contains the root initialization $p_{\text{root}} \| K \| U$, and nonce uniqueness separates that initialization from all previous encryption queries. The final root tag transcript is therefore either newly generated at a distinct point of the fixed query, or is a reuse of a value first sampled by an earlier construction computation. In the newly generated case, its τ_{root} -bit RO_{flat} output is a fresh lazy-table sample; in the reuse case, it is the same uniform sample drawn when the bridged input was first generated. $\square$

A.9 Mu-Ind Hybrid Games and Verification Accounting

The direct mu-ind adjacent-hybrid proof bounds one user-replacement step of the hybrid through an explicit game sequence, defined next, using the generic key-test and leaf-collision accounting lemmas together with the verification-specific facts below. We use the root tag output point and final root transcript classes of Definition 3; in particular, computations on a common ciphertext share the same root tag output point because $\text{leaves}(C)$ is fixed by $\|C\|$.

Definition 5 (Adjacent-Hybrid Games). Fix a flat-RO mu-ind adversary $\mathcal{B}$, the pre-sampled keys $K_1, \dots, K_D$ of Section 6.2, and a user index $d \in \{1, \dots, D\}$. On the branch where two sampled keys collide, every game below outputs the same fixed bit; the remaining clauses describe the branch with pairwise distinct keys. Every game maintains one lazy table Θ implementing RO_{flat} : $\Theta[Y]$ samples a fresh uniform infinite string at the first access to Y and returns the stored string afterwards—equivalently, Θ is sampled in full at the outset and read on demand. Direct adversary queries (Y, ℓ) are answered by $\lfloor \Theta[Y] \rfloor_\ell$; the *environment* answers users $e < d$ by (Rand, Replay) and users $e > d$ by $(\text{Enc}_{K_e}^{\text{RO}}, \text{Ver}_{K_e}^{\text{RO}})$, reading construction outputs from Θ ; and the New interface and the validity and nonce-respecting checks of Section 6.2 are common to every game. Every game maintains the flags bad_{key} , bad_{leaf} , and forge , initialized false; no oracle answer or branch depends on a flag, and bad_{key} is set whenever a direct query input decodes under KeyedTWOut to K_d and a counted output-point class of Table 11.

- Games $\mathcal{G}_d^{\text{re}}$ and $\mathcal{G}_d^{\text{rj}}$ answer user d by the shared oracle code of Algorithm 4 and differ exactly in the marked return statement: a non-replay verification whose comparison succeeds sets forge and returns accept in $\mathcal{G}_d^{\text{re}}$ and reject in $\mathcal{G}_d^{\text{rj}}$.

Algorithm 4 Shared user- d oracle code of the games G_d^{re} and G_d^{rj} (Definition 5). Θ is the shared lazy RO_{flat} table, read at bridged transcript prefixes, and W the user- d replay record. Inputs failing the validity or nonce-respecting checks of Section 6.2 are rejected beforehand, as in every game. The two games differ exactly in the marked lines 17 and 18.

```

1: procedure  $\text{RO}(Y, \ell)$  ▷ direct adversary query
2:   set  $\text{bad}_{\text{key}}$  if  $\text{KeyedTWOut}(Y) = (K_d, \mathcal{C})$  for a counted class  $\mathcal{C}$  (Table 11)
3:   return  $[\Theta[Y]]_\ell$ 
4: end procedure

5: procedure  $\text{ENC}(U, A, M)$ 
6:   run  $\text{Enc}_{K_d}^{\text{RO}}(U, A, M)$ , reading each requested duplex output as the corresponding
   projection of  $\Theta$  at its bridged transcript prefix (Section 4.2); let the result be  $C \parallel T$ 
7:   set  $\text{bad}_{\text{leaf}}$  if a final leaf transcript first created or first reached here completes a
   same-slot collision (Definition 5)
8:   record  $(U, A, C \parallel T)$  in  $W$ ; return  $C \parallel T$ 
9: end procedure

10: procedure  $\text{VF}(U, A, C \parallel T)$ 
11:   if  $(U, A, C \parallel T) \in W$  then return accept
12:   end if
13:   determine the verification transcript of  $(K_d, U, A, C)$ ; read from  $\Theta$  the  $\tau_{\text{leaf}}$ -bit leaf
   tags at its final leaf transcripts, setting  $\text{bad}_{\text{leaf}}$  when a transcript first created here
   completes a same-slot collision, and nothing at its stream output points
14:    $T' \leftarrow [\Theta[\text{flat}(R)]]_{\tau_{\text{root}}}$  at the root tag output point  $R$  of the computation (Defini-
   tion 3)
15:   if  $T' = T$  then
16:      $\text{forge} \leftarrow \text{true}$ 
17:     return accept ▷ in  $G_d^{\text{re}}$  only
18:     return reject ▷ in  $G_d^{\text{rj}}$  only
19:   end if
20:   return reject
21: end procedure

```

- Game G_d^{id} answers user- d encryption queries with fresh uniform byte strings of length $\|M\| + \tau_{\text{root}}/8$, recorded in W , and user- d verification by the replay check alone; it performs no user- d construction computation and sets neither bad_{leaf} nor forge . G_d^{id} is the hybrid H_d^* of Lemma 17, instrumented with bad_{key} .

The flag bad_{leaf} is set, in G_d^{re} and G_d^{rj} , when a user- d construction computation first creates, or an encryption computation first reaches, a final leaf transcript whose τ_{leaf} -bit projection equals that of a distinct, already created final leaf transcript sharing its (K_d, U, j) slot, at least one of the two reached by an encryption computation—the same-slot event of Lemma 11, part 2, for user d . It guards no code difference and is used only in the accounting.

Lemma 13 (Environment Neutrality). *In each game of Definition 5, on the branch with pairwise distinct keys, the environment’s construction-internal RO_{flat} lookups—made while simulating $\text{Enc}_{K_e}^{\text{RO}}$ and $\text{Ver}_{K_e}^{\text{RO}}$ for users $e > d$ —neither set a user- d flag nor generate the bridged input of a user- d counted transcript point.*

Proof. Whenever such a lookup lies in a counted keyed transcript class, Lemma 8 decodes its candidate key as K_e , which on the distinct-keys branch differs from K_d ; a lookup

outside the counted keyed language is irrelevant to the flags. By the key-field disjointness of Lemma 9, the environment's addresses are disjoint from user d 's. The flags are set only by direct adversary queries decoding to K_d (bad_{key}), by user- d final leaf transcript creations (bad_{leaf}), and by user- d verification comparisons (forge), none of which the environment performs. Ideal users $e < d$ perform no construction lookups at all. $\square$

Lemma 14 (Verification Visible-Prefix Key Uniformity). *In the stopped single-key TW128 random oracle encryption/verification interaction, or its INT-RUP extension with a plaintext-computing oracle, with hidden key K , direct access to RO_{flat} , and immediate abort on bad_{key} , order the direct RO_{flat} queries. For a direct query position i , let S_i be the set of distinct candidate keys decoded by KeyedTWOut from earlier direct-query inputs. Condition on any positive-probability visible execution prefix immediately before query i , on no earlier bad_{key} , and on S_i . Then the hidden key K is uniform on $\{0, 1\}^k \setminus S_i$. Consequently the uniformity invariant required by Lemma 10 holds in the stopped interaction.*

Proof. The visible prefix contains the adversary's oracle queries and replies: direct RO_{flat} outputs, encryption outputs and the resulting replay table, plaintext-computing replies when that oracle is present, verification bits, and the adversary state determined by these values. Fix such a prefix and two keys $K_0, K_1 \notin S_i$.

Let $\Phi_{0 \rightarrow 1} = \Phi_{K_0 \rightarrow K_1}$ be the key-renaming map of Lemma 9 and $\Phi_{1 \rightarrow 0}$ its inverse. By that lemma, $\text{flat} \circ \Phi_{0 \rightarrow 1} \circ \text{flat}^{-1}$ is a bijection $\mathcal{Y}_{K_0} \rightarrow \mathcal{Y}_{K_1}$ between the bridged hidden-key construction addresses for K_0 and for K_1 , preserving output-point classes and equality relations among addresses.

This bijection is applied online, not to a pre-existing fixed address set. Whenever a construction computation under K_0 would first query a hidden-key address Y , the coupled computation under K_1 queries $\Phi_{0 \rightarrow 1}(Y)$ and receives the same lazy-table value; repeated queries are coupled consistently through the same map. If the queried address is a root aggregation address containing leaf tag bytes, those bytes are already coupled equal because the corresponding leaf tag lookups were either repeated or first sampled earlier under the same renaming rule. Thus adaptively generated message transcripts and root aggregation transcripts are renamed step by step with the same visible ciphertext, leaf tag, and root tag bytes.

Prior direct-query inputs are not renamed: they are adversary-chosen visible strings. Since $K_0, K_1 \notin S_i$, neither $\mathcal{Y}_{K_0}$ nor $\mathcal{Y}_{K_1}$ contains a prior direct-query input (Lemma 9, direct-query avoidance). Direct-query lazy-table values are therefore coupled identically and remain disjoint from the renamed hidden construction addresses.

Use the same adversary coins, the same direct-query lazy-table values, and the online renamed construction-internal lazy-table values. Encryption outputs have the same distribution under the coupling, because every construction lookup that determines ciphertext, leaf tags, or the final root tag is paired with a distinct renamed lookup carrying the same sampled value. Plaintext-computing computations are renamed in the same online way: a plaintext reply is the submitted ciphertext block XORed with a coupled keystream projection at a stream output point (Table 11), so it carries the same visible bytes under both keys, and the recomputed tag is not returned. Replay verification decisions depend only on the visible replay table. Non-replay verification computations are renamed in the same online way as encryption computations, and the final root tag comparison therefore gives the same accept/reject bit. Verification does not reveal plaintext on rejection or acceptance; it reveals only that bit.

Thus every visible prefix with hidden key $K_0 \notin S_i$ has the same conditional probability as the corresponding execution with hidden key $K_1 \notin S_i$. Since K is initially uniform and the condition “no earlier bad_{key} ” removes exactly the keys in S_i , all keys in $\{0, 1\}^k \setminus S_i$ remain equally likely. $\square$

Foreign-key key-test claim. In each game of Definition 5, work in the original experiment, where the key-collision branch outputs a fixed bit rather than conditioning on pairwise distinct keys. Then

$$\Pr[\text{bad}_{\text{key}}] \leq \text{KeyGuess}_k(q_{\text{RO}}(\mathcal{B})).$$

Fix the foreign keys and leave their transcript addresses unchanged. Before the first user- d key hit, the visible prefix is invariant under renaming K_d among keys that have not already been tested and are not equal to a fixed foreign key, by the same online coupling as Lemma 14 and the key-field disjointness of Lemma 9. For each fresh candidate key tested by one of $\mathcal{B}$'s direct RO_{flat} queries, the contributing event is contained in $\{K_d \text{ equals that candidate}\}$ and costs at most 2^{-k} in the original pre-sampled-key experiment; a candidate equal to a fixed foreign key contributes only on the fixed-output key-collision branch and therefore contributes nothing to bad_{key} . A union bound over at most $q_{\text{RO}}(\mathcal{B})$ direct queries gives the claim.

Lemma 15 (Verification-First Root Guessing). *In the game G_d^j of Definition 5, stopped at the first occurrence of $\text{bad}_{\text{key}} \cup \text{bad}_{\text{leaf}}$ —with bad_{leaf} the same-slot event of Lemma 11, part 2, for user d —let win_{vf} be the event that some non-replay user- d verification comparison succeeds within a verification-first final root transcript class. Then*

$$\Pr[\text{win}_{\text{vf}}] \leq q_v^{(d)} / 2^{\tau_{\text{root}}},$$

where $q_v^{(d)}$ is the user- d non-replay verification cap of Section 6.2.

Proof. Throughout, a *non-replay verification submission* means a valid non-replay user- d verification query; by the validity convention, an invalid verification query returns 0 and performs no construction computation, so it realizes no bridged final root input, belongs to no final root transcript class, and contributes to no G_C below.

In G_d^j , every non-replay submission performs its construction computation and returns reject (Algorithm 4), so no comparison outcome is visible to the adversary, and each class's root tag value is read from the lazy table only by the class's own comparisons and by any later encryption computation reaching the class.

Fix a verification-first final root transcript class $\mathcal{C}$ and follow the stopped execution until the first encryption query generating the same class, if any (the cutoff). Let G_C be the set of distinct candidate tags submitted by non-replay verification queries in class $\mathcal{C}$ before that cutoff; repeated tags only shrink this set. The class's correct root tag U_C is the τ_{root} -bit projection of RO_{flat} at the class's bridged root tag input. Before the stop, no direct query has read this hidden-key input: such a query would be decoded by KeyedTWOOut to the hidden key and would set bad_{key} ; the environment reads no user- d address (Lemma 13). Before the cutoff, no encryption query has revealed U_C as an encryption tag. (When the class's root tag point is $\mathcal{R}_{\text{msg}}^+$, i.e. $n = 0$, an encryption query on a longer message that shares chunk 0 may touch this same address while closing its chunk-0 message phase, but it requests zero output there and continues into its aggregation phase, so it does not reveal U_C ; its own tag is emitted at a later $\mathcal{R}_{\text{tag}}$ point.) The comparisons against U_C return the constant reject, so the visible execution and the set G_C are functions only of the adversary's coins, visible oracle replies, and lazy-table values at inputs other than this class's root tag address. By the lazy-sampling order exchange, conditioned on those values, U_C may equivalently be sampled after G_C is fixed, and is uniform independently of G_C . The probability that some pre-cutoff comparison in class $\mathcal{C}$ succeeds—that $U_C \in G_C$ —is therefore at most $|G_C|/2^{\tau_{\text{root}}}$.

After the cutoff, if it exists, class $\mathcal{C}$ admits no successful non-replay comparison. Let the cutoff encryption query return (U_e, A_e, C_e, T_e) . Any later verification submission in the same final root transcript class has, by the final-root determinacy of Lemma 6 and Leaf-Transcript Recovery (Lemma 7) under the absence of bad_{leaf} —which supplies the

recovery hypothesis because the cutoff computation is an encryption computation—the same (U, A, C) as that encryption computation. If the submitted tag is T_e , the tuple is a recorded encryption tuple and the query is answered by the replay check before any comparison; if the submitted tag is not T_e , the comparison fails.

Each non-replay verification submission deterministically reconstructs one final root transcript at the root tag output point, so it contributes its submitted tag to at most one set G_C , and only when its class is verification-first and the submission precedes that class's cutoff. Thus $\sum_C |G_C| \leq q_v^{(d)}$, and a union bound over the verification-first classes gives the claim. $\square$

Lemma 16 (INT-RUP Root Guessing). *Let $\mathcal{B}$ be a single-key INT-RUP adversary in the TW128 random oracle model, with nonce-respecting encryption queries and the resources of Section 4.3. In its encryption/plaintext-computing/verification interaction with hidden key K , let bad_{key} be the event that some direct RO_{flat} query input Y satisfies $\text{KeyedTWOOut}(Y) = (K, C)$ for a counted class C of Table 11, and let bad_{leaf} be the same-slot event of Lemma 11, part 2, over the construction computations generated by encryption, plaintext-computing, and verification. Let forge be the event that some valid non-replay verification query accepts. Then*

$$\Pr[\text{forge before } \text{bad}_{\text{key}} \cup \text{bad}_{\text{leaf}}] \leq q_v(\mathcal{B})/2^{\tau_{\text{root}}}.$$

Proof. Throughout, a *non-replay verification submission* means a valid non-replay verification query; invalid verification queries return 0, perform no construction computation, realize no bridged final root input, and contribute to no set of guessed tags.

Work in the game stopped at the first occurrence of bad_{key} or bad_{leaf} . A final root transcript class is *tag-revealed* once an encryption query has returned the root tag for that class. If a non-replay verification submission belongs to a tag-revealed class, compare it with an encryption computation that revealed the class. The two computations' final-root transcript prefixes are well-formed typed transcript prefixes with the same bridged final root input; by Lemma 8, they have the same native flattened input. Then final-root determinacy (Lemma 6) and Leaf-Transcript Recovery (Lemma 7) under $\neg \text{bad}_{\text{leaf}}$ —which supplies the recovery hypothesis because the comparand is an encryption computation—recover the same (U, A, C) . If the submitted tag is the revealed tag, the tuple is a replay-table hit; if it is any other tag, the final comparison rejects. Thus any accepting non-replay verification submission must lie in a class whose root tag has not yet been revealed by encryption.

For tag-unrevealed classes, define an all-reject continuation: encryption and plaintext-computing queries are answered normally, replay verification queries accept normally, and each non-replay verification submission performs the same construction computation but returns reject to the adversary. Plaintext-computing queries may generate the same stream and leaf tag transcripts as decryption, but they do not reveal the final root tag value. Equivalently, the final root tag sample for a tag-unrevealed class can be deferred until it is first needed for an encryption answer or a verification comparison; this does not change any plaintext reply. By Output-Point Separation (Corollary 8) and the keystream schedule recorded at Table 11, the keystream values used to compute returned plaintext blocks are projections at stream output points, not at the final root tag output point; in particular the chunk-0 $\mathcal{R}_{\text{msg}}^+$ call of an $n \geq 1$ computation requests zero output, so no plaintext reply exposes an $\mathcal{R}_{\text{msg}}^+$ tag value.

Fix a final root transcript class C whose first construction generation is not by an encryption query; for a class first generated by an encryption query the tag is revealed from that generation onward, so every verification submission in it is covered by the previous paragraph. Follow the all-reject path until the first encryption query that reveals the class, if any (the cutoff; any such query is later than the class's first generation). Let G_C be the set of distinct tags submitted by non-replay verification queries in class C before that cutoff. Before the cutoff, no encryption query has returned the class tag and no direct

query has read the hidden-key root tag point without triggering bad_{key} . Conditioned on the all-reject path and on all lazy-table values except this final root tag projection, the correct tag for $\mathcal{C}$ is uniform in $\{0, 1\}^{\tau_{\text{root}}}$ and independent of $G_{\mathcal{C}}$. The probability that any pre-cutoff submitted tag in $G_{\mathcal{C}}$ equals the deferred class tag is therefore at most $|G_{\mathcal{C}}|/2^{\tau_{\text{root}}}$.

Consider the event that the first accepting non-replay verification submission in the stopped game belongs to class $\mathcal{C}$. Up to that global first accept, the adversary's visible answers match the all-reject continuation: every earlier non-replay submission rejects, replay and plaintext-computing queries are answered identically in both executions, and encryption answers are unchanged. The construction computations also agree at every address except that the continuation defers the final root tag value for class $\mathcal{C}$ until the comparison is needed. Therefore the first accepting non-replay submission can belong to class $\mathcal{C}$ only before the cutoff, and only if its submitted tag equals the deferred class tag, which has probability at most $|G_{\mathcal{C}}|/2^{\tau_{\text{root}}}$.

The execution-uniform count $q_{\mathcal{V}}(\mathcal{B})$ applies to the all-reject continuation. Each valid non-replay verification query contributes its submitted tag to at most one such set $G_{\mathcal{C}}$, and replay-table hits do not contribute. Thus $\sum_{\mathcal{C}} |G_{\mathcal{C}}| \leq q_{\mathcal{V}}(\mathcal{B})$, and a union bound over the classes not first generated by an encryption query, each with its pre-cutoff guess set $G_{\mathcal{C}}$, gives the claim. $\square$

A.10 Mu-Ind Adjacent-Hybrid Lemma

The multi-user indistinguishability theorem (Theorem 9) telescopes a hybrid over the D users, replacing one real user by its ideal (Rand, Replay) interface at each step. The following lemma bounds one such step through the game sequence of Definition 5: the instrumented real game G_d^{re} , the all-reject game G_d^{rj} differing from it in a single return statement, and the ideal game G_d^{id} , with the divergence events set as the flags bad_{key} , bad_{leaf} , and forge by game code.

Lemma 17 (Mu-Ind Adjacent-Hybrid Bound). *Let $\mathcal{B}$ be a flat-RO mu-ind adversary in the pre-sampled-key formulation of Section 6.2, with keys $K_1, \dots, K_D$ sampled at the start of the experiment. Fix $d \in \{1, \dots, D\}$. Let H_{d-1}^* and H_d^* be the adjacent stopped hybrids that output a fixed bit if any two sampled keys collide, and otherwise run $\mathcal{B}$ with one shared lazy RO_{flat} table as follows: users $e < d$ are answered by (Rand, Replay), users $e > d$ are answered by $(\text{Enc}_{K_e}^{\text{RO}}, \text{Ver}_{K_e}^{\text{RO}})$, and user d is answered by $(\text{Enc}_{K_d}^{\text{RO}}, \text{Ver}_{K_d}^{\text{RO}})$ in H_{d-1}^* and by (Rand, Replay) in H_d^* . If $n_{\text{leaf}}^{(d)}$ and $q_{\mathcal{V}}^{(d)}$ are the user- d resource caps, then*

$$|\Pr[\mathcal{B}^{H_{d-1}^*} \Rightarrow 1] - \Pr[\mathcal{B}^{H_d^*} \Rightarrow 1]| \leq \text{KeyGuess}_k(q_{\text{RO}}(\mathcal{B})) + \frac{n_{\text{leaf}}^{(d)}}{2^{\tau_{\text{leaf}}}} + \frac{q_{\mathcal{V}}^{(d)}}{2^{\tau_{\text{root}}}}.$$

Proof. The fixed-output branch on a key collision is identical in H_{d-1}^* , H_d^* , and the games of Definition 5, so only the branch with pairwise distinct keys contributes; work on that branch, with the keys fixed, throughout. The proof identifies H_{d-1}^* with G_d^{re} (Claim 1), relates G_d^{re} to G_d^{rj} pointwise (Claim 2) and G_d^{rj} to $H_d^* = G_d^{\text{id}}$ in distribution (Claim 3), and assembles the bound by a single first-occurrence decomposition in G_d^{rj} (Claim 4).

Claim 1: G_d^{re} realizes H_{d-1}^* . The user- d oracles of Algorithm 4 implement $(\text{Enc}_{K_d}^{\text{RO}}, \text{Ver}_{K_d}^{\text{RO}})$ with three distribution-preserving refactors. First, reading every construction output as a projection of the shared table Θ at its bridged transcript prefix is the lazy implementation of RO_{flat} (Section 4.2); whether Θ is sampled in full at the outset or entry by entry on first access is immaterial. Second, VF reads Θ only at leaf tag and root tag output points. This is sound because decryption absorbs the submitted ciphertext, so the verification transcript is determined by (K_d, U, A, C) independently of any stream output (Lemma 5, Section 3.5), and verification discards the recovered plaintext, so the keystream projections a real $\text{Ver}_{K_d}^{\text{RO}}$

computation reads influence neither a visible answer nor the value of any other table entry; leaving those entries unread until some later computation first accesses them leaves the joint distribution of all oracle answers unchanged. Third, the flags are instrumentation: no oracle answer or branch depends on them. Hence $\Pr[\mathcal{B}^{G_d^e} \Rightarrow 1] = \Pr[\mathcal{B}^{H_{d-1}^*} \Rightarrow 1]$.

Claim 2: G_d^e and G_d^j agree until forge. Run the two games on the same adversary coins, environment coins, and table realization Θ . The games share all code, including the comparison that sets `forge`, and differ in exactly one statement, the marked return line of Algorithm 4, which executes only when `forge` has just been set. The two executions are therefore identical—every oracle answer, table access, record in W , and flag update—up to and including the query at which `forge` is first set, and the first differing value is that query’s returned bit. In particular, on executions in which G_d^j never sets `forge`, the two runs coincide, outputs included, and the flags `badkey` and `badleaf` are set at the same queries in both games up to the first `forge`.

Claim 3: G_d^j agrees with H_d^* until `badkey`. Couple $H_d^* = G_d^{\text{id}}$ to G_d^j as follows: run G_d^j on the same adversary and environment coins, give the G_d^{id} coordinate the same visible oracle answers for every query answered while `badkey` is unset, and let the G_d^{id} coordinate continue independently according to its own law from the query that sets `badkey` onward. We claim the resulting G_d^{id} coordinate is distributed exactly as G_d^{id} . The answer rules coincide query by query except for user- d encryption: non-replay user- d verification returns reject in G_d^j and under Replay alike, and replay queries are answered from the record W , which the coupling keeps equal in both coordinates; the environment runs identical code in both games; and direct queries read table entries that agree in both games before `badkey`—an entry written by a user- d construction computation is read by a direct query only if that query decodes under `KeyedTWOut` to K_d and a counted class (Lemma 9), which sets `badkey`, while entries written by the environment are produced identically in both games (Lemma 13).

It remains to show that, before `badkey`, each user- d encryption answer of G_d^j is, conditioned on the entire prior visible transcript, a fresh uniform byte string of length $\|M\| + \tau_{\text{root}}/8$ —the law of `Rand`. Fix such a query (U, A, M) . Before `badkey`, no direct query has read the bridged input of a user- d hidden-key counted point, so the hypothesis $\neg\text{hit}_{\text{fresh}}$ of Encryption Transcript Marginal Uniformity (Lemma 12) holds, and every counted output point first generated by this query is pairwise distinct and receives a fresh uniform sample. The stream output points are always first generated by this query: user- d verification computations read no stream points (Algorithm 4), nonce-respecting encryption and Transcript Injectivity (Lemma 6) separate them from every earlier user- d encryption computation, and the environment writes only foreign-key addresses (Lemma 13). The ciphertext body C is therefore a fresh uniform string. The internal leaf tags are either fresh samples or reuse entries written by user- d verification computations; they are never returned and influence the answer only through the root tag address. The root tag T is read at the computation’s root tag output point (Definition 3), an address distinct from every stream point of the same query by Output-Point Separation (Corollary 8). If that entry is first generated here, T is a fresh uniform sample. Otherwise the entry was written by an earlier user- d verification computation—before `badkey`, the only other writer of user- d addresses—whose reads of it produced only the constant reject answers of G_d^j ; since no oracle answer or branch depends on the flags, the visible transcript is a function of the coins and of table entries other than this one, and conditioned on the visible transcript the entry remains uniform. In either case (C, T) has the `Rand` law, and at most one user- d encryption reaches any root tag address, since two would share the nonce U , contradicting nonce-respecting encryption. By induction over the query sequence, the coupled G_d^{id} coordinate has exactly the law of H_d^* , and its visible transcript agrees with

2680 G_d^{rj} 's while bad_{key} is unset.

2681 **Claim 4: assembly.** Place the three games on one probability space: shared adversary
 2682 and environment coins and one table realization Θ drive G_d^{re} and G_d^{rj} pointwise, and the H_d^*
 2683 coordinate is coupled to G_d^{rj} as in Claim 3. Until the first occurrence of $\text{forge} \vee \text{bad}_{\text{key}}$ in G_d^{rj} ,
 2684 the three visible transcripts coincide: G_d^{re} 's by Claim 2, since forge has not yet been set,
 2685 and H_d^* 's by Claim 3, since bad_{key} has not. When neither flag is ever set, the final outputs
 2686 of $\mathcal{B}$ coincide as well, so the outputs can differ only on the event $\text{forge} \vee \text{bad}_{\text{key}}$. With
 2687 Claim 1 identifying $\Pr[\mathcal{B}^{H_{d-1}^*} \Rightarrow 1] = \Pr[\mathcal{B}^{G_d^{\text{re}}} \Rightarrow 1]$ and Claim 3 identifying $\Pr[\mathcal{B}^{H_d^*} \Rightarrow 1]$
 2688 with the coupled coordinate's output probability,

$$2689 \quad |\Pr[\mathcal{B}^{H_{d-1}^*} \Rightarrow 1] - \Pr[\mathcal{B}^{H_d^*} \Rightarrow 1]| \leq \Pr_{G_d^{\text{rj}}}[\text{forge} \vee \text{bad}_{\text{key}}].$$

2690 Decompose by first occurrence, inserting bad_{leaf} —which guards no code difference and
 2691 enters only this accounting:

$$2692 \quad \Pr_{G_d^{\text{rj}}}[\text{forge} \vee \text{bad}_{\text{key}}] \leq \Pr[\text{bad}_{\text{key}}] + \Pr[\text{bad}_{\text{leaf}} \text{ before } \text{bad}_{\text{key}} \vee \text{forge}] \\ + \Pr[\text{forge} \text{ before } \text{bad}_{\text{key}} \vee \text{bad}_{\text{leaf}}].$$

2693 The three terms are bounded in G_d^{rj} . For the first, the foreign-key key-test claim above gives
 2694 $\Pr[\text{bad}_{\text{key}}] \leq \text{KeyGuess}_k(q_{\text{RO}}(\mathcal{B}))$. For the second, G_d^{rj} restricted to user d is a hidden-key
 2695 game with nonce-respecting encryption and all-reject verification; before bad_{key} no direct
 2696 query has read a user- d hidden final leaf point—such a query would decode to K_d and
 2697 its $\mathcal{L}_{\text{tag}}(j)$ class and set bad_{key} —and the environment creates no user- d leaf transcript
 2698 (Lemma 13), so the same-slot case of Lemma 11 gives $\Pr[\text{bad}_{\text{leaf}} \text{ before } \text{bad}_{\text{key}} \vee \text{forge}] \leq$
 2699 $n_{\text{leaf}}^{(d)}/2^{\tau_{\text{leaf}}}$. For the third, before $\text{bad}_{\text{key}} \vee \text{bad}_{\text{leaf}}$ every successful comparison lies in a
 2700 verification-first class: a class first generated by a user- d encryption computation admits
 2701 none, since by the final-root determinacy of Lemma 6 and Leaf-Transcript Recovery
 2702 (Lemma 7)—whose hypothesis $\neg \text{bad}_{\text{leaf}}$ supplies because the comparand is the class's
 2703 encryption computation—any non-replay submission in the class carries the same (U, A, C)
 2704 as that encryption and either matches the recorded tuple, in which case the replay
 2705 check answers it before any comparison, or fails the comparison. Lemma 15 then gives
 2706 $\Pr[\text{forge} \text{ before } \text{bad}_{\text{key}} \vee \text{bad}_{\text{leaf}}] \leq q_v^{(d)}/2^{\tau_{\text{root}}}$.

2707 Summing the three bounds gives

$$2708 \quad |\Pr[\mathcal{B}^{H_{d-1}^*} \Rightarrow 1] - \Pr[\mathcal{B}^{H_d^*} \Rightarrow 1]| \leq \text{KeyGuess}_k(q_{\text{RO}}(\mathcal{B})) + \frac{n_{\text{leaf}}^{(d)}}{2^{\tau_{\text{leaf}}}} + \frac{q_v^{(d)}}{2^{\tau_{\text{root}}}},$$

2709 with the user- d key-test event bad_{key} charged a single time for the adjacent hybrid. $\square$

2710 B The Flat Sponge Lift

2711 This appendix supplies the flat sponge lift machinery (Lemmas 20 and 22 and Corollary 9)
 2712 invoked by the tag hash bounds of Section 5, the committing consequences of Corollary 5,
 2713 and the baseline AEAD bounds of Section 6. The machinery applies uniformly to the
 2714 games named once here.

2715 The lift has two steps in the win-event games and three in the real-vs-ideal games.
 2716 First, the [BDPVA08] indistinguishability theorem replaces the real permutation used by
 2717 the TW128 construction and by direct primitive queries with the simulator over a flat
 2718 random oracle, at cost $\text{Lift}_c(N_{\text{tot}})$, where $N_{\text{tot}} = N + N_{\text{prim}}$ bounds the operational primitive
 2719 trace: construction-internal duplex calls induced by construction queries and deterministic

challenger computations, together with direct primitive queries. The simulator-side construction trace is then identified with the flat-RO construction interface by the operational trace lemma below. Second, the simulator-world adversary is converted into a flat-RO adversary whose direct random oracle query count is at most N_{prim} . Thus the random oracle bounds are instantiated at $q_{\text{RO}} \leq N_{\text{prim}}$, while the public permutation lift pays $\text{Lift}_c(N + N_{\text{prim}})$. Third, in the real-vs-ideal games, whose ideal world exposes the real permutation rather than the simulator (Section 4.1), the ideal-side simulator introduced by the first step is exchanged back for (π, π^{-1}) at cost $\text{Lift}_c(N_{\text{prim}})$ (Lemma 21); the exchange is possible because the ideal construction oracles are independent of the primitive.

Definition 6 (Single-Stage TW128 Games). The *single-stage TW128 games* are the public-permutation games of Sections 5.1, 6.1 and 6.2 in which one adversary interacts with the challenger in a single stage, so that a [BDPVA08] replacement applies to an entire world’s public-permutation transcript at once, including deterministic challenger checks after the adversary’s final output. They are the H_{TW128} tag hash games (collision and preimage games, standard and everywhere), CMTDs-4, CMTs- ℓ for $\ell \in \{1, 3, 4\}$, INT-RUP, all-in-one AE, and mu-ind. The win-event games among these have one world; the real-vs-ideal games—all-in-one AE and mu-ind—have two, and the lift below treats their real world by the full replacement and their ideal world by the simulator–permutation exchange of Lemma 21.

The derived adversary forwards the following game data to the corresponding flat-RO experiment.

Game family	Forwarded data
H_{TW128} tag hash games	final structured hash game output
CMTDs / CMTs	final openings and deterministic challenger checks
AE / mu-ind / INT-RUP	construction queries, including New where present and replay state

B.1 Simulator Convention

Throughout this appendix, $(\mathcal{S}^{\text{RO}_{\text{flat}}}, \mathcal{S}^{-1, \text{RO}_{\text{flat}}})$ denotes the public primitive simulator supplied by [BDPVA08, Theorem 2] — the indistinguishability theorem for the simple padded sponge over a random permutation — after applying the [BDPVA11, Theorem 3] bridge of Lemma 8 to TW128’s native `pad10*1` transcript inputs. The superscript records that both simulator directions share the same internal lazy RO_{flat} table.

The simulator is a proof device: each public-permutation game of Sections 5.1, 6.1 and 6.2 states its advantage over the real permutation, and the reduction to the corresponding random oracle model adversary passes through intermediate simulator worlds. In those worlds, the construction interface is the one specified by the game $((\text{Rand}, \text{Replay}), \text{instantiated per user in mu-ind, on the ideal side})$ and the public primitive interface is $(\mathcal{S}^{\text{RO}_{\text{flat}}}, \mathcal{S}^{-1, \text{RO}_{\text{flat}}})$. The RO_{flat} table in this notation is internal to that system and is not an extra interface exposed to the application adversary.

Lemma 18 (At Most One Simulator RO Call). *For the simulator $(\mathcal{S}^{\text{RO}_{\text{flat}}}, \mathcal{S}^{-1, \text{RO}_{\text{flat}}})$, every primitive-interface query induces at most one simulator-side RO_{flat} call. More precisely, every inverse primitive-interface query and every forward primitive-interface query violating one of the following conditions induces no simulator RO_{flat} call. The symbols s , p , $\mathcal{R}$, $\mathcal{O}$, and $\mathcal{C}$ are the local internal notation of the BDPV simple-padded sponge simulator, not TW128 resource notation:*

(C1) New edge. *The simulator’s internal graph has no outgoing edge from the input node s .*

2763 **(C2) Unsaturated.** $\mathcal{R} \cup \mathcal{O} \neq \mathcal{C}$.

2764 **(C3) Rooted input.** *The input node is rooted.*

2765 **(C4) Paddable path.** *The constructed path p decomposes uniquely as $p = p' \parallel 0^r$ with*
 2766 *p' not ending in 0^r , and the prefix p' is a valid padded sponge input: equivalently,*
 2767 *$p' = x \parallel \text{pad10}^*[r](|x|)$ for an unpadded BDPV08 random oracle input x .*

2768 *When all four conditions hold, the forward primitive-interface query induces one simulator*
 2769 *RO_{flat} call. Consequently, each primitive-interface query induces at most one simulator-side*
 2770 *RO_{flat} call.*

2771 *Proof.* This is the branch structure of the [BDPVA08] simple-padded permutation simulator.
 2772 The simulator invokes the random oracle only on the forward-query branch where the input
 2773 node is rooted, unsaturated, paddable to an unpadded random oracle input x , and not
 2774 already extended by an outgoing edge in the simulator graph. Forward queries violating
 2775 any of these conditions and all inverse queries return outputs sampled or read from the
 2776 simulator's internal graph without invoking the random oracle. Under the [BDPVA11,
 2777 Theorem 3] bridge fixed in Section B.1, that simulator random oracle invocation is the
 2778 simulator-side RO_{flat} call used throughout this appendix. $\square$

2779 **Lemma 19** (Operational Duplex Trace). *In any execution of a single-stage TW128 game*
 2780 *(Definition 6), the construction computations counted by N admit an operational primitive*
 2781 *trace of cost at most N : each root or leaf duplex call contributes one forward primitive*
 2782 *query, and the requested output bits are the corresponding projection of the state returned*
 2783 *by that query. When this trace is run against the [BDPVA08] simulator $(\mathcal{S}^{\text{RO}_{\text{flat}}}, \mathcal{S}^{-1, \text{RO}_{\text{flat}}})$*
 2784 *in an execution whose combined primitive trace has cost less than 2^c , the construction*
 2785 *outputs are the flat-RO values $\text{RF}(X) = \text{RO}_{\text{flat}}(\text{flat}(X))$ at the same well-formed TW128*
 2786 *transcript prefixes X .*

2787 *Proof.* A TW128 duplex call absorbs a single suffixed input block, padded by pad10^*1 ,
 2788 applies the permutation once, and returns at most r rate bits from the resulting state.
 2789 Thus the public-permutation implementation of any root or leaf construction computation
 2790 is already a primitive trace with one forward query per duplex call. The definition of
 2791 N in Section 4.3 counts exactly these root initialization, associated data, message, leaf
 2792 initialization, leaf message, and aggregation duplex calls, including the deterministic
 2793 challenger checks of the tag hash and committing games.

2794 Lemma 5 is the [BDPVA11, Lemma 3] duplex-to-sponge identity specialized to TW128:
 2795 the output of each duplex call is the output of the sponge after absorbing the accumulated
 2796 flattened duplex-call transcript, and no additional permutation call is made after the
 2797 current absorption block when the requested output length is at most r . The bridge of
 2798 Lemma 8, fixed in Section B.1, maps these pad10^*1 transcript prefixes to the [BDPVA08]
 2799 padded-sponge input domain without changing the output bits at $r = r_{\text{max}}$.

2800 When the same operational trace is run against $(\mathcal{S}^{\text{RO}_{\text{flat}}}, \mathcal{S}^{-1, \text{RO}_{\text{flat}}})$, every construction-
 2801 internal forward query lies on a rooted, paddable path determined by a well-formed TW128
 2802 prefix. In an execution whose combined primitive trace has cost less than 2^c , the simulator
 2803 cannot have saturated: each primitive query adds at most one capacity value to the
 2804 simulator's internal $\mathcal{R} \cup \mathcal{O}$ set, so fewer than 2^c queries cannot fill the 2^c -element capacity
 2805 space. The simulator's sponge-consistency invariant [BDPVA08, Lemma 2] therefore makes
 2806 the rate part returned at that node agree with the corresponding lazy value $\text{RO}_{\text{flat}}(\text{flat}(X))$.
 2807 Thus the simulator-side operational construction has the same construction-output law as
 2808 the flat-RO construction interface used in Section 4.2. $\square$

2809 **Lemma 20** (Flat Sponge Lift Bound). *Let G be a single-stage TW128 game (Definition 6).*
 2810 *Let $\mathcal{A}$ be a public permutation G adversary with construction interface cost at most*

N and at most N_{prim} direct primitive-interface queries. In the [BDPVA08] single-stage replacement, implement the TW128 construction computations by the operational duplex trace of Lemma 19 and keep the adversary's direct primitive-interface queries as primitive queries. This primitive transcript has [BDPVA08] query cost at most

$$N_{\text{tot}} = N + N_{\text{prim}}.$$

Consequently replacing the real public permutation and TW128 construction by the simulator-world primitive interface $(\mathcal{S}^{\text{RO}_{\text{flat}}}, \mathcal{S}^{-1, \text{RO}_{\text{flat}}})$ and the corresponding flat-RO construction interface costs at most $\text{Lift}_c(N_{\text{tot}})$.

Proof. The [BDPVA08] replacement is applied before the derived-adversary construction. View the entire single-stage game, including its construction interface and deterministic challenger checks, as a primitive-only distinguisher that runs the TW128 algorithms internally. By Lemma 19, the construction side contributes at most N forward primitive queries: one per duplex call. This includes plaintext-computing computations, verification computations whether they accept or reject, the per-user sum of construction calls in mind, the deterministic challenger decryption or encryption checks charged in the committing games, and the tag hash games' challenger encryptions—the collision game's s submitted-input encryptions, the standard preimage game's source encryption and post-output check, and the everywhere-preimage game's post-output check.

Each direct adversary query to (π, π^{-1}) contributes one primitive-interface query and is counted in N_{prim} . These two counts are disjoint in the public permutation experiment: N counts construction-internal use of the primitive, while N_{prim} counts only the adversary's public primitive interface queries.

Construction work is modeled by the operational duplex trace, rather than by a collection of standalone [BDPVA08] construction queries at every flattened duplex prefix. The standalone-query model would assign a nested-prefix transcript the sum of the individual sponge-evaluation costs, while the operational trace is the same stateful computation the construction actually performs, with the output of each call read from the state just returned by that call. Hence the execution-uniform caps bound the primitive-only query cost by $N + N_{\text{prim}} = N_{\text{tot}}$. If $N_{\text{tot}} \geq 2^c$ then $\text{Lift}_c(N_{\text{tot}}) \geq 1$ and the claimed bound holds trivially, so assume $N_{\text{tot}} < 2^c$, satisfying the hypothesis of [BDPVA08, Theorem 2]. Applying that theorem to this primitive-only distinguisher gives the real-to-simulator replacement, and Lemma 19 identifies the simulator-side construction outputs with the flat-RO construction interface in this nontrivial case. The theorem gives distinguishing or success-probability cost at most the random-permutation differentiating bound $f_P(N_{\text{tot}})$ of [BDPVA08, Lemma 5]. The [BDPVA08, Lemmas 4 and 5] product bounds give the random-transformation bound $f_T(N_{\text{tot}}) = 1 - \prod_{i=1}^{N_{\text{tot}}} (1 - i/2^c)$, and the factor of $f_P(N_{\text{tot}})$ corresponding to the f_T factor at index i is that factor divided by a quantity $1 - (i-1)/2^{r+c} \in (0, 1]$, hence at least as large, so $f_P(N_{\text{tot}}) \leq f_T(N_{\text{tot}})$. With $N_{\text{tot}} < 2^c$ every factor lies in $[0, 1]$ and

$$f_P(N_{\text{tot}}) \leq f_T(N_{\text{tot}}) = 1 - \prod_{i=1}^{N_{\text{tot}}} \left(1 - \frac{i}{2^c}\right) \leq \sum_{i=1}^{N_{\text{tot}}} \frac{i}{2^c} = \frac{N_{\text{tot}}(N_{\text{tot}} + 1)}{2^{c+1}} = \text{Lift}_c(N_{\text{tot}}),$$

the second inequality by the Weierstrass product inequality $\prod_i (1 - x_i) \geq 1 - \sum_i x_i$ for $x_i \in [0, 1]$. $\square$

Lemma 21 (Simulator–Permutation Proximity). *Let $\mathcal{D}$ be a distinguisher with access only to a two-directional primitive interface, making at most q queries. Then*

$$|\Pr[\mathcal{D}^{\mathcal{S}^{\text{RO}_{\text{flat}}}, \mathcal{S}^{-1, \text{RO}_{\text{flat}}}} \Rightarrow 1] - \Pr[\mathcal{D}^{\pi, \pi^{-1}} \Rightarrow 1]| \leq \text{Lift}_c(q),$$

where $\pi \leftarrow \$ \text{Perm}(\{0, 1\}^{1600})$ and RO_{flat} is the lazily sampled random oracle internal to the simulator.

Proof. Consider the indistinguishability distinguisher $\mathcal{D}'$ that runs $\mathcal{D}$, forwards its primitive-interface queries, and never queries the construction interface. The [BDPVA08] query cost of $\mathcal{D}'$ is at most q : each forwarded query costs one, and no construction queries arise. Because $\mathcal{D}'$ ignores the construction interface, its two views are exactly $\mathcal{D}^{\pi, \pi^{-1}}$ in the real pair and $\mathcal{D}^{S^{\text{RO}_{\text{flat}}}, S^{-1, \text{RO}_{\text{flat}}}}$ in the ideal pair of [BDPVA08, Theorem 2], applied under the [BDPVA11, Theorem 3] bridge fixed in Section B.1. If $q \geq 2^c$ then $\text{Lift}_c(q) \geq 1$ and the claim is trivial; otherwise that theorem bounds the displayed difference by the random-permutation differentiating bound $f_P(q)$ of [BDPVA08, Lemma 5], and the derivation in the proof of Lemma 20 gives $f_P(q) \leq \text{Lift}_c(q)$. $\square$

Lemma 22 (Derived-Adversary Execution Contract). *Let G be a single-stage TW128 game (Definition 6). Let $\mathcal{A}$ be a simulator-world G adversary whose public primitive interface is answered by $(S^{\text{RO}_{\text{flat}}}, S^{-1, \text{RO}_{\text{flat}}})$, with at most N_{prim} direct primitive-interface queries. Define $\mathcal{B}_{\text{derived}}(\mathcal{A})$ to run $\mathcal{A}$ internally, answer each primitive-interface query by running the same simulator against direct finite-output queries to its own RO_{flat} oracle, forward construction interface queries unchanged in games with construction oracles, and forward the adversary's final structured output unchanged in the output-checked H_{TW128} tag hash and committing games. Then the simulator-world execution of $\mathcal{A}$ and the random oracle execution of $\mathcal{B}_{\text{derived}}(\mathcal{A})$ have identical joint distributions, and*

$$q_{\text{RO}}(\mathcal{B}_{\text{derived}}) \leq N_{\text{prim}}.$$

In this coupling, simulator-induced RO_{flat} calls and construction-internal RF lookups use the same lazy table. Therefore, if a simulator-induced query is the bridged input $\text{flat}(X)$ of a well-formed TW128 transcript prefix X and a forwarded construction computation later reaches the same prefix X , both interfaces read the same $\text{RO}_{\text{flat}}(\text{flat}(X))$ value, projected to the requested output length. Moreover, $\mathcal{B}_{\text{derived}}$ inherits the construction-side caps of $\mathcal{A}$:

$$\begin{aligned} q_v(\mathcal{B}_{\text{derived}}) &= q_v(\mathcal{A}) \quad (AE, \text{mu-ind}, \text{INT-RUP}), \\ n_{\text{leaf}}(\mathcal{B}_{\text{derived}}) &\leq n_{\text{leaf}}(\mathcal{A}) \quad (\text{all games above}). \end{aligned}$$

In the mu-ind game these resource inheritances hold per user, and the New-query transcript is unchanged.

Proof. Couple the two executions by using the same lazy RO_{flat} table for the simulator and for all construction-internal $\text{RF}(\cdot)$ lookups. In the simulator-world execution, $\mathcal{A}$ sees construction answers from the selected game interface and public primitive answers from $(S^{\text{RO}_{\text{flat}}}, S^{-1, \text{RO}_{\text{flat}}})$. In the derived random oracle execution, $\mathcal{B}_{\text{derived}}$ runs the same $\mathcal{A}$, forwards construction interface queries or the final structured output required by the selected game unchanged, and answers primitive-interface queries by invoking the same simulator code with the same lazy RO_{flat} table. The construction interface, tag hash challenger, or committing challenger therefore receives the same forwarded queries or final output, the simulator receives the same primitive queries in the same order, and the lazy table supplies the same values. The full views of $\mathcal{A}$ in the simulator-world execution and of $\mathcal{B}_{\text{derived}}$ in the random oracle execution are identically distributed.

The simulator's random oracle calls may be on arbitrary [BDPVA08] H -input strings, not only bridged TW128 transcript inputs. These calls are the direct RO_{flat} queries of $\mathcal{B}_{\text{derived}}$ and are counted in $q_{\text{RO}}(\mathcal{B}_{\text{derived}})$. Construction-internal $\text{RF}(\cdot)$ lookups made by Enc_K^{RO} , Dec_K^{RO} , $\text{PDec}_K^{\text{RO}}$, Ver_K^{RO} , or a committing challenger are forwarded to the same RO_{flat} table but are not counted in q_{RO} , exactly as in Section 4.3.

By Lemma 18, each primitive-interface query induces at most one direct RO_{flat} query by $\mathcal{B}_{\text{derived}}$, and inverse queries or non-RO-touching forward queries induce none. Since $\mathcal{A}$ makes at most N_{prim} direct primitive-interface queries, this gives $q_{\text{RO}}(\mathcal{B}_{\text{derived}}) \leq N_{\text{prim}}$.

Lemma 8's bridge places TW128 construction restrictions and simulator-induced RO_{flat} calls in a single lazy table. If the simulator invokes RO_{flat} on $\text{flat}(X)$ for a well-formed

TW128 transcript prefix X , and a forwarded construction computation later reaches X , the construction lookup $\text{RF}(X) = \text{RO}_{\text{flat}}(\text{flat}(X))$ reads the same table entry and returns the requested projection. Since TW128 uses $r = r_{\text{max}}$, the [BDPVA11, Theorem 3] output post-processing does not alter TW128 output bits. Thus simulator-induced direct queries and construction-internal lookups at the same bridged input are transcript-consistent. In the AE-style games, a simulator call whose input is accepted by KeyedTWOut for the hidden key is still a direct RO_{flat} query and is covered by the same bad_{key} accounting used by the direct random oracle mu-ind proof. The derivation step itself does not prove a key-guessing statement; it only supplies the derived adversary and the query bound at which the random oracle bounds of Sections 5 and 6 are instantiated.

Finally, the execution-uniform caps of Section 4.3 hold along every oracle-answer trace of $\mathcal{A}$, including traces generated by the simulator on the ideal public primitive interface. The derived-adversary construction does not add construction interface encryption, plaintext-computing, or verification queries and forwards $\mathcal{A}$'s construction interface queries unchanged along each trace. $\mathcal{B}_{\text{derived}}$ therefore inherits the relevant construction resources: $q_v(\mathcal{B}_{\text{derived}}) = q_v(\mathcal{A})$ in the AE, mu-ind, and INT-RUP games, and $n_{\text{leaf}}(\mathcal{B}_{\text{derived}}) \leq n_{\text{leaf}}(\mathcal{A})$ for the distinct construction-generated final-leaf-transcript cap. In mu-ind, construction interface forwarding preserves the New calls and these caps user by user. $\square$

B.2 Lift Application

The lift bound of Lemma 20, the simulator-permutation proximity bound of Lemma 21, and the derived-adversary construction of Lemma 22 combine into a single template, stated once and then instantiated by the tag hash bounds of Section 5, the committing consequences of Corollary 5, and the baseline AEAD bounds of Section 6. Every public-permutation bound is obtained by adding $\text{Lift}_c(N_{\text{tot}})$ —plus $\text{Lift}_c(N_{\text{prim}})$ in the real-vs-ideal games—and substituting $q_{\text{RO}} \leq N_{\text{prim}}$ in the corresponding random oracle theorem.

Corollary 9 (Lift Application). *Let G be a single-stage TW128 game (Definition 6). For every public permutation G -adversary $\mathcal{A}$ against TW128 with the resources of Section 4.3, the derived adversary $\mathcal{B} = \mathcal{B}_{\text{derived}}(\mathcal{A})$ of Lemma 22 is a random oracle G -adversary with $q_{\text{RO}}(\mathcal{B}) \leq N_{\text{prim}}$ and the construction-side caps of Lemma 22—in particular $q_v(\mathcal{B}) \leq Q_v$ and $n_{\text{leaf}}(\mathcal{B}) \leq N_{\text{leaf}}$, where these apply—such that, when G is a win-event game,*

$$\text{Adv}_{\text{TW128}}^{\mathsf{G}}(\mathcal{A}) \leq \text{Lift}_c(N_{\text{tot}}) + \text{Adv}_{\text{RO}}^{\mathsf{G}}(\mathcal{B}),$$

and, when G is a real-vs-ideal game (all-in-one AE or mu-ind),

$$\text{Adv}_{\text{TW128}}^{\mathsf{G}}(\mathcal{A}) \leq \text{Lift}_c(N_{\text{tot}}) + \text{Adv}_{\text{RO}}^{\mathsf{G}}(\mathcal{B}) + \text{Lift}_c(N_{\text{prim}}).$$

If moreover $\text{Adv}_{\text{RO}}^{\mathsf{G}}(\mathcal{B}) \leq \varepsilon_{\mathsf{G}}(q_{\text{RO}}(\mathcal{B}), n_{\text{leaf}}(\mathcal{B}))$ for some ε_{G} nondecreasing in each argument, then $\varepsilon_{\mathsf{G}}(N_{\text{prim}}, N_{\text{leaf}})$ may replace $\text{Adv}_{\text{RO}}^{\mathsf{G}}(\mathcal{B})$ in the corresponding bound.

Proof. By the flat sponge lift bound (Lemma 20), replacing the real public permutation and TW128 construction by $(\mathcal{S}^{\text{RO}_{\text{flat}}}, \mathcal{S}^{-1, \text{RO}_{\text{flat}}})$ and the flat-RO construction interface costs at most $\text{Lift}_c(N_{\text{tot}})$, where $N_{\text{tot}} = N + N_{\text{prim}}$ counts both the construction calls of N —including any construction oracle exposed during $\mathcal{A}$'s run and, in the win-event games, the challenger's construction computations, pre- and post-output (in the standard preimage game, the source encryption computing T^* as well as the post-output check)—and the N_{prim} direct primitive queries. The derived-adversary contract (Lemma 22) expresses the resulting simulator-world execution as the identically distributed random oracle execution of $\mathcal{B} = \mathcal{B}_{\text{derived}}(\mathcal{A})$, with $q_{\text{RO}}(\mathcal{B}) \leq N_{\text{prim}}$ and the stated construction-side inheritances. For a win-event game this gives the first bound.

For a real-vs-ideal game, chain four worlds: W_1 , the real construction interface with (π, π^{-1}) ; W_2 , the flat-RO real construction interface with the simulator; W_3 , the ideal

construction interface with the simulator; and W_4 , the ideal construction interface with (π, π^{-1}) . The real-vs-ideal games of Sections 6.1 and 6.2 compares W_1 with W_4 . The step from W_1 to W_2 costs at most $\text{Lift}_c(N_{\text{tot}})$ as above, and under the coupling of Lemma 22 the pair (W_2, W_3) is precisely the real/ideal pair of the random oracle game for $\mathcal{B}$, so $|\Pr[\mathcal{A}^{W_2} \Rightarrow 1] - \Pr[\mathcal{A}^{W_3} \Rightarrow 1]| \leq \text{Adv}_{\text{RO}}^G(\mathcal{B})$. For the step from W_3 to W_4 , the ideal construction oracles—(Rand, Replay), instantiated per user in mu-ind, together with the New interface and the validity checks—are independent of the primitive, so a distinguisher can answer them from its own coins, run $\mathcal{A}$, and forward the at most N_{prim} primitive-interface queries; Lemma 21 bounds this step by $\text{Lift}_c(N_{\text{prim}})$. The triangle inequality combines the three estimates into the second bound.

The final claim follows by the monotonicity of ε_G . When G is parameterized by a challenge fixed before $\mathcal{A}$ runs—the everywhere-preimage target tag, or the standard preimage source X^* , whose tag $T^* = H_{\text{TW128}}(X^*)$ each coupled world computes—the coupling holds for each fixed challenge and the challenge-independent bound holds for the maximum or average over challenges defining $\text{Adv}_{\text{TW128}}^G(\mathcal{A})$. $\square$

C Vectorized Kernel Details

This appendix details the optimized cores summarized in Section 8.1: their shared batching structure, chunk 0 fusion, instruction selection, register layout, remainder handling, and data-independent addressing.

The optimized paths share the same logical decomposition but differ in their batch widths, remainder strategy, and state writeback points:

Path	Cores and writeback
amd64 AVX-512	Wide: one eight-state ZMM group. Remainder: masked 2–7 state AVX-512 core. Single duplex: AVX-512 permutation. Chunk 0: all active states are written back through <code>state8</code> .
amd64 AVX2	Wide: two four-state YMM groups. Remainder: 2–7 states with dummy inactive lanes. Single duplex: general purpose register permutation. Chunk 0: all active states are written back through <code>state8</code> .
arm64 NEON+SHA3	Wide: four two-state NEON pairs. Remainder: two-state pair core. Single duplex: NEON/SHA-3 permutation. Chunk 0: pair (0, 1) is written back in the wide core, and both states are written back in the pair core.

Shared batching structure. The implementation separates the sequential root duplex from the independent leaf duplexes. The root duplex normally uses a single-duplex ($\times 1$) core, while complete leaf chunks are grouped into eight-instance batches. Every complete leaf chunk is exactly $\beta = 49\rho$ bytes, so all eight instances in a batch follow the same sequence of 49 full ρ -blocks: the first 48 close with `MSG_MORE`, and the last closes with `MSG_LAST`. A single public block index therefore drives the whole batch. The batched state is lane-major: for each of the 25 Keccak lanes, the kernel stores the corresponding 64-bit word of each active instance together in vector registers. Equivalently, for Keccak lane i the vector register holds $(S_0[i], \dots, S_7[i])$ across the active duplex states, so one vector operation updates the same lane in several states.

Chunk 0 fusion. Chunk 0 is part of the root transcript, but its message phase has the same kernel-visible block schedule as a complete leaf chunk. The optimized paths exploit this by placing the post-AD root state into lane 0 of the first batched call and placing the first leaf chunks in the remaining lanes. When the message has at least seven complete leaf chunks, this fused call uses the $\times 8$ core; below that, it uses the same remainder machinery

that drains incomplete batches. The output state from lane 0 is written back to the root duplex, which then absorbs the leaf tags produced by the same call.

Remainder and tail handling. Messages are otherwise decomposed into full eight-chunk batches plus a remainder. On **amd64**, a register-resident remainder kernel handles two to seven complete chunks in one call, reading each active chunk in place; on **arm64**, a 2-wide kernel drains the remainder two chunks at a time. A single leftover complete chunk, a partial trailing chunk, and the count-aggregation tail use the single-duplex core. The **amd64** assembly chunk kernels handle the 48 full `MSG_MORE` blocks and leave the final `MSG_LAST` block plus leaf tag extraction to shared Go code, keeping byte-granular tail logic out of assembly. The **arm64** batched kernels process the complete chunk and store back the states needed for chunk 0 fusion: pair (0, 1) in the $\times 8$ kernel, and both active states in the 2-wide kernel.

amd64. The **amd64** implementation provides AVX-512 and AVX2 cores and selects between them once at startup from an AVX-512 feature probe requiring OS support for YMM/ZMM state, **AVX512F**, and **AVX512VL**. The single-duplex permutation has two variants: a general-purpose-register core adapted from the standard library’s **KECCAK- p** permutation (using the lane-complement trick so that χ requires no separate negation), and an AVX-512 variant; the root duplex uses whichever matches the selected core.

The eight-way AVX-512 core packs the lane-major state into 25 ZMM registers, one per lane with all eight instances in the register’s eight 64-bit slots, and performs the permutation with no cross-lane shuffles: rotations use **VPROLQ**, and **VPTERNLOGQ** fuses the θ D -term formation and the χ and-not into single three-input logic instructions. The fused chunk kernel reads the eight chunks with contiguous loads, transposes each rate block into the lane-major layout in registers (**VPUNPCKLQDQ**/**VPUNPCKHQDQ** and **VSHUFI64X2**), absorbs on the lane-major state, and transposes each output block back for a contiguous store. Since $\rho = 167 = 20 \cdot 8 + 7$, the first 20 rate lanes use this transposed path and only the trailing 7 bytes in lane 20 use a masked gather. These sequential accesses are prefetcher-friendly, whereas per-lane gathers and scatters at the chunk stride β (**VPGATHERQQ**/**VPSCATTERQQ**) would stall on cold memory and cap the long-message rate.

The register-resident kernel that drains a two-to-seven-chunk AVX-512 remainder instead uses that per-lane gather/scatter form under a mask, reading the active chunks in place: the transpose’s shuffle work is constant regardless of the chunk count and would be spent partly on inactive lanes, whereas the gathers scale with the active element count.

The eight-way AVX2 core, lacking 512-bit registers, runs the eight instances as two groups of four (four 64-bit lanes per YMM register), rotating by shift-or and computing χ with **VPANDN**. The AVX2 path drains remainders with a variant of the grouped-by-four kernels that aims each inactive instance’s loads and stores at a dummy block inside the kernel frame with a zero advance stride, so inactive instances touch no memory beyond the remainder and the hot loop stays branch-free.

arm64. The **arm64** path uses NEON plus the Armv8.2 SHA-3 extension (**FEAT_SHA3**), available on Apple Silicon and on Neoverse V1/V2 and N2. The permutation is expressed with the extension’s fused instructions: **EOR3** (three-input XOR) for the θ column parities, **RAX1** (rotate-by-one and XOR) for the θ D -terms, **XAR** (XOR then rotate) to fuse the θ accumulation with the ρ rotation, and **BCAX** (bit-clear and XOR) for χ . The single-duplex core stores each Keccak lane in one 64-bit D lane across the 25 vector registers. The eight-way core packs two instances per 128-bit register and processes the batch of eight as four such pairs, using **ZIP1** and duplicating moves to interleave and extract the two instances of each lane in the fused chunk kernel. The same pair machinery backs a 2-wide kernel that drains a leaf-chunk remainder two chunks at a time—one instance per 64-bit

half of a 128-bit register—at roughly twice the single-duplex per-chunk rate. Both batched kernels store the permutation state back into the batch after the closing block—the $\times 8$ kernel for its first pair, the 2-wide kernel for both instances—which is the writeback that permits the root duplex to occupy lane 0.

Data-independent addressing. The optimized kernels contain no secret-dependent branches and no secret-dependent memory or vector indices. The only indexed table is the round-constant array, addressed by the public round number. The AVX-512 steady-state transpose uses contiguous loads and stores; the remaining gathers and scatters, for the partial rate lane and register-resident remainder kernel, use fixed public chunk strides under masks determined by the public chunk count. The AVX2 remainder kernel redirects inactive instances to a dummy block in its stack frame with a zero advance stride, again selected only by the public chunk count. The **arm64** pair kernels use fixed lane interleaving and extraction patterns. None of these addresses depends on the key, nonce, plaintext, or decrypted plaintext.

D Test Vectors

The following selected known-answer test vectors exercise TW128 encryption and are intended for implementation conformance checks against the construction in Section 3. The complete vector set, including full byte strings for the long cases and the regeneration harness, is maintained with the reference implementation at <https://github.com/codahale/treewrap>.

All byte strings are written in hexadecimal. For each vector we list the key K , nonce U , associated data A , message M , ciphertext C , and authentication tag T ; the sealed output of encryption is $C \parallel T$. The empty string is written ε . The parameters used below are $\rho = 167$ bytes, $\beta = 8183$ bytes, and 32-byte keys, nonces, and tags. The leaf index is encoded as a little-endian 64-bit integer and the phase suffixes are appended least significant bit first.

To keep the longer vectors compact, any value exceeding 32 bytes is given by its length in bytes together with its SHA-256 digest rather than in full. For generated associated data and message inputs, byte i is defined as $x_i = (17i + 31) \bmod 256$ for $i = 0, 1, \dots$, so each such input is determined by its length alone and can be reconstructed and checked against the stated digest; a ciphertext given by its length and digest can likewise be checked by encrypting the corresponding message and hashing the result.

Vector 1. Empty associated data and empty message: the associated data and aggregation phases are both elided, and the root tag is emitted by the single closing message call with no leaves.

K	000102030405060708090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f
U	202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f
A	ε
M	ε
C	ε
T	da65bbda0b20b1e936f5326458166633d1e5a3f7aebc0b318d24b051a4efa697

Vector 2. Multi-block associated data with an empty message: the associated data spans two rate blocks and no message bytes are processed.

K	000102030405060708090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f
U	202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f
A	168 bytes, SHA-256 4c7d125a3488a7603e2f403f0010a6060b8321b60b0fd2471120ae627bbe759a
M	ε
C	ε
T	2093878418f3fca96ad3a8d1748a9be4d2a8b518e917d988fd703dcb7bf72173

Vector 3. Empty associated data and a single-block message: the root message phase only, no leaves.

3074 *K* 000102030405060708090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f
U 202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f
A ϵ
M 1f30415263748596a7b8c9daebfc0d1e2f405162738495a6b7c8d9eafb0c1d2e
C 676a1441e20daea0f1ee4571ee4543842c946d59e0d3df6f1dd2f92dc4f876aa
T 1873332d359ff2fe5197488224af7bf6ad821875fdbd964100de424aa77342be

Vector 4. Empty associated data and a two-block message: the root keystream is chained across two rate blocks.

3075 *K* 000102030405060708090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f
U 202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f
A ϵ
M 168 bytes, SHA-256
4c7d125a3488a7603e2f403f0010a6060b8321b60b0fd2471120ae627bbe759a
C 168 bytes, SHA-256
df3f0a98a91a69ed474edbc52e0dc1d8e54bfd6fc514966ce7b878d48be0bf9a
T aab2413074a202cfadd5eb784663b1831eb330025972d163d03e9ed5a964b765

Vector 5. A message of exactly one chunk: the root is full, there is still no leaf, and the aggregation phase is elided so the closing root message call emits the tag.

3076 *K* 000102030405060708090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f
U 202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f
A ϵ
M 8183 bytes, SHA-256
6532a370a4ab6b5f4094dd8a6016512ab8e8416abe910bc1a0b532979224151d
C 8183 bytes, SHA-256
8d58f8aa3413ea0ba1b8cc3ed4e9a334521d8dfe9694a0661fb8251049907e9
T c2201637f1116c1303b8982622756770d23acb85384f4658fcb63c37e4b629dd

Vector 6. A message of one chunk and one further byte: one root chunk and one leaf, exercising leaf tag aggregation.

3077 *K* 000102030405060708090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f
U 202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f
A ϵ
M 8184 bytes, SHA-256
c4c383bca0808e2ce44d481ddfe56efdc7a147dcd5a14c898ef5423f2604cd7
C 8184 bytes, SHA-256
ef1b2f7607ade503ab497fc61e62cef2b44c6328fa06e499718eb7da5fdd04d3
T c96bea48e07a8431fd19721ffbe8afe6a73047f96d953e3281f10126eeca06a1

Vector 7. A multi-leaf message of two chunks and one further byte: the full tree path with two leaves.

3078 *K* 000102030405060708090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f
U 202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f
A ϵ
M 16367 bytes, SHA-256
17e31f3700d0602aab8a686024ecc05b8192e80a531483c67dd90d5a26557c5e
C 16367 bytes, SHA-256
6d83d8fbdca2a7c2ef590d29a36527a45019e50d1f8c8db037959081685a46908
T cdc2e50ac5380419f06772a2f3af2146050bd0340b35f61b67f839351767804e

Vector 8. Associated data spanning multiple blocks with a short message: exercises associated data absorption before message encryption.

3079 *K* 000102030405060708090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f
U 202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f
A 337 bytes, SHA-256
04ca7b24930bc6b5fe8b95ff1eb0933907657c1283cc3a7e69c5f7f02220f281
M 1f30415263748596a7b8c9daebfc0d1e
C Odd615a0a05ed82a78c3999b9e80cbcd
T 22e4e6725f548ad0660041d66cf2f18dc45c69b447cdaae1f147d33a3b6aa1a9

Vector 9. An all-ones key and nonce with a multi-chunk message.

<i>K</i>	ff
<i>U</i>	ff
<i>A</i>	168 bytes, SHA-256 4c7d125a3488a7603e2f403f0010a6060b8321b60b0fd2471120ae627bbe759a
<i>M</i>	8184 bytes, SHA-256 c4c383bca0808e2ce44d481ddfe56efdcd7a147dcd5a14c898ef5423f2604cd7
<i>C</i>	8184 bytes, SHA-256 49818c320ab282296639798d840cbd60751e1623f449dcbf9423371d44145dfb
<i>T</i>	310c9fff20e9651f8ed47eed9bb182a086ed1af1b3f1799e35307a4139865fc7
